

possible pair of genes to determine whether a functional relationship exists between them. This is determined using a metric, such as Euclidean distance between the genes (when genes are positioned in a multidimensional space with each experiment as a separate axis), Pearson's correlation coefficient (assumes a linear model of control between the two genes), or mutual information (with no assumptions as to the model of control). The end result of this analysis is a ranked list of hypotheses of pairs of genes that work together. A taxonomy of techniques divided in this manner would look like this:

1. **Unsupervised:** Analysis looking for characterization of the components of the data set, without a *priori* input on cases or genes.

a. **Feature determination:** Determine genes with interesting properties, without specifically looking for a particular pattern determined *a priori*.

i. Principal components analysis: Determine genes explaining the majority of the variance in the data set.

b. **Cluster determination:** Determine groups of genes or samples with similar patterns of gene expression.

i. Nearest neighbor clustering: The number of clusters is decided first, the clusters are calculated, then each gene is assigned to a single cluster.

A. Self-organizing maps: Groups are found by iteratively moving each cluster centroid toward a randomly chosen data point; the center of each cluster is a hypothetical gene expression pattern.

B. *K*-means clustering: Similar to self-organizing maps, except there is no map structure for the centroids.

ii. Agglomerative clustering: Bottom up method, where clusters start as empty, then genes are successively added to the existing clusters.

A. Dendrograms: Groups are defined as sub-trees in a phylogenetic-type tree created by a comprehensive pairwise dissimilarity measure, such as the correlation coefficient.

B. Two-dimensional dendrograms: Both genes and samples are clustered separately.

iii. Divisive or partitional clustering: Top-down method, where large clusters are successively broken into smaller ones, until each subcluster contains only one gene.

A. Dendrograms.

B. Two-dimensional dendrograms.

c. **Network determination:** Determine graphs representing gene-gene or gene phenotype interactions.

i. Boolean networks: Gene expression measurements are binarized; nodes in the graph represent an entire expression state, and links represent transitions from one expression state to another.

ii. Bayesian networks: Nodes in the graph represent features (genes or phenotypic measures), and links between features are determined probabilistically.

iii. Relevance networks: Nodes in the graph represent features (genes or phenotypic measures); links between features are calculated with pairwise

application of a dissimilarity measurement, and networks represent agglomerations of these pairwise associations.

2. **Supervised:** Analysis to determine ways to accurately split into or predict groups of samples or diseases.

a. **Single feature or sample determination:** Find genes or samples that match a particular *a priori* pattern

i. Nearest neighbor: Find genes or samples closest to another pattern, using a dissimilarity measure, such as Euclidean distance or the correlation coefficient.

A. Comparison to hypothetical pattern: Find genes or samples closest in similarity to a hypothetical pattern, determined with *a priori* information.

B. Comparison to actual pattern: Find genes or samples closest in similarity to one of the genes or samples in the data set.

C. Comparison to clusters: First, remember the training set of data and labels; then, when presented with a test gene or sample, find the element of the training set that is closest and use its label.

ii. Student's *t*-test: Find genes where the expression measurements are statistically different between groups of samples

b. **Multiple-feature determination:** Find combinations of genes (i.e., more than one) that match a particular *a priori* pattern.

i. Decision trees: Use the training set of genes or samples to construct a decision tree to help classify test samples or test genes.

ii. Support vector machines: First take the set of measured genes, then create a richer feature set with combinations of genes, then find a hyperplane that linearly separates groups of samples in this larger multidimensional space.

The hierarchy above is meant to serve as a way to categorize bioinformatics techniques, and the examples under each branch of this hierarchy are incomplete; newer methodologies in each branch are shown in section 4.10.

This hierarchy just touches upon one of the important goals of the bioinformatics arm of functional genomics. That is, to ascertain gene regulatory networks from gene expression data sets. Few of the algorithms listed above find their way into this endeavor for reasons that we address here.

One crude definition of a regulatory network is a set of genes *G* and a set of relations *R* that relates any two elements from *G*. The relation may specify either a positive or a negative interaction. For example (*A*, *B*, +) may indicate gene *A* interacts with gene *B* in a positive manner, such that the level of expression of gene *B* increases as the level of expression of gene *A* increases. Completely missing are alternative models other than these simple linear models of gene–gene interactions. Any element of translation into protein, the actual actions of proteins in DNA binding, any sense of temporality, and gene and protein degradation are missing. However, despite all that is missing, there is enough captured in this simple model to provide material to construct a biological hypothesis linking the two genes. [177]

In the above hierarchy, we do not consider a dendrogram to be a member of the network determination category of algorithms. Why is this? One typically constructs dendrograms from a gene expression data set to survey the types of expression patterns, and to determine the

neighbors of genes to be classified. Although dendrograms are computed from a comprehensive pairwise calculation of gene–gene expression pattern similarity, as detailed below, these gene–gene links are not shown in the output. Besides the first gene–gene pair, each subsequent gene is added to the dendrogram when its expression pattern matches an expression pattern of a branch. These branch expression patterns represent the average expression pattern of the descendant genes, and may not match any one of these genes. Furthermore, gene–gene interactions with negative correlations are not represented in dendrograms, thereby eliminating a large class of biological regulatory behavior.

In the same vein, why do we not consider a self–organizing map as part of the network determination category of algorithms? Self–organizing maps are typically made to survey the types of expression patterns seen in a gene expression data set. Like dendrograms, self–organizing maps do not display gene–gene interactions; instead, these typically output a centroid expression pattern (essentially, an average expression pattern not matching anyone of the genes in the data set) and output the expression patterns of the genes surrounding the centroid. There is no representation of a gene–gene interaction, nor are negative interactions considered.

^[1]For example, human expert–derived labels.

4.5 Data Reduction and Filtering

Clustering algorithms can easily be biased if certain assumptions they hold are not met. Each dissimilarity measure has its own set of assumptions and requirements. For example, the reliability of the correlation coefficient not only assumes that two data sets are normally distributed but that they are easily biased by outlying points (see figure 4.1). Another example is that genes with expression measurements that are constant across samples may still show variation due to measurement noise, and if these genes are not filtered out, such measurement noise can be amplified by normalization and can appear as a true signal. Four common methods for filtering genes meeting these degenerate cases are given here.

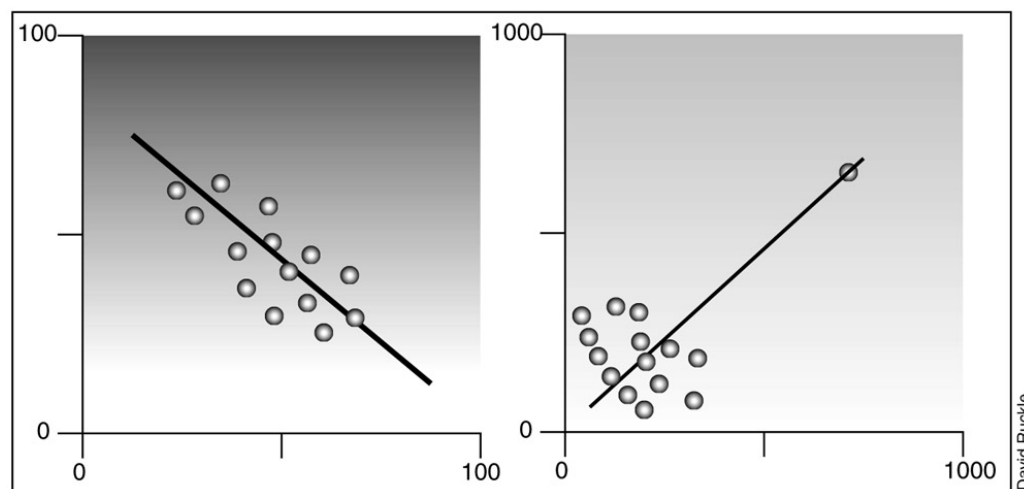


Figure 4.1: Example of how a single point can distort overall correlation. On the left: the scatterplot shows a negative correlation. The scatterplot on the right is identical, save for the additional point added. If the values of even a single point are high enough, the correlation coefficient can be altered (though the variance of the correlation coefficient, if calculated, would be higher). This is primarily because using the correlation coefficient assumes values distributed normally.

4.5.1 Variation filter

One way to prevent degenerate clustering is to remove genes or samples that are changing "too little." A variation filter is used to ensure that genes or samples have a wide enough dynamic range of measurements. Dynamism can be measured in both absolute and relative terms. For example, a gene could be removed from further analysis if there is less than a two fold difference in the measurements across the samples (a relative term) or if there is less than 35 expression units of difference in the measurements across the samples (an absolute term).

```
make a new empty list called post_filter
loop through each gene G to be filtered
  find the minimum expression level for G
  find the maximum expression level for G
  set the flag keep_gene to false
  if maximum - minimum is over 35
    set keep_gene to true
  if maximum / minimum is over 2
    set keep_gene to true
  if the flag keep_gene is true
    add this gene G to the post_filter list
end loop
```

There are those investigators who will object to this kind of filter because there may well be biologically important changes in expression that are very small and will be removed by this variation filter. However, if functional genomics experiments are seen as a funnel (see figure 1.6) for determining which are the high-yield hypotheses to pursue for biological testing, then the high false-positive rate of genes hypothesized wrongly to be biologically relevant may prove to be costly (see figure 2.9 in section 2.1.4).

4.5.2 Low entropy filter

The opposite problem can also occur with gene expression patterns. Genes can demonstrate spiking behavior, where low expression levels are seen in all samples except one. The single high expression can dominate a pairwise analysis using correlation coefficients, for example (see figure 4.1 for an example) [94].

An entropy filter can be used to remove genes that demonstrate spiking behavior, or, in other words, that are not well distributed over its range of values. Entropy is a measure of the amount of disorder in a variable. If measurements of a variable are equally distributed across its dynamic range, then very little information is known about the expected value for the variable. Once the measurement is known, the amount of information about the variable increases greatly, meaning such a variable has high entropy. Compare this to a variable that can be measured at only two values. A great deal of *a priori* information is already known about the expected value for this variable; thus, once the measurement is actually known, the amount of information increase is less. This indicates the variable has lower entropy (see figure 4.2).

$$H(x) = - \sum_i p(x_i) \log_2(p(x_i))$$

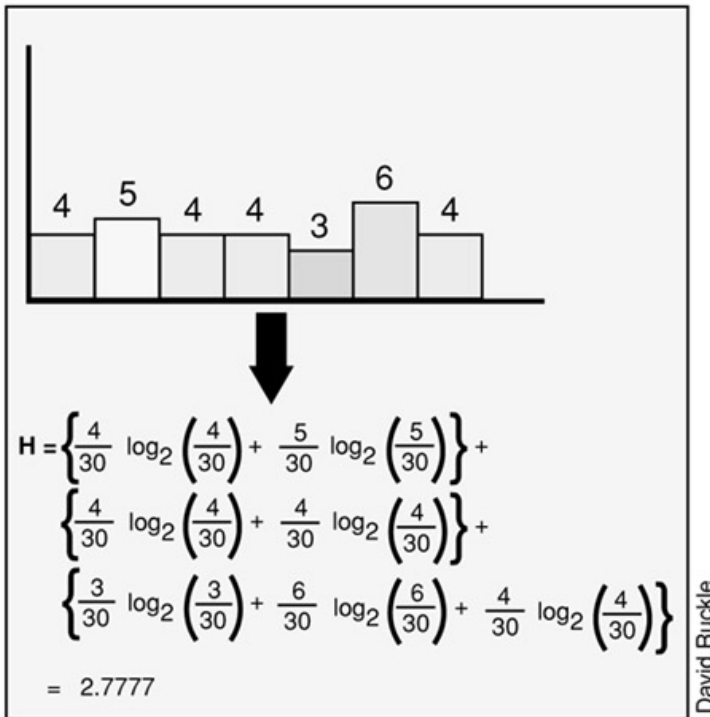
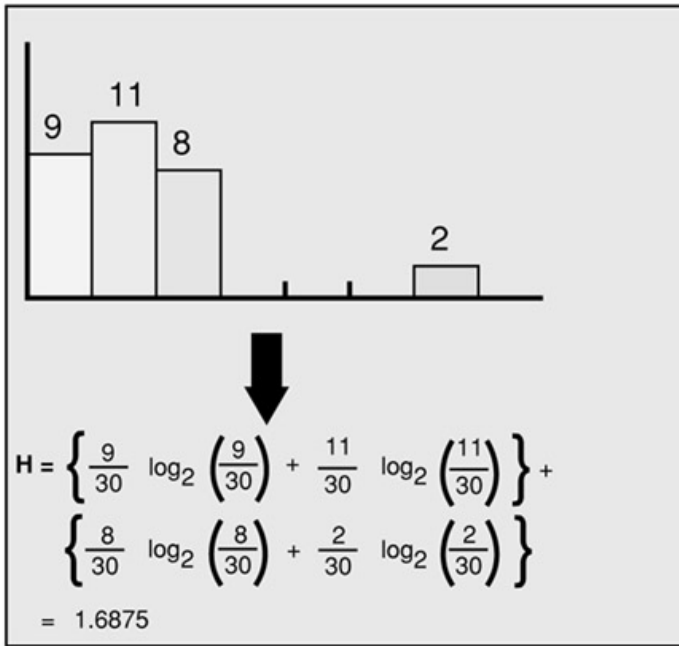


Figure 4.2: Detecting spikes by calculating entropy. The top graph shows a hypothetical gene with "spiking" behavior; *i.e.*, the gene expression is markedly higher in two samples, compared to the other samples. The bottom graph shows a second hypothetical gene with gene expression measurements that are more distributed across the dynamic range. Distribution of the gene expression measurements are not just a characteristic of the gene; it is also connected to the samples in which expression was measured.

where x is the variable whose entropy H is being calculated, $\log_2$ is base 2 logarithm, and $p(x_i)$ is the probability a value^[2] of x was within quantile i of that feature. For example, if one were using 10 quantiles, and a gene with the expression amounts 20, 22, 60, 80 and 90 would have deciles 7 units wide, with two values in the first decile, one in the sixth decile, and one in the ninth and tenth decile, making $H = .92$. This pseudocode accomplishes the same.

```

define number_bins as the number of bins to use in entropy
  calculations
make a new empty list called post_filter
loop through each gene G to be filtered
  find the minimum expression level for G
  find the maximum expression level for G
  set expression_range to maximum - minimum
  set interval to expression_range divided by the number_bins
  make a new empty array called deciles, with size = number_bins
  set deciles[1] to interval
  loop through the rest of deciles[2..number_bins] using the
    index i
    set deciles[i] to deciles[i-1] plus interval
  end loop
make a new empty array called decile_count, with
  size = number_bins
loop through all the expression measurements for G
  set the flag bin_found to false
  loop through deciles[1..number_bins] using the index i
    if the expression measurement is <=
      deciles[i] and not bin_found
      add one to decile_count[i]
      set the flag bin_found to true
    end loop
  end loop
set entropy to zero
loop through decile_count[1..number_bins] using the index i
  if decile_count[i] is over zero
    set probability_in_bin to decile_count[i] divided by
      the number of expression measurements in G
    add to entropy (probability_in_bin times
      log-base-2( probability_in_bin ) )
  end if
end loop
set entropy to -1 times entropy
end loop

```

After entropy is calculated for each gene or sample, a threshold low entropy (e.g., at the lower 5th percentile) can be chosen and used to filter out genes that do not display a sufficient range of values. In using this technique, however, one needs to note that the calculation of entropy does directly depend on the number of quantiles used, and the use of 10 deciles is arbitrary.

As an alternative method, Heyer *et al.* [94] proposed the use of the jackknife correlation coefficient to counter the spiking problem. The jackknife correlation coefficient is an alternative dissimilarity measure to the standard (Pearson's) correlation coefficient. To compute this measure for two genes measured in n samples, the technique involves computing n different correlation coefficients, each time with one of the samples removed. The jackknife correlation coefficient is then the minimum of the separate correlation coefficients. Use of this dissimilarity measure will effectively remove single outliers. However, it is not clear whether removing a single outlier is enough, especially when samples from many disparate tissues are used, and several spikes are seen. It is also questionable whether removing spikes is even desirable, because when two genes demonstrate spiking behavior in the same samples, it may reflect something biologically interesting.

It is important to note that both variation filters and low entropy filters remove entire genes from downstream analysis. If the subsequent analysis is being used for hypothesis generation, this means that hypotheses containing the filtered genes will not be created or considered. Yet again,

the cost–benefit analysis of section 2.1.4 should be considered.

4.5.3 Minimum expression level filter

As described elsewhere in this book, it is commonly believed that low gene expression measurements are less reliable than high gene expression measurements. Empirically, this means one's confidence interval around a measurement differs as a function of the expression level. If the low expression levels are particularly noisy, this can cause artifacts in the downstream clustering process, or worse, cause the creation of incorrect clusters. One solution to this is to ignore those gene expression measurements with a measurement under a threshold value. Threshold minimum expression values can be calculated arbitrarily at a percentile of the distribution of expression measurements (*e.g.*, at the lower 5th percentile), determined empirically at an ideal point for clustering sensitivity and specificity [40]. Here, we show pseudocode that removes genes using the first of these three methods.

```
loop through each gene G to be filtered
  find the minimum expression level for G
  find the maximum expression level for G
  set expression_range to maximum - minimum
  set lowest_5th_percentile to expression_range times 0.05
  loop through all the expression measurements for G
    if the expression measurement is under lowest_5th_percentile
      remove that measurement from G
    end loop
  end loop
end loop
```

Note that with this type of filter, we are removing gene expression measurements, not entire genes, from the downstream analysis. For example, when genes A and B are compared using a dissimilarity measure, each sample normally has two measurements associated with it: the expression level of gene A in that sample, and the expression level of gene B in that sample. By removing gene expression measurements using a minimum expression level filter, we are removing one or both of those gene expression measurements from the sample. Thus, when genes A and B are compared using the dissimilarity measure, there will be fewer samples used in calculating the measure.

4.5.4 Target ambiguity filter

Another seemingly straightforward filter has not been used as commonly. Given the accelerating rate of new annotations in DNA databases, it is not uncommon that a probe or probe set used on an older microarray is no longer specific for a single gene. This can happen if the probe was either designed without knowledge of the entire genome or designed against a piece of chromosome instead of an expressed sequence. As an example, probes made against GenBank accession AF053356 might represent the expression of any of 12 expressed products, since this sequence in GenBank represents a piece of chromosome 7. At the time of the microarray construction, however, this region may have been thought to represent only a single gene.

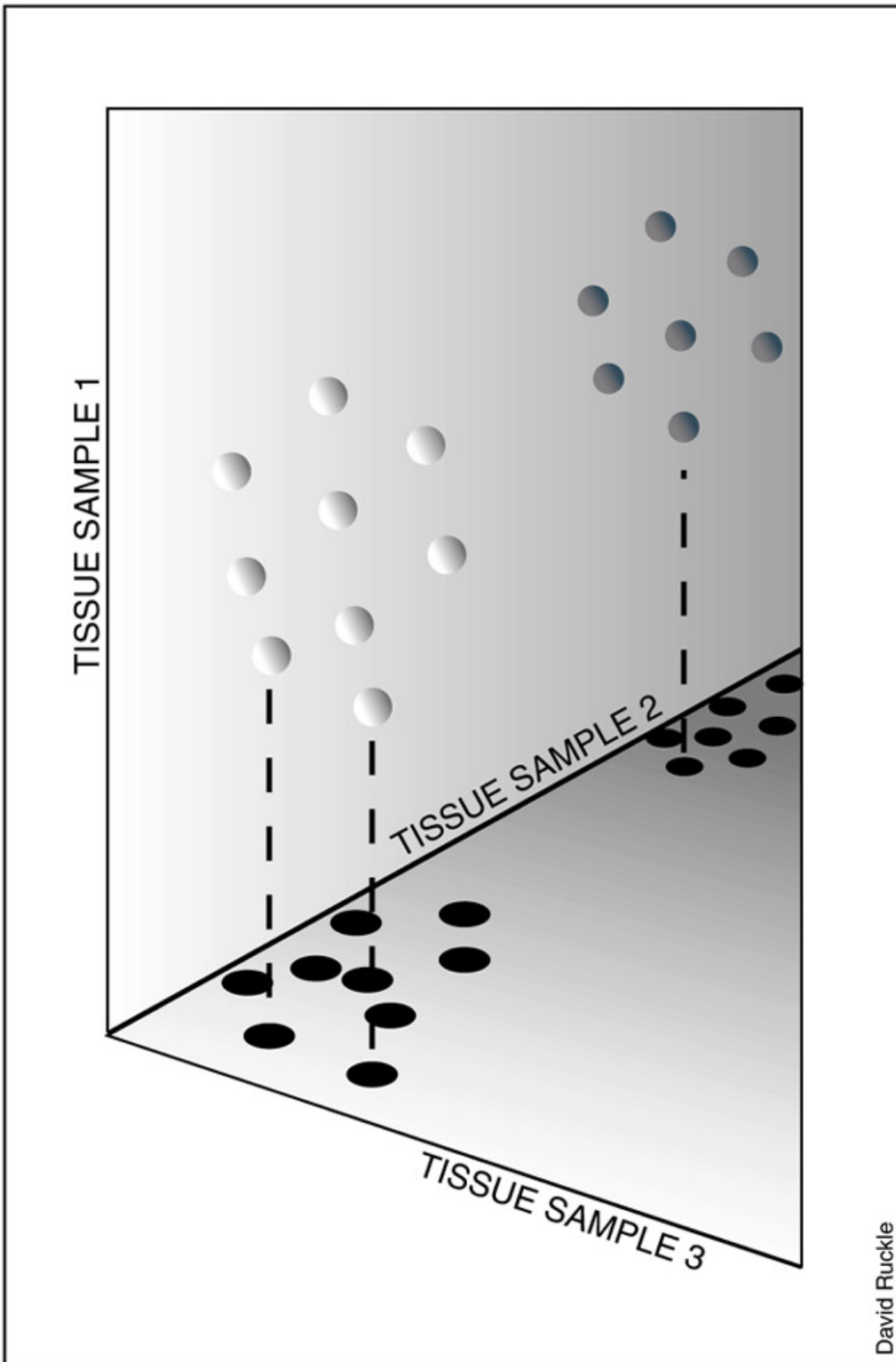
Since without knowledge of the exact probe sequences it is not clear what an expression measurement on a probe set made from a piece of chromosome means, one can argue for removal of these types of probe sets from analysis at the start. However, this requires the use of probe databases early in the analysis, which is not easily performed.

^[2]An expression value if the variable whose entropy is being calculated is a measure of gene expression, but any kind of variable, even a noncontinuous or categorical variable such as eye color, can be used in an entropy calculation.

4.6 Self-Organizing Maps

The use of self-organizing maps (SOM) in functional genomics was first described by Tamayo *et al.* [175]. The underlying hypothesis in this technique is the standard functional genomics dogma articulated in section 2.2: genes with similar expression patterns are functionally similar. Thus, if one can find groups of genes that are all near each other, such a group is likely to be related functionally.

As typically used, this methodology starts by rescaling each gene into a standard distribution, *e.g.*, with mean = 0 and variance = 1, though this is not required. Each measured gene is mapped into a k -dimensional point, such that each of the k dimensions corresponds to a different tissue sample or microarray. The i th coordinate axis represents the i th sample. Each gene is evaluated by its vector of measurements^[3] and is represented by a point in the map. Coordinates for these points therefore represent the expression levels in various experiments. This is easy to visualize for a gene measured in three experiments (see figure 4.3). The multidimensional space would be represented as a hypercube. Each side of the cube represents a different tissue sample or microarray. Each gene would be a multidimensional point within the cube space, with each coordinate equal to its measurement.



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Figure 4.3: Genes can be represented as points in a multidimensional space. Each sphere represents a gene measured in three tissue samples. Each of the expression levels is considered as a measure along a three-dimensional coordinate. One can imagine that genes with similar expression levels in all three samples may be grouped as clusters in the three-dimensional space. These clusters can be found using an unsupervised technique, such as self-organizing maps. The analysis starts by choosing a map or topology of nodes N (called centroids), typically arranged in a grid.^[4] Each node is in the same dimensional space as the genes, or objects being clustered. Each centroid starts at a random position in the k -dimensional space as follows:

```

define number_centroids as the number of centroids we will start with
define number_axes as the number of samples or experiments for which
    we have data
define vector as an array of floating point numbers,
    with size = number_axes
make a new empty array of vectors called centroids,
    with size = number_centroids
loop through centroids using the index i
    loop from 1 to number_axes, using index j
        find the minimum expression level across all the
            genes measured in sample j
        find the maximum expression level across all the
            genes measured in sample j
        set coordinate j for centroid i to be a random number
            between the minimum and maximum
    end loop
end loop

```

The centroids are subsequently mapped to a position in this k -dimensional space and correspond to the center of a cluster. The mapping process is iterative, meaning that the map is formed over several cycles. During each cycle, a random data point is chosen and all of the centroids are moved toward that point, as illustrated in figure 4.4. The distance each centroid is moved is typically a function of both the iteration number and the distance between the point and centroid, such that the distance moved decreases as more iterations have passed. The distance moved is also less for those centroids farther away from the randomly chosen point.

```

define number_centroids as the number of centroids we will start with
define number_genes as the number of genes we are working with
set continue_loop to true
set iteration to zero
loop while continue_loop
    set cumulative_moving to a zero vector
    calculate learning_rate, as a function of iteration
    pick a random gene G from the list of all genes we have
    loop through centroids using the index c
        calculate dissimilarity measure (i.e. distance) from
            centroid to G
        calculate vector v by taking centroid c and subtracting G
        move centroid c along vector v as a function of
            learning_rate
        add the amount moved to cumulative_moving
    end loop
    add one to iteration
    set average_moved to cumulative_moving divided by
        number_centroids
    if average_moved is under a predetermined threshold
        set continue_loop to false
    end if
end loop

```

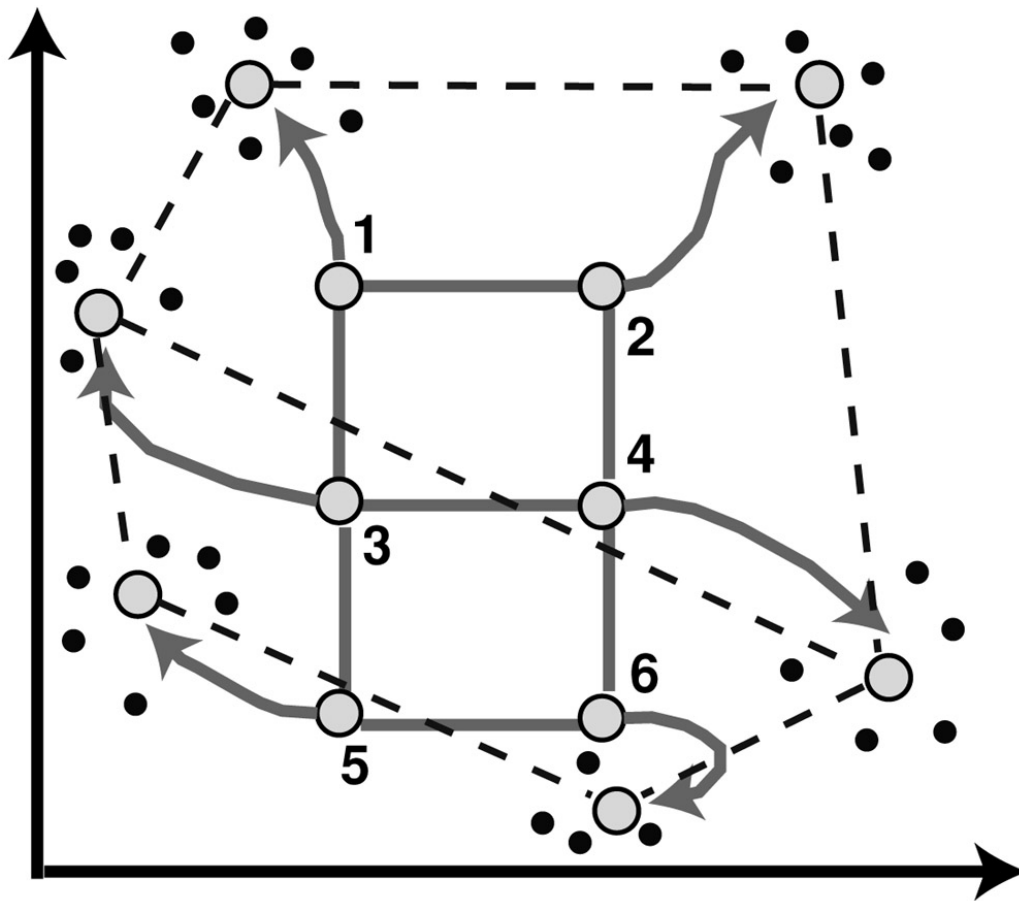


Figure 4.4: Principle of self-organized maps. Centroids start in an arbitrary topology; then, as the method progresses, each moves toward a randomly chosen gene during each iteration. After proceeding for enough time, each centroid will be in the middle of a cluster. (From Tamayo et al. [175].)

The algorithm continues until either fixed endpoints are reached, or after there is no movement of nodes above a threshold.

As most commonly used, self-organizing maps employ Euclidean distance as the dissimilarity measure. However, there is nothing that would prevent the use of the correlation coefficient or mutual information as the dissimilarity measure instead. Nonetheless, we describe this algorithm using the Euclidean distance because that is the way it is typically presented in the most well-known applications in functional genomics [78].

Figure 4.5 explains the conventional graphical representation of self-organizing maps.

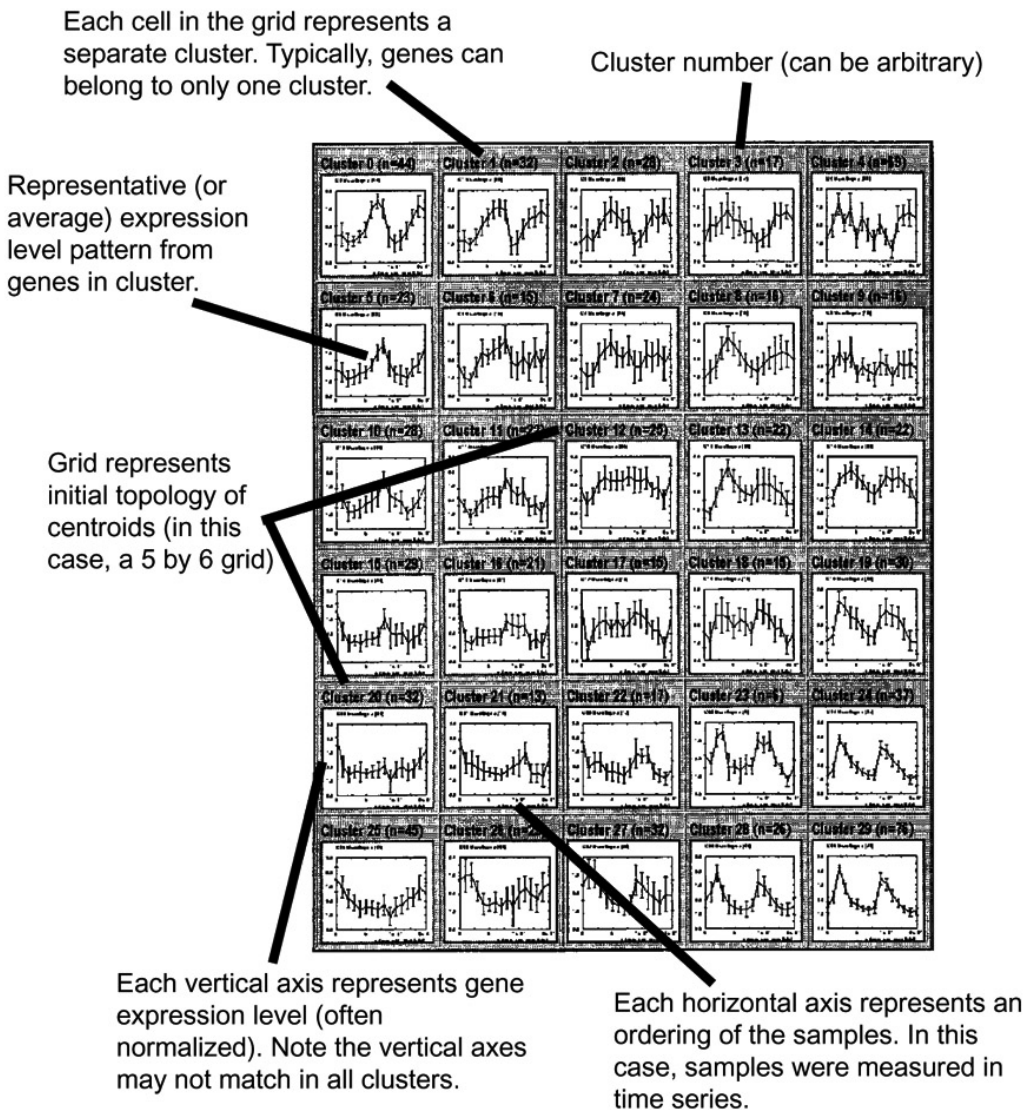


Figure 4.5: Graphical convention for self-organizing maps. (Derived from Tamayo *et al.* [175].)

4.6.1 K-means clustering

The k -means clustering algorithm is very similar to the self-organizing map methodology. Centroids are initialized to randomly chosen genes, then each gene is mapped to its closest centroid. In an iterative manner, the centroids are then updated with the average expression pattern of the genes assigned to that cluster; then the genes are again mapped to their closest centroid. This continues until all genes remain mapped to the same centroids.

Published examples

- Tamayo *et al.* [175] used self-organizing maps to functionally cluster genes into various patterned time courses in HL-60 cell macrophage differentiation.
- Toronen *et al.* [180] used a hierarchical self-organizing map to cluster yeast genes responsible for diauxic shift.
- Soukas *et al.* [165] used a k -means approach to cluster genes involved in leptin signaling.
- Tavazoie *et al.* [176] used k -means to determine clusters of genes based on function, for which 52 upstream regions in the genome were searched in an effort to find common regulatory sequences.

Advantages

- Easy visualization in two dimensions of the range of variation seen in the data: Since the map will be spread across the multidimensional space of experiments, by displaying the expression patterns represented at each node, one can quickly see the variety of gene expression patterns. For example, if there was not much variety in the gene expression patterns, one would quickly find the representative gene expression patterns from the many nodes of the self-organizing map appearing the same.
- Computational requirements: Since only the centroids are moving in each iteration, the number of distances to be computed is linear with respect to the number of nodes. After the map is made, gene assignment to a cluster may require the computation of distances from each gene to each centroid. On the face of things, this is still much less computationally expensive than the initial precomputation of comprehensive pairwise dissimilarity measures required by the other techniques, such as dendrograms and relevance networks. However, this gain in performance is tempered by the large (but finite) number of iterations that may be required to reach stability.

Disadvantages

- Arbitrary map topology: There is nothing to indicate exactly how to create the initial map geometry. The k -means algorithm requires the number of nodes to be specified as a priori information. The self-organizing map technique also requires the initial geometry of the nodes to be specified.
- Misses negative correlations: Negative correlations include gene interactions such as those demonstrated by tumor suppressor genes. As an example, *p53* suppresses the expression of several other genes. This means that the higher the expression of *p53* seen, the less expression of other genes is expected. Negative correlations will be missed in this type of approach.
- Influenced by spiking in gene expression: Without adequate filtering, spiking (isolated high or low expression in single samples) can prevent proper normalization, which impairs self-organizing map formation.
- Each experiment or coordinate is treated independently and with the same weight: each axis is treated as orthogonal to the others. This may not be the actual case, however. Consider the extreme example of a situation where two replicated measurements are made on three patients, making six microarray measurements. In each of the three patients, the pair of replicated measurements should not differ. However, if each of the six measurements is treated as a separate, orthogonal axis, too much weight may be given to the replicates, which may have measurements close together. In other words, genes may appear closer simply because they have many similar measurements across the replicated experiments.
- Further analysis is needed to determine the actual boundaries of each cluster: After the map is made, all the genes in the multidimensional space may be assigned to a cluster by choosing the centroid closest to each gene. However, it is not immediately clear where the boundaries of each cluster lie. First, this is important in determining which genes lie within the cluster. Second, this is crucial because it is important to distinguish those genes that are far away from all centroids. If too many genes are too distant from all the clusters, it can be an indication that additional centroids need to be added at the start of analysis to gain coverage in that area of the multidimensional space. Thus, it is important to determine what the shape and size of the clusters are, but this requires prior assumptions (the space of the centroids can be round or oval, the diameter can be large or small, or equal among all the clusters, etc.) for which there may be little domain or expert knowledge.
- Membership of a gene may be limited to a single cluster: If a gene resembles the centroid pattern seen in two clusters, it may be assigned to one of the two clusters but not both,

unless precautions are taken to scan for this situation.

- Missing data: If a single gene expression measurement is missing, it becomes difficult to determine where to place that gene in the multidimensional space. This positioning may be crucial to determining which centroid is closest to that gene.
- Stochastic technique: Since the initial conditions are determined randomly, the results may not be repeatable. Validation techniques can be performed by repeating the algorithm using several random starting positions, but it is not clear how to merge the resulting, differing maps.

^[3]Each measurement of that gene in each experiment corresponds to one element of that vector.

^[4]Note that this topology is selected by the investigator sometimes on the basis of prior knowledge (the number of expected natural clusters of the data) and sometimes empirically by iterating across several possible topologies and inspecting the results.

4.7 Finding Genes That Split Sets

As suggested by the machine-learning taxonomy above (section 4.4), nearest neighbor analysis has been used in both supervised and unsupervised learning methods. In a supervised application, the technique is traditionally used to find genes that match an externally specified pattern. These patterns may be "ground truths" in independently validated biological knowledge (*e.g.*, a gene known to be involved in the process being studied) or empirical (*e.g.*, a gene that is observed to be highly expressed in certain samples and expressed at a lower level in other samples). In an unsupervised manner, the technique is used to find clusters of genes that share similar expression patterns.

Let us start with the supervised method. We may have a specified gene of interest in our data sets. Alternatively, we may specify a desired or hypothetical gene expression pattern which may not exist in the data set. Such a pattern could be made based on the desired properties of a gene in particular samples or patients. For example, one may want to find a gene that was upregulated in patients with disease and downregulated in nondiseased patients.

To get a list of genes related to either query pattern, we must again define what it means to have similar patterns. Although any dissimilarity measure can be used as the judge of whether expression patterns are similar, Euclidean distance and the correlation coefficient have been used traditionally. Iteration through the data set will quickly find and rank genes by the degree of similarity to the query pattern.

```
define number_genes as the number of genes we are working with
define query_pattern Q as a vector across all the samples,
    representing the gene with the expression pattern we are most
    interested in
make a new empty array called distances, with size = number_genes
loop through all the genes, with index G
    calculate dissimilarity measure (i.e. distance) between Q to G
    store the measure in the appropriate position in distances
end loop
```

Published examples

- Using a hypothetical pattern, Golub *et al.* [78] used this technique to find those genes that best split samples of acute lymphocytic leukemia from acute myelogenous leukemia.
- Ben-Dor *et al.* [19] used a nearest neighbor classifier to split normal colon samples from

cancer samples from [4], as well as other data sets.

Advantages

- Computational and memory requirements: When used in a supervised manner, nearest neighbor analysis is considered a "lazy classifier," in that the algorithm requires only that the data set be kept in memory. The genes "nearest" to the search pattern are found when required, minimizing unnecessary comparison operations. This is in contrast to other techniques, which may require the initial expensive computation of pairwise dissimilarity measures.
- Quickly finds genes or features that most significantly split the labeled sets. These can be validated biologically, then developed into diagnostic tools, for example.

Disadvantages

- Only gross differences are found. In other words, genes with differences in absolute expression level are typically chosen. Differences in gene–gene interactions may be missed where the expression level of the genes involved in the interaction may not be different (see figure 4.6).

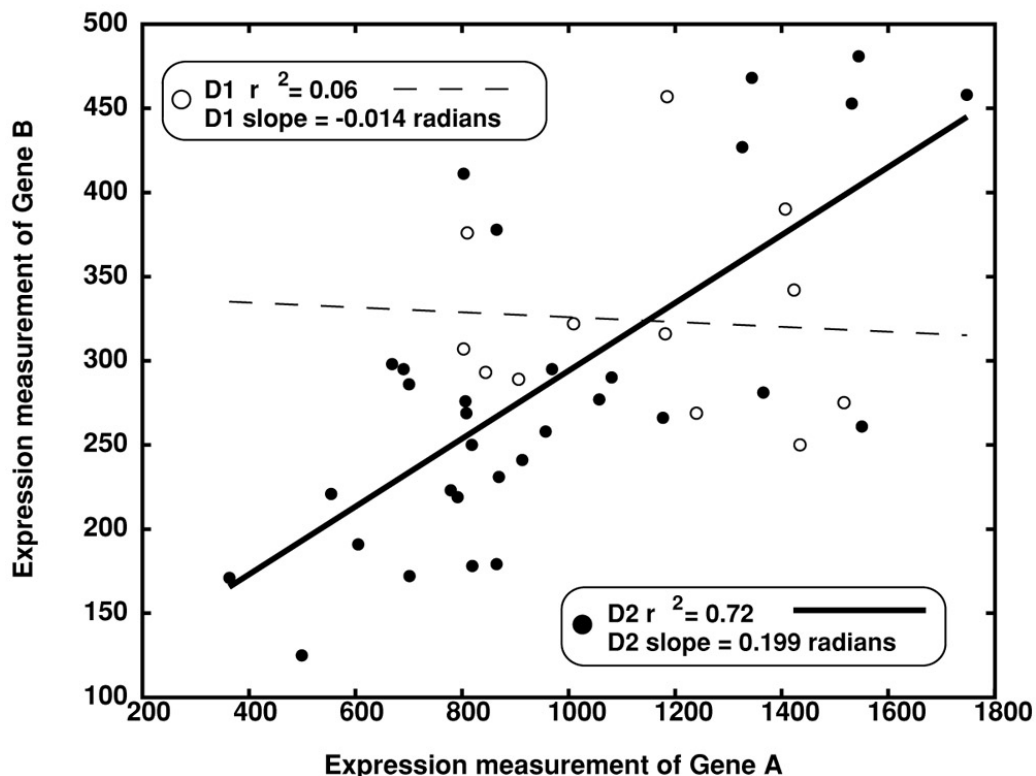


Figure 4.6: Genes can have a difference in interaction, but not in expression level.

Scatterplot of gene A and gene B, measured in samples from disease 1 (open circles) and disease 2 (closed circles). Note that expression measurements from neither gene A nor gene B can be used to separate disease 1 from disease 2. However, the linear regression model of gene expression levels from disease 2 is different from disease 1.

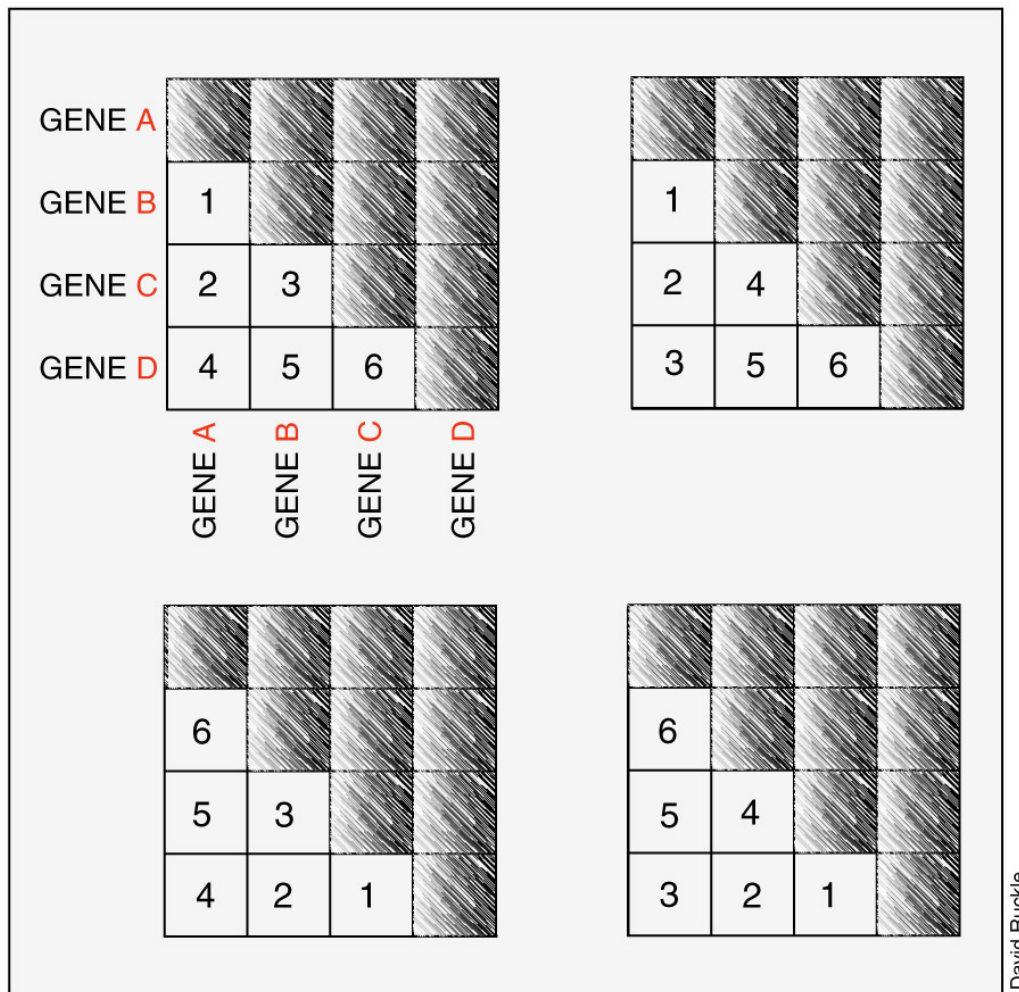
- The genes or features that best split two sets may not necessarily be the most significant or biologically causative. For instance, the genes selected may be the most obvious but distant "downstream" effects of other genes which are primarily responsible for the difference in states.
- Because many more features (*i.e.*, genes) are measured compared to cases (*i.e.*, samples

or experiments), it is almost always possible to find genes that split samples into labeled sets. If one considers combinations of genes as features (*i.e.*, with support vector machines), then one can find even more features that successfully split the sets. One has to be careful to look for simple models that split sets, even if they are not as accurate [30]. Otherwise, these models may lose biological relevance.

4.8 Phylogenetic-Type Trees

A third methodology is phylogenetic-type^[5] tree construction, otherwise known as *dendrograms*. Although there are several ways to construct this data structure, this technique typically first involves comprehensively comparing all genes against each other using a dissimilarity measure, such as the correlation coefficient. Eisen *et al.* [63] took expression levels at various time points and created a vector for each gene. They then compared all genes against each other and recorded the correlation coefficient for each pair of vectors, then constructed a phylogenetic-type tree with branch lengths proportional to the correlation coefficients.

Let us take this process one step at a time. First, enough memory must be allocated to store the comprehensive pairwise comparisons. If n genes are being compared with each other, there are $((n*n)/2) = n/2$ possible pairwise comparisons. Typically, when maximum efficiency is used, dissimilarity measures are stored as real values taking 4 bytes each. Clearly, as n increases, the necessary memory increases on the order of n^2 . Next, an iterative method must be used to perform the pairwise comparisons. Since associativity holds with the correlation coefficient and Euclidean distance, once gene A is compared with gene B, the reverse comparison does not need to be performed. There are four logical ways to order these comparisons, as shown in figure 4.7, and one needs to be chosen arbitrarily. Regardless of how we are calculating or storing the measures in this chapter, we refer to the comprehensive set of measures for a collection of variables or features as a *measure-triangle*.



David Ruckle

Figure 4.7: Four possible ways to order comprehensive pairwise comparisons. We define the term *measure-triangle* as the comprehensive pairwise comparison between each of a set of features. The comparisons can be performed and stored in any of four different ways. Although there are many efficient ways to store the results of each comparison, *e.g.*, using matrices, we purposely use a less efficient, but easier-to-understand method here. We store the result of each comparison in a hash table, using the two genes (sorted) as the key. That is, the dissimilarity measure can be retrieved from the hash table using the unique pair of genes.

```

define number_genes as the number of genes we are working with
make a new empty two-dimensional array called measure-triangle,
  with size = number_genes by number_genes
loop from 2 to number_genes, with index i
  loop from 1 to i - 1, with index j
    calculate dissimilarity measure (i.e., correlation
      coefficient) between gene[i] and gene[j]
    store measure in measure-triangle[i][j]
  end loop
end loop

```

For each possible pair, the dissimilarity measure must be applied. Eisen *et al.* [63] essentially used the correlation coefficient in pairwise comparisons. Wen *et al.* [189] used Euclidean distance as a dissimilarity measure.

In addition to the measure-triangle used to store the dissimilarity measures, we need a data structure to hold the developing tree. A tree can be thought of as a set of internal nodes and external nodes (or leaves). Internal nodes serve as branchpoints linking together genes or subbranches. External nodes or leaves are exclusively genes. Because both genes and branches can be connected by branches, we will make that fact explicit by stating that they both can be branchable.

```

define number_genes as the number of genes
define number_samples as the number of samples
define an object called branchable, which contains
    (1) an array of expression measurements of size number_samples
    (2) a link called parent, to another object of type branchable
    (3) a list called children, to other objects of type branchable
define
    an object called branch-branch,

    which is derived from branchable
define an object called branch-gene, which is derived from branchable
make a new empty list called remaining-branchable-objects
make a new empty list called used-branchable-objects
loop through all the genes, with index g
    make an object of type branch-gene for each gene
    set the list of children for this object to null
    set the parent for this object to null
    add this object to the list remaining-branchable-objects
end loop

```

We are ready to begin building the tree. Iterations continue until every node has been attached to the tree. The tree begins with a single branch with two leaves, connecting the two genes with the strongest connection (*e.g.*, either the highest correlation coefficient or the shortest Euclidean distance). We will store the branches

```

loop until the size of remaining-branchable-objects reaches zero
    find the strongest measure in measure-triangle
    determine the two branchable objects, called a and b,
        that were compared to make this measure
    make a new object of type branch-branch, called c
    add the branchable objects a and b as children of c
    set the parent of branchable objects a and b to c
    set the expression measurements of c to the
        average expression measurements of a and b
    remove objects a and b from remaining-branchable-objects, if
        they were in that list
    add objects a and b to used-branchable-objects
    remove all comparisons involving a and b in measure-triangle
    compare object c to all the objects in remaining-branchable
        -objects and add those measures to measure-triangle
    if size of remaining-branchable-objects is zero
        set main-tree-trunk to c
    end if
end loop

```

The two genes are removed from the measure-triangle, or specifically, all the measures involving any one of these two genes are removed from the measure-triangle.^[6] In place of the two genes, a

new entity is created representing the branch with the two genes. The gene expression pattern for this new branch is the average pattern of the two gene expression patterns being linked. Notably, the new gene expression pattern for the branch does not have to be the average pattern. In computing the distance between a gene and a branch, or a branch and a branch, one could use a minimum, maximum, or average dissimilarity measure. New measures need to be calculated comparing the branch against all the remaining genes, and these are stored back in the measure-triangle. At every iteration, the next strongest measure is found and the process repeats, removing two entities (genes or branches) and replacing them with a new one, until only one entity remains.

At this point, the construction of the dendrogram is complete. However, the results still need to be displayed. The constructed dendrogram is typically used to order and display genes with similar clusters. At every branch, the display ordering can be one of two possibilities: Order one branch before the other, or *vice versa*. This means for any tree with n branch points, there are 2^n possible display orderings of the data. Other information needs to be used to determine a consistent way to order the data. For example, position along a chromosome or average expression level can be used to order genes. Since there are no loops present in dendrograms, we can use a depth-first search to output the results. The following pseudocode will traverse the dendrogram and output the leaves of the tree (*i.e.*, the genes) in order by average expression level.

```
define a new procedure called print-tree, that takes a branchable
  object b as a parameter
if this branchable object b is a branch-gene (i.e., and not a
  branch-branch)
  print the name/label of b
end if
if the list of children of b is not empty
  move the left margin left 1 cm
  sort the list of children by expression level
  call procedure print-tree recursively on each child
    of b separately
  move the left margin back 1 cm
end if
end procedure

start the printing by calling print-tree on main-tree-trunk
```

The result is a dendrogram, typically drawn with the graphical conventions shown in figure 4.8.

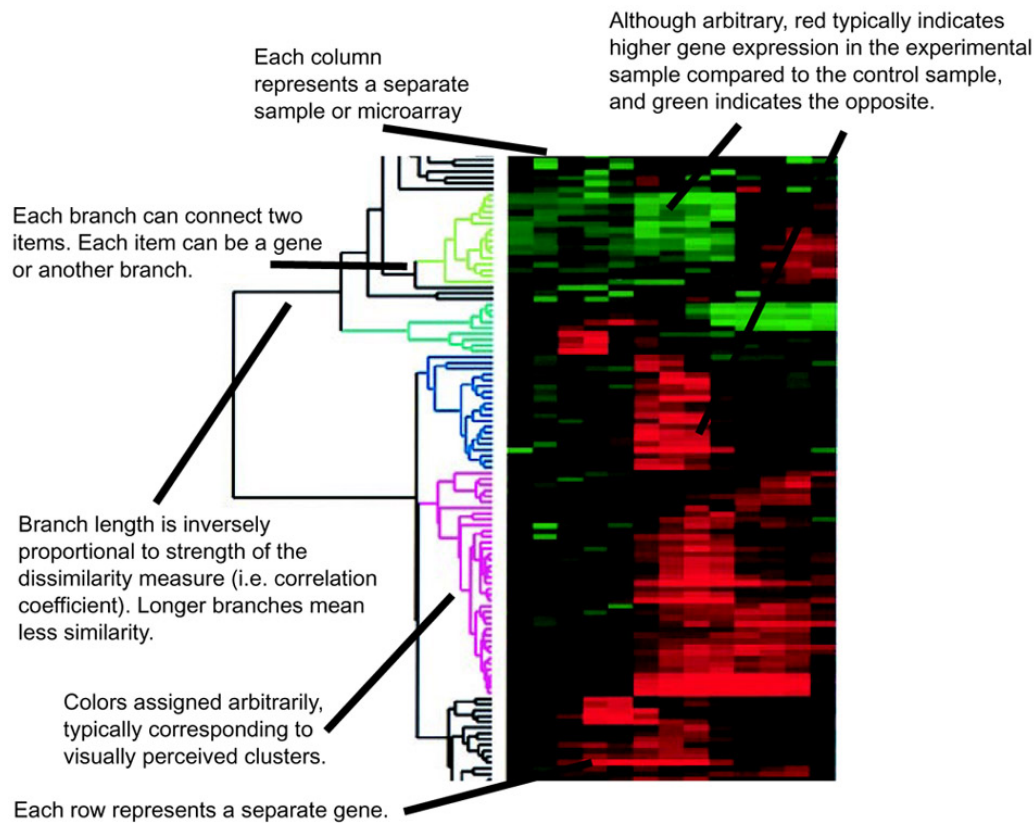


Figure 4.8: Graphical convention for dendrograms.

4.8.1 Two-dimensional dendrograms

Using the technique above, one creates a dendrogram and the position of the branches suggests the presence of clusters. As mentioned above, the results still have to be displayed, and one can use a secondary key to order the data, such as average expression level, chromosomal position, *etc.*

We have not yet answered how to order the data horizontally; *i.e.*, how should the experiments be ordered, rather than the genes? For ordered experiments, such as time series data collection, gene expression is typically shown from earliest to latest acquisition. However, when the samples have no ordering, a second dendrogram is typically constructed to cluster the samples. This was the approach used by Ross *et al.* [154] in clustering both various samples of cancer cell lines and the genes measured by microarray, as shown in figure 4.9. The algorithm to construct such a two-dimensional dendrogram is identical to constructing each of the dendrograms individually.

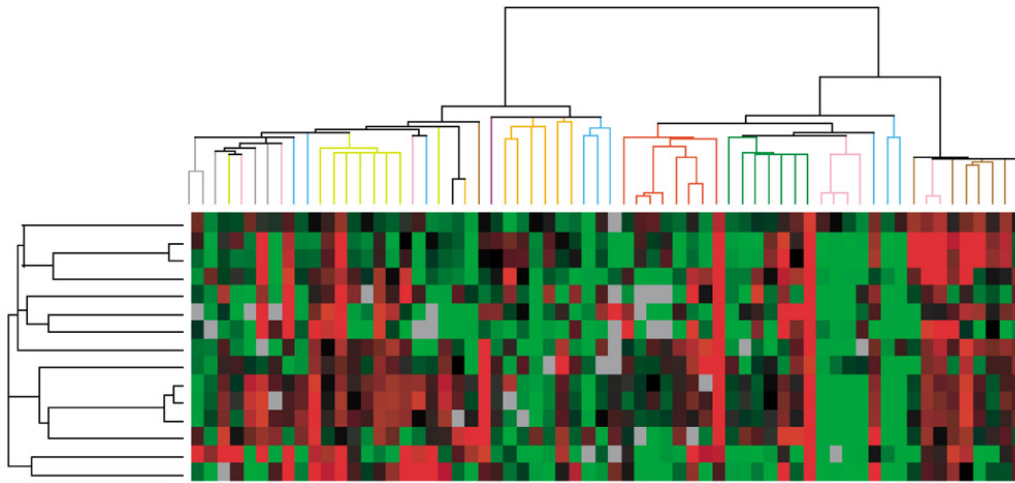


Figure 4.9: Two-dimensional dendrogram. Dendrograms are constructed for both the genes and samples individually, and are combined in the display. (Derived from Ross *et al.* [154].)

Published examples There are many examples of dendrograms applied to gene expression measurements—too many to accurately list all of them. We will cite the earliest publications and the ones most cited.

- Spellman *et al.* [167] constructed dendrograms from a yeast expression data set constructed under multiple conditions.
- Eisen *et al.* [63] described the application of dendrograms to multiple yeast expression data sets, 1998.
- Alizadeh *et al.* [3] applied this technique to discover two subgroups within samples from the single disease B-cell lymphoma, and showed a statistically significant difference in survival curves of the patients from whom the samples were obtained.
- Ewing *et al.* [66] applied this technique after linking EST measurements with gene expression measurements.
- Bittner *et al.* [26] classified genes measured in malignant melanoma.
- Ross *et al.* [154] classified gene expression measurements from 60 cancer cell lines.

Advantages

- Category number and size: One can estimate the number or size of different expression patterns in the data set. For example, in viewing a self-organizing map, the user may not be sure there was enough coverage of the multidimensional space.
- Quickly visualize overall pattern similarity: One can quickly see whether most of the genes or samples in the data set are similar to or different from each other.
- Parallelizability: The most time-consuming aspect in the construction of dendrograms is the initial comprehensive precomputation of the dissimilarity measure across the set of features. This computation can be easily parallelized. The construction of the actual dendrogram requires the serial creation of branches, and is not trivial to parallelize.
- Display requirements: Optimally displaying dendrograms requires only a narrow column, in which gene names, symbols, and expression patterns (typically displayed in a color representation) can fit.

Disadvantages

- Misses negative correlations: Negative correlations include gene interactions such as those demonstrated by tumor suppressor genes. As an example, *p53* suppresses the expression of several other genes. This means that the higher the expression of *p53* seen, the less expression of other genes is expected. Negative correlations need to be explicitly searched

for.

- Each gene is connected to the overall structure by only one link: the genes in a dendrogram are stored as leaves of the tree. Clearly, each leaf can only be connected to the tree by one stem. However, multiple links from a gene may be of interest. For example, one may wish to distinguish between a transcription factor responsible for the regulation of 10 genes and a structural protein whose gene is controlled by a single transcription factor. Dendrograms do not allow for a variable number of links from each gene.
- All genes are clustered, regardless of connection strength: the algorithm continues until every gene is connected. By definition, certain connections may be weaker than others. It may be a disadvantage to see all genes connected, when particular links are much weaker (*i.e.*, perhaps less likely to be validated) than others.
- Single structure: Although a dendrogram is a single structure, intuitively one can use a fixed distance from the root or a set of long branches to determine where to cut the branches in order to delineate the subgroups. This may or may not have a statistical basis.
- Single metric type: Dendrograms can only be constructed using a single dissimilarity measure. For example, it is not practical to construct a tree with genes connected by both Euclidean distance and correlation coefficient. Only one dissimilarity measure may be used at a time.
- Single data type: As stated at the start of this chapter, although we have many objects in functional genomics that may be clustered, it is only practical to cluster one data type at a time using dendrograms. For instance, genes may be clustered, and samples may be clustered. If both need to be clustered, this is performed as a two-dimensional construct using coupled two-way clustering [76]. Nonetheless, on occasion it does make sense to cluster genes and phenotypic measures along the same axis. A "systolic blood pressure" measurement might appear in one part of the dendrogram, while a "weight" may appear in a different part. It would be difficult to determine which of the genes in the immediate surrounding branches might be closest in profile to the phenotypic measure.
- It becomes difficult to follow the branching in a dendrogram, when the number of genes or features is high [93].
- As each new branch is added to the forming tree, the average gene profile representing a higher branch may not match any of the specific gene profiles within the subtree [162].
- These are not globally optimal trees: The algorithm described here falls under the category of "greedy algorithms" in computer science, as the best match is always used at every step, regardless of the later consequences it may have for forming the tree. Early "bad" decisions in the formation of branches cannot be corrected later.
- Entire vector is used: This means that for each gene, the entire set of samples are used in calculating the dissimilarity measurements. However, there may be very little variation in most of the samples, and high variation in only a few of the samples. Two genes might be scored as being similar, when they may actually differ greatly in one of the many samples.
- Computational requirements: This technique requires the comprehensive precomputation of the dissimilarity measure for all pairs of genes, which grows on the order of n^2 where n is the number of genes being studied. This can be a time-consuming computation.

^[5]They do not describe the distance between the genomes of organisms as in standard phylogenetic trees and therefore are referred to as "phylogenetic-type" trees.

^[6]That is, all the elements of the hash table for which one of the two genes is a key.

4.9 Relevance Networks

Relevance networks offer a method for constructing networks of similarity, with the principal advantages being the ability to (1) include features of more than one data type, (2) represent multiple connections between features, and (3) capture negative as well as positive correlations. Like dendrogram construction, this algorithm begins by evaluating the similarity of features by

comprehensively comparing all features with each other in a pairwise manner over the same cases. Several dissimilarity measures have been used in this methodology, including mutual information and the correlation coefficient. In this section, we describe the construction of relevance networks using $\hat{r}^2$. This is equivalent to r^2 , except the sign (positive or negative) from the r , or Pearson's correlation coefficient, is applied. This is crucial in maintaining information about negative correlations. After the measure-triangle is completed, a threshold measure is chosen. The entire measure-triangle is scanned and any measures below the threshold are set to zero.

```
define threshold as the strongest link we are going to rule out
loop through all the measures in the measure-triangle
  if the measure is under threshold
    set the measure to zero in the measure-triangle
  else
    leave the measure alone
  end if
end loop
```

The measure-triangle must then be scanned to determine the networks. First, for the connections that remained because of measures that were greater than the threshold, each of the two nodes participating in the connection is listed as being used. An array stores the number of associations in which each gene is participating.

```
make a new empty array called node-connections, with
  size = number_genes
loop through all the measures in the measure-triangle
  if the measure is at or over threshold
    find the two genes A and B that were compared to make
      this measure
    add one to gene A's entry in the array node-connections
    add one to gene B's entry in the array node-connections
  end if

end loop
set number-nodes_used to zero
loop through the entries in the array node-connections
  if the entry is greater than zero, increment number-nodes-used
end loop
```

Next, each of the networks needs to be found and assigned a number. This is done using a combination linear-recursive methodology. A counter for the network number to be assigned is first set to 1. The list of features is iterated on, until a feature is found that is used in the relevance networks. That feature is assigned to the first relevance network. The algorithm then recursively descends to each of the connected features, and assigns them the same network number. This continues until all the features have either been assigned to a network, or have been eliminated.

```
define a new procedure called assign-network, that takes a
  current-gene-index and a current-network-number
set G to be the gene indexed by current-gene-index in the
  array all-genes
if G's entry in the array node-connections is greater than zero
  (i.e., that node is being used in a connection)
if G has not been assigned a network number
  assign it the current-network-number
else
```

```

        just in case we find a gene that was already
        assigned a number,
        if G's current network number is larger
            than current-network-number change G's
            network number to current-network-number
        end if
    end if
if we have assigned or changed G's network number, we need to
do the same to G's connected genes
loop through all of G's active connections, using the
measure-triangle, using index H
    if gene H's network number does not equal G's network number
        call assign-network recursively on gene H
        using current-network-number
    end if

end if

end if
end procedure

set the current-network-number to 1
to start the assignment, loop through all the genes, using index G
    if G's entry in the array node-connections is greater than zero
        (i.e., that node is being used in a connection) and G has
        not been assigned a network number
        call assign-network on G using current-network-number
        increment current-network-number
    end loop
end loop

```

Finally, the results need to be displayed graphically. There are several standard formats for graph descriptions, and the specifics of these are beyond the scope of this book. We have successfully used both the GMF format used by Tom Sawyer Software, Inc. (Oakland, CA) in their Graph Editor Toolkit, as well as the DOT format used by AT&T Research Labs in their Graphviz package. For the purposes of this chapter, we will demonstrate how one of these networks may look in DOT format, primarily because software to format the graphs is freely available for academic use along with the source code.^[7] One can output this type of file by first looping through all the nodes and genes in use and printing out their characteristics, then looping through and printing all the connections. A sample output file in this format is shown here.

```

graph RN {
    size="7.5,10";
    ratio="auto";
    node [ fontname = "Helvetica", fontsize = 8 ];
    node001 [ shape=box, height=0.2, width=1.6, label="1318_at",
        style="setlinewidth(1.5)"];
    node002 [ shape=box, height=0.2, width=0.8, label="36262_at",
        style="setlinewidth(0.1)"];
    node003 [ shape=box, height=0.2, width=0.8, label="422_s_at",
        style="setlinewidth(3.0)"];
    node001 -- node002 [weight=1,style="setlinewidth(0.1)"];
    node001 -- node003 [weight=1,style="setlinewidth(3)"];
}

```

After installing the Graphviz package, this sample graph can be stored in a file called "test.dot." Executing the following commands on a standard Linux computer will first translate the test.dot into a PostScript file called test.ps, which can then be processed to make an Adobe Acrobat PDF file

called test.pdf.

```
dot -Tps -o test.ps test.dot
ps2pdf test.ps
```

The output PDF file from this example will contain three nodes connected with two lines. Alternatively, a commercial software package can be used to lay out more sophisticated graphs, such as the Graph Layout Toolkit. Using this package, we have constructed relevance networks such as those in figure 4.10.

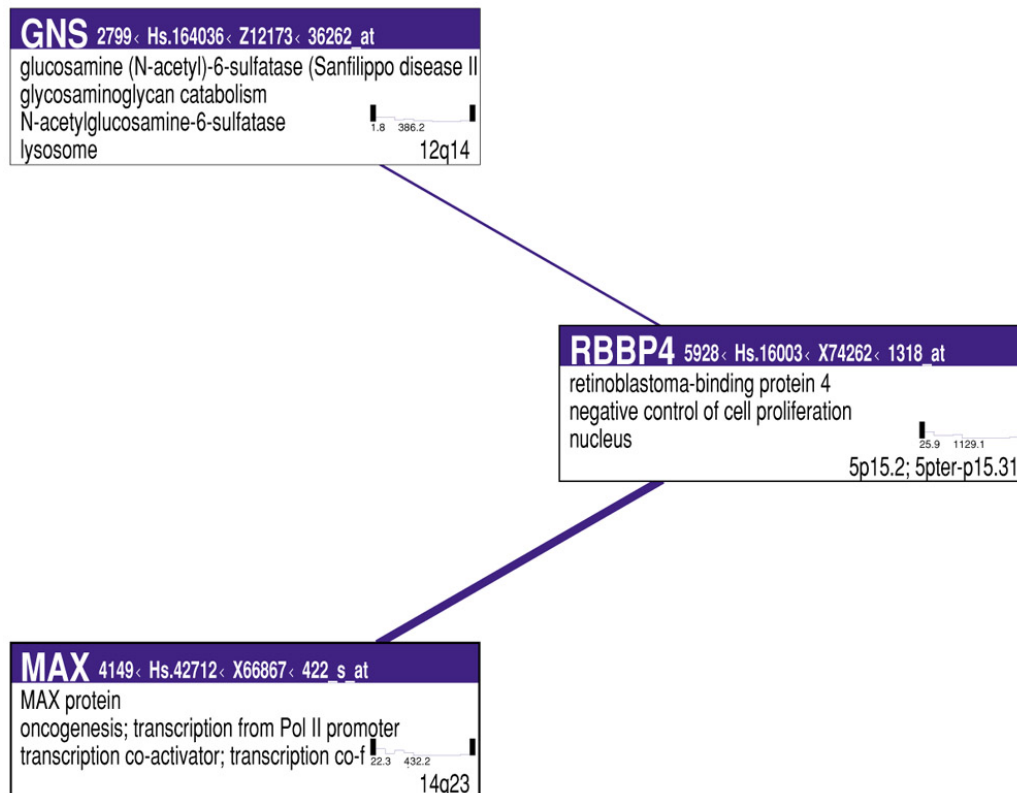


Figure 4.10: Sample relevance network. One of 78 networks formed by aggregating mouse samples from six different experiments. Threshold $\hat{r}^2$ was set to .99, meaning that the "thinnest" line seen represented an $\hat{r}^2$ of .99. The legend for this type of diagram is shown in figure 4.11. The hypothesis generated here is that *RBBP4* expression correlates with *GNS* and *MAX* gene expression. The miniature histograms for each gene show no "spiking" in the expression measurements (see section 4.5.2). Each of these genes appears at a different chromosomal location. By including information from LocusLink (see section 5.5.1) and Gene Ontology (see section 5.1.1), we see that *RBBP4* is known to control cell proliferation, and *MAX* is involved in oncogenesis.

Typically, several graphical conventions are used to highlight the findings in relevance networks, as shown in the figure 4.11. Since each line represents a strong connection or putative association, the thickness of each line is proportional to the strength of the dissimilarity measure, so thicker connections are stronger (and thus easier to find on the diagram). The width of each box representing a feature is proportional to the number of features to which it is connected, and the edge thickness of each box is proportional to the average strength of all the connections entering that box. The length of the connection lines is not significant, and is left variable so that the graph layout algorithms can more easily place the networks in a visually appealing manner (e.g., minimizing edge crossings).

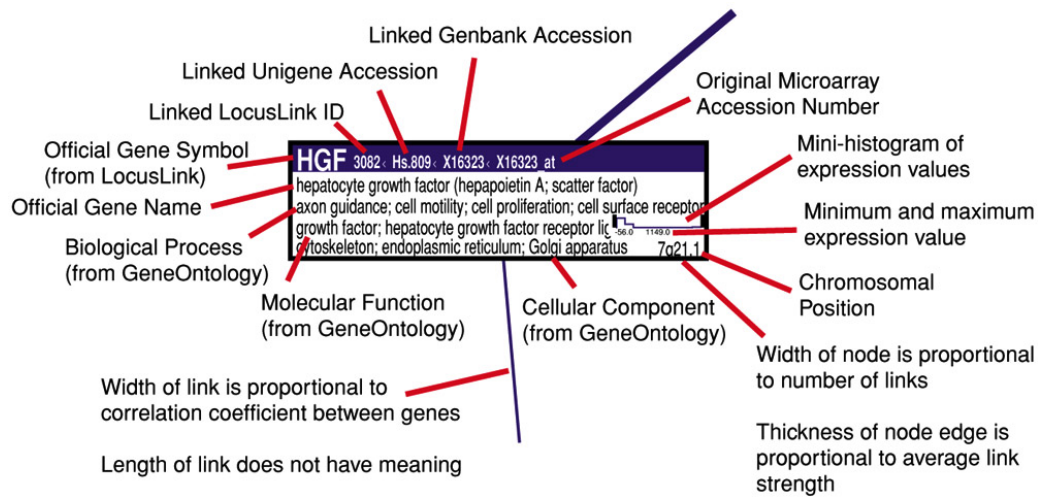


Figure 4.11: Graphical convention for relevance networks. To optimize the hypothesis generation process, relevance network display software contains ties from microarray accession number to GenBank, UniGene, LocusLink, Gene Ontology, and other databases (see section 5.5 for details of this process).

A question immediately arises: Where does one set the threshold? One way to approach this question is to plot the distribution of dissimilarity measure scores, then recompute the scores using repeatedly randomly permuted data sets (see section 4.12.1). For instance, if one randomly permutes the data set 100 times and finds the highest dissimilarity measure score is x , then one could set the threshold to be greater than or equal to x . This approach is shown in figure 4.15. However, an important point to raise is that the threshold does not have to be a fixed measurement, but instead can be "dialed" up and down. In other words, one can set the threshold to be arbitrarily high to start with and the resulting relevance networks can be viewed. If the associations are strong *but are already known and no longer worthy of pursuit*, the threshold can be "dialed down" slightly until new genes and connections are added, creating potentially novel hypotheses. Of course, one must be careful not to drop the threshold so low as to allow potentially noisy and inaccurate associations. The permutation testing can help with this.

Published examples

- Butte and Kohane described the methodology and first applied it to clinical laboratory data (*i.e.*, phenotypic measurements) in [37], then applied it to yeast expression measurements using mutual information as a dissimilarity measure in [38].
- Butte *et al.* [39] linked expression measurements in cancer cell lines to phenotypic measurements on those cell lines (*i.e.*, susceptibility to pharmaceutical treatment).
- Reis *et al.* [150] showed how dynamics (*i.e.*, the slope or change in expression levels over a time series) could be used as a dissimilarity measure in creating relevance networks.
- Klus *et al.* [110] constructed relevance networks from a breast cancer expression data set and linked the results to chromosomal position.

Advantages

- Captures negative correlations: Relevance networks can simultaneously display both positive and negative correlations. Negative correlations are seen commonly in biology, where greater expression of one gene (*e.g.*, a tumor suppressor gene) is associated with decreased expression of other genes.
- Genes can be connected in more than one way (genes can be connected to more than one other gene). This gives rise to three advantages: First, in biology, certain genes (*e.g.*, transcription factors) are responsible for the expression of several other genes. This type of

multiple connection can be modeled using relevance networks. Second, certain features may display weak similarity to two different groups of features, and using relevance networks, these features can be seen to bridge the two groups. For example, certain anticancer drugs can perform as both topoisomerase I and topoisomerase II inhibitors, and the bridging action of these drugs between the two classes can be modeled using relevance networks.

- Multiple structures created: Certainly in biology, some gene regulatory networks are larger or smaller than others, and this variation in size can be modeled using this technique.
- Multiple measures allowable simultaneously: In theory, a relevance network could be drawn with the highest associations from multiple dissimilarity measures. For example, two genes could be linked by r^2 , while another two genes could be linked by mutual information.
- Multiple data types allowable simultaneously: As shown in [39], gene expression measurements can be combined with phenotypic measurements in the same data set and the resultant hypotheses easily viewed. For example, a "systolic blood pressure" measurement may appear linked to the specific genes whose expression measurement correlate with that measurement.
- Parallelizability: The most time-consuming aspect of the construction of relevance networks is the initial comprehensive precomputation of the dissimilarity measure across the set of features. This computation can be parallelized.

Disadvantages

- Excessive complexity at lower thresholds: Although it is tempting to drop the thresholds to reach a potentially novel hypothesis, empirically the number of links between the nodes already present in the graph quickly grows to an unmanageable number. Depending on the graph layout software used, these typically end up looking like large circular networks with hundreds of connections between them, such as figure 4.12. In these cases, it is advantageous that dendrograms have each gene attached to the overall structure by only a single link. One can display relevance networks in a similar manner by taking each network separately and calculating the *minimum spanning tree* by iterating through all the links from weakest to strongest and deleting those links that do not cause the removal of a node in the network. Fundamentally, however, relevance networks are most useful for identifying the subgroup interactions that are linked the strongest.

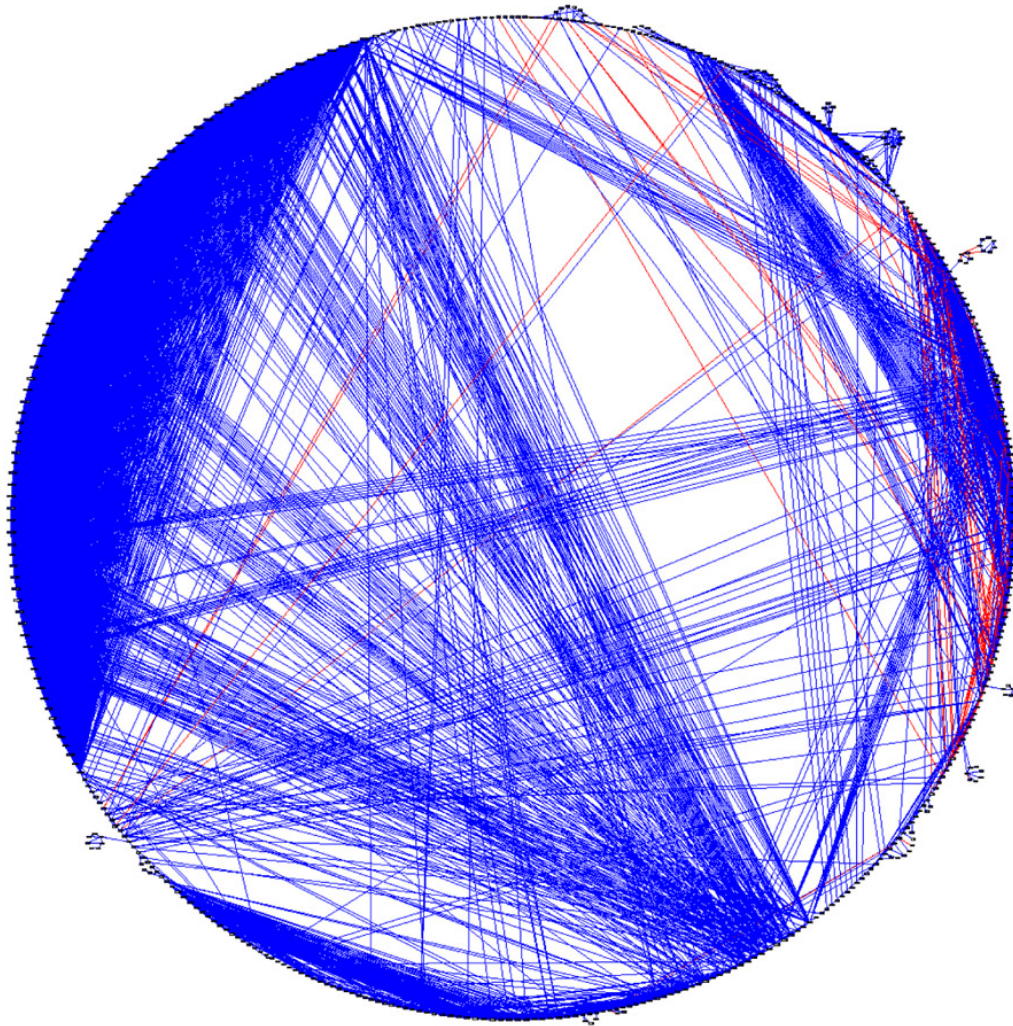


Figure 4.12: Relevance network laid out as a circle. Relevance networks with too many genes may be rendered in circles that minimize overlapping nodes and edge crossings. However, these may be impossible to visualize.

- Obscures large-scale partitional structure. Because all links that are subthreshold are discarded, some large-scale functional groupings will not be detected.
- Larger cliques cannot be found efficiently: Those portions of the relevance networks that are completely connected (*i.e.*, all the nodes in the graph subset are fully connected to each other) are known as *cliques* in graph theory, and are significant in that they represent the most "believable" part of the relevance networks. In effect, if gene A and gene B are in a clique together, it means that not only is there a strong direct association between gene A and gene B but there are also indirect associations, where gene A and gene B both are associated with gene C. These cliques are useful structures, but finding them in an automated manner is an "NP-hard" problem (*i.e.*, the size of the graph quickly overwhelms the ability of any processor to find them using brute force).^[8]
- Entire vector is used: Like other analytic methods, this means that for each gene, the entire set of samples are used in calculating the dissimilarity measurements. However, there may be very little variation in most of the samples, and great variation in only a few of the samples. Two genes might be scored as being similar, when they may actually differ greatly in one of the many samples.
- This is not a globally optimal network: Early bad decisions cannot be corrected later as the threshold drops. Like dendrograms, relevance networks fall under the category of "greedy algorithms" in computer science, as the strongest links are always used at every step, regardless of the later consequences this may have for forming the networks.

- Computational requirements: This technique requires the comprehensive precomputation of the dissimilarity measure for all pairs of genes, which grows on the order of n^2 , where n is the number of genes being studied. This can be a time-consuming computation. However, compared to dendrogram formation, after the precomputation, no further dissimilarity measures need to be computed, and thus with proper programming, the network formation can be performed without much memory.
- Ternary and higher comparisons: As described here, this technique performs pairwise comparisons. However, there are several examples in biology where a combination of events is required for downstream action to take place. For example, several transcription factors may need to combine to form a larger complex before transcription is initiated. Performing pairwise comparisons is an expensive operation; performing three-way, four-way and higher comparisons comprehensively quickly becomes intractable.
- Display requirements: Optimally displaying relevance networks requires a large two-dimensional area (compared to a smaller columnar area for dendrograms) and requires the use of sophisticated graph layout algorithms for optimal node placement. (See figure 4.12 for an example of a particularly difficult network to visualize.)

^[7]<http://www.research.att.com/sw/tools/graphviz/>.

^[8]See sections 1.4.1 and 5.2 for more thoughts on "NP-hard" problems.

4.10 Other Methods

Many other methods have also been used in functional genomics analysis. Only a short list is included here, with references to published biomedical works incorporating each technique, and using the hierarchy proposed at the start of this chapter:

1. Unsupervised.

a. Feature determination.

- i. Principle component analysis and singular value decomposition [189, 95, 68, 149, 5].

b. Cluster determination.

- i. Nearest neighbor clustering.
- ii. Agglomerative clustering.
 - A. Fitch dendrogram algorithm [189].
 - B. Self-organizing tree algorithm [92].
- iii. Divisive or partitional clustering.
 - A. Matrix incision tree [108].
 - B. Two-way clustering binary tree [4].
 - C. Coupled two-way clustering [76].
 - D. Cluster affinity search technique: [20].
 - E. Gene shaving [87].

c. Network determination.

- i. differential equations [44].
- ii. Bayesian networks [71].
- iii. Hybrid petri networks [128].
- iv. Dynamic bayesian networks [134].
- v. Boolean regulatory networks [119, 197, 172, 1, 2].

2. Supervised.

a. Single feature or sample determination.

- i. Naive bayes classifier [21, 46].
- ii. Naive Bayes global relevance [133].

b. Multiple–feature determination.

- i. Decision trees: Supervised and multiple–feature determination [59].
- ii. Support vector machines [72, 33, 46].
- iii. Tree harvesting [86].

c. Boosting [19].

4.11 Which Technique Should I Use?

Each of the techniques described above has its own set of advantages, disadvantages, and proper uses. It is useful at this point to revisit our initial list of potential hypotheses and determine which bioinformatics technique would be most helpful to apply in each case.

- *What uncategorized genes have an expression pattern similar to these genes that are well characterized?* One could iterate through the group of well–characterized genes and apply the supervised nearest neighbor technique to find genes with similar patterns. Similarity in this context could be defined using any of the established dissimilarity measures (starting with the correlation coefficient and Euclidean distance).
- *How different is the pattern of expression of gene X from other genes?* Starting with all possible gene–gene pairs, one can first comprehensively compute a dissimilarity measure across all the pairs (e.g., measuring the Euclidean distance between all pairs of genes). This set of measures will form a distribution, and very possibly a bell–shaped distribution. One can then find where the dissimilarity measures for gene X are in this distribution. If, for instance, the dissimilarity measures for gene X are in the top 5th percentile of the overall distribution, it would suggest that gene X is more dissimilar than other genes are.
- *What genes closely share a pattern of expression with gene X?* One can use a supervised nearest neighbor approach to find those genes with expression patterns most similar to gene X.
- *What category of function might gene X belong to?* Here, one can create a dendrogram, then find gene X in the dendrogram. As one progresses up the tree starting from gene X, one finds those genes with expression patterns similar to gene X. Using a "guilt by association" approach, if most of the neighbors of gene X belong to a particular functional category, then one should hypothesize that gene X belongs to that category [63]. Obviously, this approach is only useful when the functional categories are known for the majority of the genes. An alternative, simpler approach than constructing a dendrogram is to compute the nearest neighbors to gene X and ascertain what their functions are.
- *What are all the pairs of genes that closely share patterns of expression?* Although constructing a dendrogram starts with the comprehensive pairwise computation of a dissimilarity measure, these pairs are not clearly displayed in the results. Instead, one can use relevance networks to find those pairs of genes that score highest with a dissimilarity measure.
- *Are there subtypes of disease X discernible by tissue gene expression?* To answer this question, one can easily construct a dendrogram across the samples, not across the genes. For instance, depending on the mixture of samples collected, one might find that half of the samples fall in one half of the dendrogram, and the other half of the samples fall in the other

half of the dendrogram. One can then proceed to determine if there are phenotypic differences between these branches. This was most dramatically demonstrated by Alizadeh *et al.* [3], where the two major branches in the gene expression dendrogram obtained were found to correspond to patients with significantly different mortality (see figure 4.13). Of course, if the sample mixture is not diverse enough, one might miss the true biological subsets and might instead overfit normal variance or noise in the measurements. This is a matter of experimental design addressed in chapter 2.

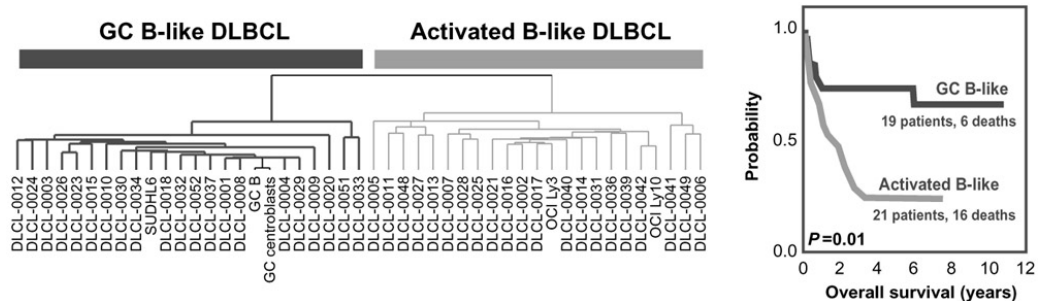


Figure 4.13: Subcategories of B-cell lymphoma determined by microarrays correspond clinically to duration of survival. On the left is a dendrogram constructed across the samples of B-cell lymphoma, using an unsupervised technique. The top branch essentially defines an even split between the categories GC B-like DLBCL and Activated B-like DLBCL, but this distinction was never before made clinically. On the right are Kaplan-Meier survival curves of the patients from whom the samples were obtained. Patients whose cancer matched the Activated B-like DLBCL gene expression profile had a significantly worse prognosis. (From Alizadeh *et al.* [3].)

- *What tissue is this sample tissue closest to?* One can create a dendrogram of the tissues and determine where the unknown tissue lies. Alternatively, one can use a supervised nearest neighbor approach to determine the "closest" pattern of gene expression.
- *What are all the different patterns of gene expression seen?* Although dendrograms can provide a categorization and ordering of genes based on expression pattern, self-organizing maps can easily categorize and summarize the expression patterns present. Further analysis can be run on these clusters, including determining whether the genes fall in the same functional category, or searching for common sequences in the 52 upstream region from each gene.
- *Which genes have a pattern that may have been a result of the influence of gene X?* Answering this question requires the proper biological experiment to be performed. Thus, this question is often asked when gene X is being over- or under-expressed in a particular sample. One can treat the controlled expression level of gene X as an environmental factor, then, using a supervised nearest neighbor approach, one can find whether any of the genes correlate with the controlled expression level. However, the controlled expression level may not be quantifiable, but instead may be discrete (*e.g.*, gene X is "absent" in a sample from a knockout animal, but is "present" in a sample from a wild-type or normal animal). In this case, one can create a hypothetical gene expression matching this information (*e.g.*, high expression levels in the one set of samples, low expression levels in the other set), then use a supervised nearest neighbor approach to find genes with a similar pattern.
- *What are all the gene-gene interactions present among these tissue samples?* Relevance networks provide a network of gene-gene interactions based on dissimilarity measures.
- *Which genes best differentiate two known groups of tissues?* Because the two groups of tissues are known, this calls for a supervised approach. Classic methods can be used to address this question, such as the Student's *t*-test. Alternatively, one can construct a hypothetical gene expression pattern across the samples with the desired behavior (*e.g.*, high expression levels in one tissue, low expression levels in the other tissue), then use a supervised nearest neighbor approach to find genes with a similar pattern. [78] Newer

approaches to answer this question involve the use of support vector machines [33, 49, 72]. It is important to note that simplifying or pruning the list of genes using a technique such as principal components analysis does not give one the genes that best split two sets. A classic example of this is shown in figure 4.14.

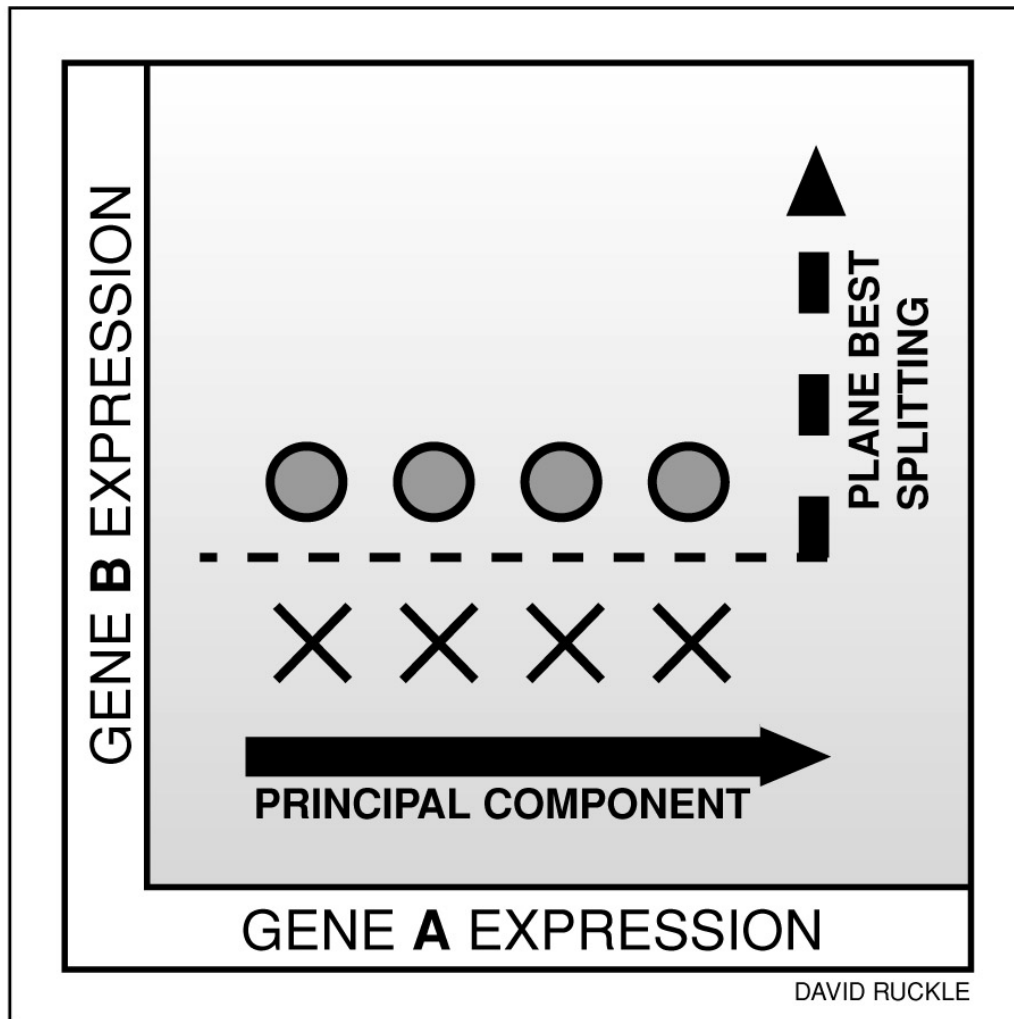


Figure 4.14: Feature reduction with principal components analysis. Two genes are measured in eight samples, four from one disease (marked by X) and four from another disease (marked by O). The expression measurements of gene A are represented on the x-axis and the expression measurement of gene B are represented on the y-axis. Note that the first principal component (*i.e.*, the vector that captures the most variance) is parallel to the x-axis, which corresponds to gene A. However, the line that best splits the two diseases is described by its orthogonal vector along the y-axis, which corresponds to gene B. This means that although the variance seen in gene A best explains the overall variance seen in gene expression measurements in each disease and both together, the variance seen in gene B is best used to split the two diseases. Although this is easy to distinguish in this simplified two-dimensional case, it is not so easily visualized in a real-world, multidimensional data set.

- *Which gene–gene associations best differentiate these two groups of tissue samples?* Answering this question requires a data set with sufficient measurements from the two types of tissues, and the analysis involves a combination of approaches. First, for each of the two tissue samples, one can first comprehensively compute a dissimilarity measure for all the gene–gene pairs. Then one can use a supervised approach to compare the dissimilarity measure for the two tissue types.^[9] For example, if the expression pattern correlation

coefficient between gene A and gene B was .95 in breast cancer, but was .25 in lung cancer, one might make a biological hypothesis that there is a difference in this interaction between the two diseases.

- *Which genes have a pattern similar to this organismal, phenotypic, or environmental factor?* If one can treat the external factor as a hypothetical gene expression pattern, then one can apply the supervised nearest neighbor technique to determine genes most similar to that pattern.

^[9]That is, we compare the measure–triangles for each tissue. The elements of the measure–triangle that are most different will suggest different relationships between the gene pairs corresponding to each of these dissimilar elements of the measure–triangle. Because the whole measure–triangle is used, the result is comprehensive.

4.12 Determining the Significance of Findings

One traditional way to gauge whether the findings for a particular microarray analysis are significant is to establish whether statistical significance is present. Traditionally, this involves the use of particular algorithms or formulas to determine the likelihood that that particular finding was governed simply by random chance. For example, in traditional clinical trials, when a study finding is noted to have a statistical significance, or p value of $<.05$, the assumption is that the likelihood of the findings in that particular analysis being due to random chance is less than or equal to 5 chances in 100.

These analyses are problematic when applied to functional genomic data sets. Suppose we have 10 tissue samples from patients with disease and 10 samples from patients without the disease. For a single gene, we can estimate how likely it is that the observed difference in the means of the expression of that particular gene in the two sets of samples could be attributable to chance. A simple lookup in the probability density function for that statistical test (e.g., a t -test) will provide this estimate. However, if the p value is found to be $<.05$ for that particular gene, then in a gene expression microarray measuring the expression of 10,000 genes, this would mean that we would expect to find the observed difference between the mean expression values across the two sets of 10 samples for 500 genes solely due to chance.^[10] Determining how to obtain the significance threshold for all the different bioinformatics techniques we have described above is an active area of research but has yet to make its way into the mainstream of published microarray analyses. Instead, pragmatic tests of the robustness of the results found have been adopted of the form described below.

4.12.1 Permutation testing

With the large number of comparisons being made, a valid criticism should be raised that perhaps high correlation coefficients or extreme values in other dissimilarity measures are found due to chance. One way to counter these arguments is to estimate how often random chance would generate correlation coefficients of the magnitude obtained in the original analysis. Typically, this is done through use of permutation testing.

One can take the entire array of cases and features (*i.e.*, samples and genes, respectively) and randomly shuffle the values in each column of features, thus breaking the links between the features. The mean and standard deviation expression level for each gene would remain the same. One can then repeat the entire analysis (e.g., dendrogram or relevance network) that one is performing 10 or 100 times or more to determine the strongest dissimilarity measure calculated.

For example, let us assume we are analyzing a microarray data set with 60 cancers and 2000 genes measured in each of those cancers. We want to perform a dendrogram analysis on this data set. As before, we would start by calculating the comprehensive pairwise comparisons. Within this

large set of comparisons, are high correlation coefficients likely to be spuriously generated? One way to answer this is to take each of the genes and randomly shuffle the gene expression measurements among the cancers and then repeat the entire comprehensive pairwise calculations. This permutation and calculation of the measure–triangle is repeated several times. For each iteration, the distribution of $\hat{r}^2$ values is stored. After all the iterations we will have a distribution of correlation coefficients from the original data set as well as several or an averaged distribution of correlation coefficients in the permuted and randomly shuffled sets. This is shown in figure 4.15.

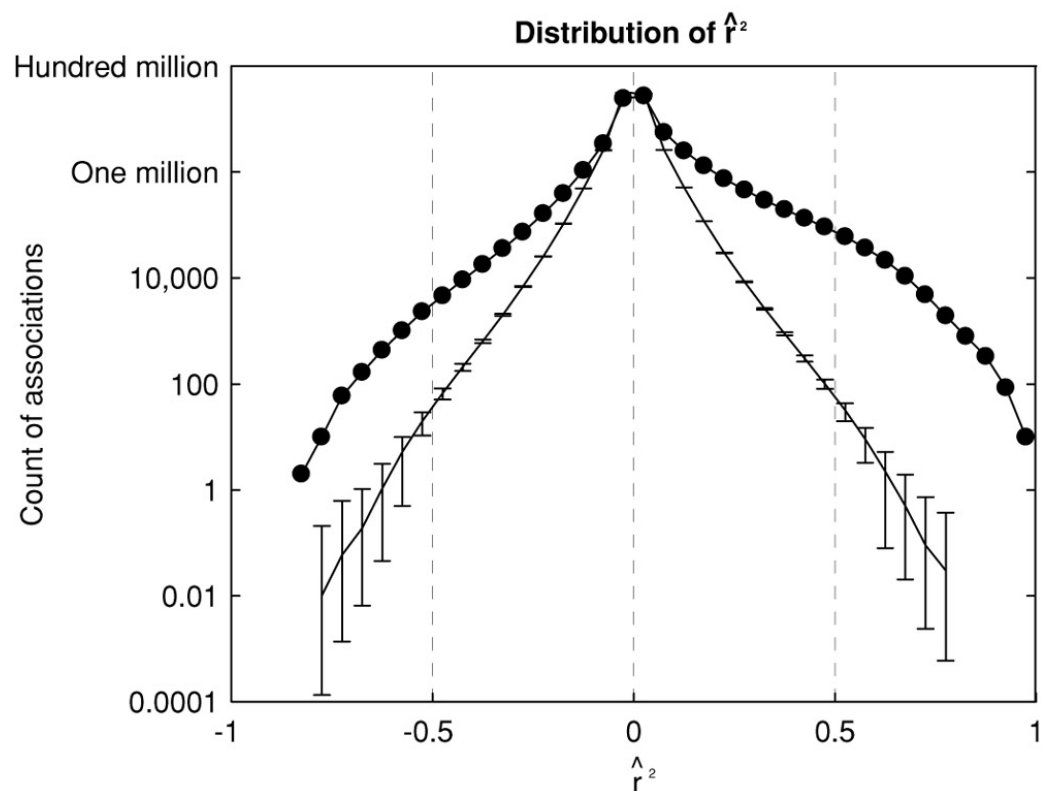


Figure 4.15: Distribution of $\hat{r}^2$ in the original versus permuted gene expression data set. The distribution of $\hat{r}^2$ calculated using an original gene expression data set is shown with solid circles. For each gene, expression measurements were independently randomly shuffled 100 times. The average distribution of $\hat{r}^2$ is shown with error bars covering 2 SD. In this example, random permutation was unable to create an association with $\hat{r}^2 > 0.80$ or $< .85$. (From Butte *et al.* [39].) If there is structure in the expression data set that is more robust than what would be expected by chance, the distribution of correlation coefficients from the original analysis should be wider than the distributions of the randomly permuted sets. We view this difference in area–under–the–curve between the two distributions as a measure of the signal–to–noise in the analysis. If the distribution of correlation coefficients in the randomly permuted data sets is as wide as the distribution correlation coefficients from the original unshuffled data set, it is a clear indication that the high correlation coefficients being seen could very well be due to random chance.

This permutation testing does not substitute for a p value, as traditionally used in tests for statistical significance, but it is one generally accepted way (in the functional genomics literature) to test a novel clustering algorithm and its application to determine whether findings are otherwise significant.

4.12.2 Testing and training sets

Perhaps the best way to test a model, whether constructed in a supervised or unsupervised manner, is to evaluate that model on a new set of inputs. This is typically done by determining a

priori those inputs that will make up the training set (*i.e.*, the input samples used to create the model), and those that make up the testing set (*i.e.*, those that will be used to test the model). It is important to determine whether population-measurable characteristics are similar between these two sets. For instance, one may want to train a system using samples from one type of tissue, and test it using another type of tissue.

This type of testing is crucial in that it prevents the generation of models that overfit the original input data. It is easy to describe overfitting with an example. Imagine a thought experiment where one takes a list of tyrosine kinases (*e.g.*, insulin receptor, IGF-1 receptor) and phosphatases (*e.g.*, PP1, PTP1B). Each one is written on a flashcard, along with one of the two category names. Now, one shows the flashcards to a child repeatedly. When later quizzed with cards without the category names, the child may be able to properly assign the function just by noticing that within the input set, whenever the name contained the word "receptor," the card was referring to a tyrosine kinase. Needless to say, the "rule" this child generated is quite specific to the particular set of input samples and their representation on flashcards.

Because so many features are typically present in microarray data sets, and because there are so few categories into which the samples have to be classified, it becomes very easy for a machine-learning mechanism to generate large numbers of "rules" specifying how each feature can accurately select the proper category. However, the characteristic of these genes may be specific only to the input sample set at hand, and those rules may not be generalizable to other input samples. Thus, it becomes important to test these rules using independent inputs.

At the risk of redundancy, we emphasize that finding test and training sets that are highly comparable is of paramount importance. In the discussion of the sources of noise in microarray experiments (section 3.2), we demonstrated that we could cluster a set of leukemia experiments into their original test and training sets because of systematic differences in the measurements in the test and training sets (figure 4.16). In the presence of such systematic biases, the inaccuracy or accuracy of a clustering or classification technique on a test set may not be a meaningful measure of its performance.

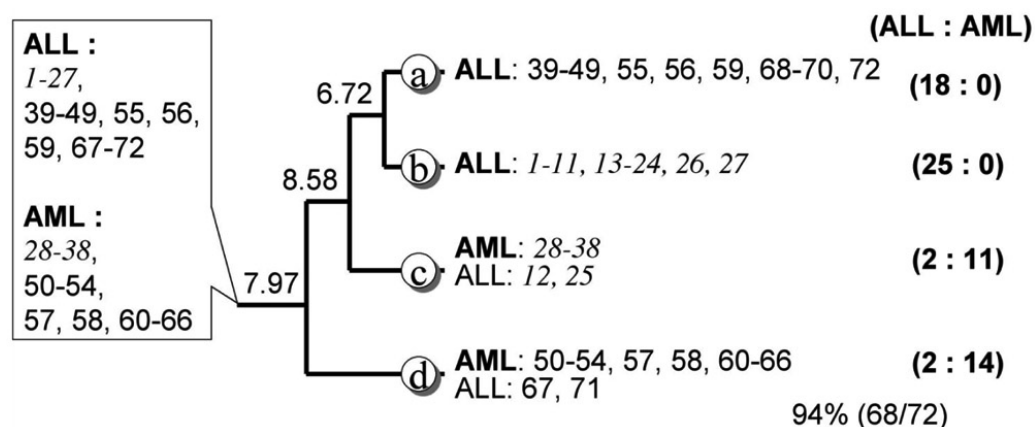


Figure 4.16: Identifying differences between the test and training sets through clustering. The matrix incision tree algorithm from [108] was applied to the leukemia classification problem published by Golub *et al.* [78]. The algorithm correctly clustered 64 of the overall 72 cell lines (94%) in the data set, placing the acute myelogenous leukemia (AML) samples in branches (c) and (d), the acute lymphocytic leukemia (ALL) samples in branches (a) and (b), and misclassifying four ALL samples into branches (c) and (d). However, the matrix incision tree also successfully revealed the distinction between the published training set (cases 1–38, italic, placed in branches (b) and (c)) and the test set (cases 39–72, placed in branches (a) and (d)) with 100% accuracy. It is unlikely that the biology of the test set of the AML and ALL cell lines were that different from the cell lines in the

training set. It is much more likely that systematic changes in the hybridization conditions, or different sample preparation, or the use of a different batch of microarrays were responsible for the clustering of test versus training sets.

Cross-validation testing Cross-validation is a classic approach to testing for robustness in models generated by machine-learning algorithms, built on the idea of having both a testing and training set, but for use when the total number of inputs is small. This technique works by repeatedly partitioning the available input into subsets for repeated trials. For each of these subsets, the algorithm or machine-learning technique is run and the output model is saved. After all the trials, the set of output models can be analyzed to determine those characteristics or features that are consistently represented across the trials. As an example, if in 10 trials, gene A is always found to accurately classify a set of samples between two diseases, but gene B accurately classifies in only 6 of the 10 trials, then gene A may be viewed as a more robust classifier, in that it may be less influenced by particularities of a single input set.

The traditional way this is performed is using *n*-way *leave-one-out cross-validation*, where *n* is the number of input samples: *n* input subsets are created, each missing the *n*th sample, then the machine-learning technique is applied to each of these input subsets. Cross-validation testing is easier to apply to supervised learning methods than to unsupervised, as it is difficult to determine accurate metrics for deciding on the strength of the generated unsupervised models. This approach has been used in several published works, including [19, 33, 46, 60, 72, 78, 86, 90].

4.12.3 Performance metrics

In evaluating genes or features that are meant to decide one category versus another, other relevant performance metrics include sensitivity, specificity, positive predictive value (PPV), accuracy, and area under the receiver operating characteristic (ROC) curve [79].

Let us refer to the state of disease or any category of interest as the "disease state" and all other categories or states as "nondisease states." The sensitivity of a test measures the proportion of correctly categorized disease cases out of the total number of actual disease cases, while specificity measures the proportion of correctly categorized nondisease cases out of the total number of actual nondisease cases. Positive predictive value is the fraction of cases correctly categorized as disease over the number of all disease cases, regardless of categorization. Accuracy is calculated by the ratio of correctly categorized cases over all the cases.

To further illustrate these metrics, figure 4.17 shows a contingency table in which the two classes are called 0 (non-disease) and 1 (disease). The vertical groupings of cases are gold standard-labeled^[11] cases belonging to each class. The horizontal groupings may be cases automatically categorized into each class. For example, a cases and c cases were actual class 0 cases, though a cases and b cases were labeled by a particular model as class 0. Similarly, b cases and d cases were actual class 1 cases, while c cases and d cases were labeled by a particular model as class 1. The relevant performance metrics in equation form are then as follows:

		Actual class:	
		0	1
Classified as:	0	a	b
	1	c	d

Figure 4.17: Contingency table illustrating performance metrics for classification algorithms.

$$\begin{aligned}
 \text{Sensitivity} &= \frac{d}{b + d} \\
 &= \frac{\text{Number of correctly categorized event cases}}{\text{Total number of event cases}}
 \end{aligned}$$

$$\begin{aligned}
 \text{Specificity} &= \frac{a}{a + c} \\
 &= \frac{\text{Number of correctly categorized nonevent cases}}{\text{Total number of nonevent cases}}
 \end{aligned}$$

$$\begin{aligned}
 \text{PPV} &= \frac{d}{c + d} \\
 &= \frac{\text{Number of correctly categorized event cases}}{\text{Number of categorized event cases (correct and incorrect)}}
 \end{aligned}$$

$$\begin{aligned}
 \text{Accuracy} &= \frac{a + d}{a + b + c + d} \\
 &= \frac{\text{Number of correctly categorized and nonevent cases}}{\text{Total number of cases}}
 \end{aligned}$$

4.12.4 Receiver operating characteristic curves

The ROC curve is a plot of sensitivity versus one minus specificity. Because sensitivity and specificity can be inversely varied by altering the threshold at which a case is categorized as one class or the other, the *area* under the ROC curve more effectively describes a classification algorithm's discriminatory ability. Stated differently, for a particular implementation of a classification algorithm, the trade-off between sensitivity and specificity is fixed as specified by the ROC curve. To find a different trade-off, one must obtain a different classification algorithm implementation that produces a different ROC curve. This is elaborated upon in figure 4.18.

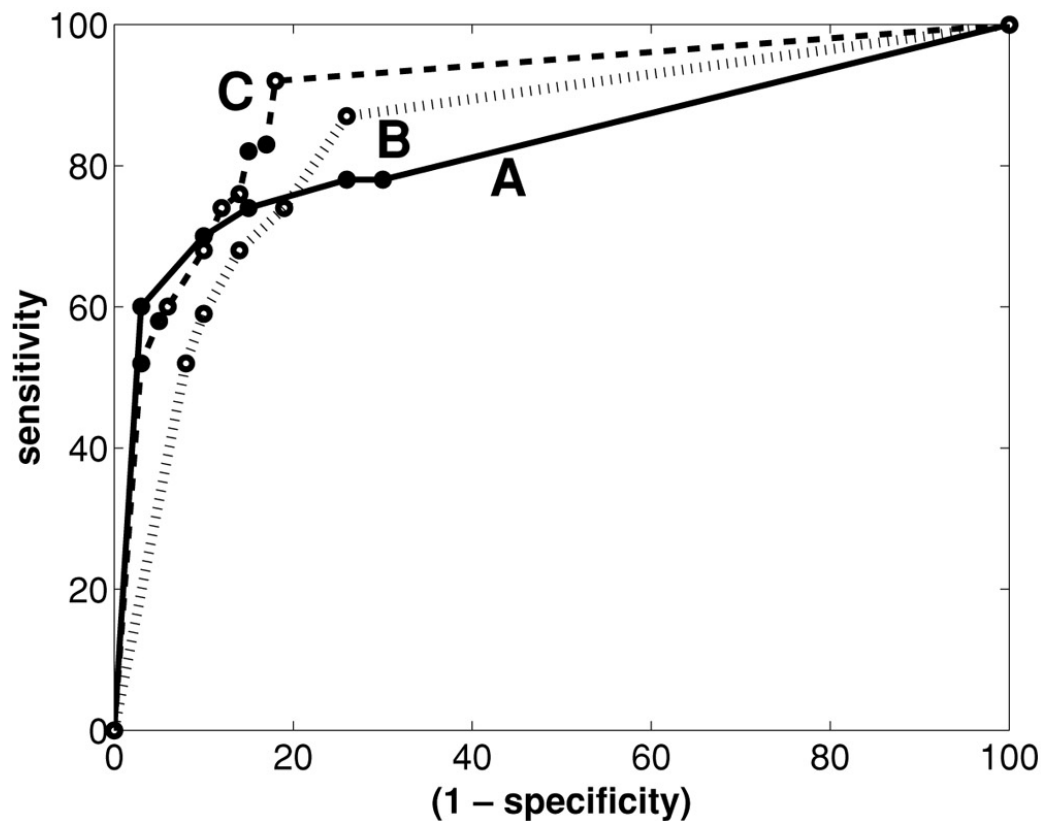


Figure 4.18: Sensitivity and specificity trade-offs are defined by the ROC curve. ROC curves A, B and C describe the trade-offs in sensitivity and specificity for the same data set using three different classification algorithms. C is clearly the best algorithm because the area under the curve is the highest. That is, it provides a better set of sensitivities and specificities for all values of these. In contrast, B and C provide trade-offs inferior to A and for some range of sensitivities, B provides better performance than C, and *vice versa*.

To calculate the ROC curve for a single continuous variable^[12] all that is required is to obtain the sensitivity and specificity at a sufficient number of thresholds. The procedure is somewhat more complex with classification models that involve different thresholds on several variables. For example, for decision trees, ROC curves are determined by first assigning to each tree leaf the probability of being an event for a set of derived values that percolates to that point. These probabilities are based upon the ratio of events to nonevents that fall into each leaf during training. The threshold for considering a case to be event or nonevent is then set at each leaf probability value. The resulting sensitivity–specificity pairs, when plotted on a grid of sensitivity versus (1–specificity), gives the ROC curve.^[13]

^[10]Many assumptions go into this estimate, including clearly erroneous ones such as the independence of all genes in their expression patterns.

^[11]For example, defined classes of two different types of leukemia as judged by an expert panel.

^[12]For example, if gene A > threshold, then the prediction is that the sample is from a patient with the disease, and if gene A ≤ threshold, then the prediction is that the sample is from a patient without a disease.

^[13]However, because the number of leaves in a decision tree are finite, the number of points that can be ascertained for that ROC curve is correspondingly finite and therefore the curve appears as a segmented trajectory of straight lines.

4.13 Genetic Networks

It is a testament to the ambitions of bioinformaticians that among the principal early goals they had for the use of microarray data was nothing less than to *reverse-engineer* the genetic networks responsible for maintaining cellular function. Although these ambitions may have been somewhat premature, we include them here because the goal of elucidating genetic networks remains a major challenge and problem of interest in this field.

4.13.1 What is a genetic network?

As covered in section 1.5, every human cell has on the order of 10^4 genes. Typically, not all the genes are expressed equally at any one time. At different stages of development, or as a response to extracellular perturbation, different genes of a cell are switched on at different intensity levels and at various time points. The genes and their associated, generally time dependent, levels interact with one another in a sophisticated regulatory web of cause and effect known as a genetic network. Restated, genetic regulatory networks are the set of mutually activating and repressing genes and gene products and their interactions [64].

It is generally difficult, if not impossible, to come up with the full set of parameters which completely characterize a genetic network. These parameters may be *intracellular* in nature, such as the mRNA concentration of another gene; *intercellular*, such as a hormone secreted by one organ to communicate a signal to another; or *extracellular*, such as an environmental condition like heat shock. Uncovering and understanding these interactions may potentially lead to new applications and insights on developmental disorders and diseases such as diabetes, cancer, and others.

The expression level of a gene is manifest by the presence of mRNA corresponding to its DNA transcript. Qualitatively, it is commonplace to say that a particular gene is highly expressed when high levels of its complementary mRNA are detected, whether by Northern blot, PCR, or microarray. Quantitatively, the reader may ask the obvious and more delicate question of how much mRNA should be present in a system in order that the gene be critically expressed so as to effect a biological state transition. In practice, this question is made more difficult by the realities of detection device sensitivity, repeat measurement variations, and how one defines a measurable biological state transition.

With the advent of microarrays, it has become possible to measure, in a relatively accurate and easy way, the expression levels of a great number of genes simultaneously. As a consequence, it is now possible to investigate topics such as time course development and drug intervention within the framework of the coordinated action of an entire transcriptome—the cellular genetic network—with the objective of reverse-engineering or unraveling the system's internal interdependencies.

4.13.2 Reverse-engineering and modeling a genetic network using limited data

Presently, microarray technology cannot yet measure the expression levels of all genes of an arbitrary biological system. Furthermore, there is the measurement technology-independent complication noted earlier such that it is generally impractical to obtain the full set of parameters that completely determine the state transitions of the system. Here, in essence, the reverse-engineering of a typical genetic network would mean uncovering dependencies and causal relationships between genes using a limited data set or knowledge base. Data in this context include empirical measurements of the respective components, in particular the mRNA levels, as well as a *priori* biological facts about the system under investigation. The information is limited in the sense that we

do not have corresponding data from every gene present in the system and, furthermore, the data we do have inevitably contain measurement or experimental errors. An example: Suppose that a researcher wants to study the interaction between two mouse genes, gene A_1 and gene A_2 , in liver from the day of birth until 1 week later. The researcher will use a robotically spotted microarray with probes for genes A_1 and A_2 . As samples, she or he harvests liver cells at N uniformly spaced time points from postnatal day 1 to day 7. Let us consider the possibilities of interaction or regulation between A_1 and A_2 , as discussed in [164]:

1. A_1 regulates A_2 , or *vice versa*. This interaction may be indirect, propagating through an intermediate cascade of other genes which are not being assayed.
2. A_1 and A_2 are co-regulated by another gene or a subset of genes.
3. A_1 and A_2 do not interact with one another nor do they share a common coregulator.

The term *regulate* here carries causal implications. One way to translate the biological statements above into mathematical formalism is to say that (1) and (2) are equivalent to saying that A_1 and A_2 are not stochastically independent, whereas (3) says that A_1 and A_2 are stochastically independent. An immediate question that comes to mind is whether it is technically possible to infer causal relationships between genes by their expression level data alone. For such a scenario, the most typical mathematical tool that our researcher could apply is the linear or rank correlation coefficient, r —provided that the number of time point measurements at which a gene is assayed, N , is large enough,^[14] for if $N = 2$ then any two genes are perfectly (and vacuously) correlated. As noted in [164], the observation of correlation between pairs of variables is commonly used in biology to predict causal relationships which have to be validated biologically. Indeed, that observation drives much of the data-mining machinery discussed in this chapter.

From a purely mathematical standpoint, correlation cannot be used to rigorously prove or disprove the existence of such relationships. Recall from our earlier review section on correlation (section 3.6.1) the basic rule of thumb that while independence implies zero correlation, the converse is not generally true. In spite of this, a high enough correlation (say $|r| > 0.9$, by no means a formal threshold) between genes A_1 and A_2 is a reasonably strong argument for further laboratory investigation of the pair, after taking into account the size of N , measurement errors in the primary data, and established facts about the pair's biological background.

Another approach to analyzing a genetic network is to view the whole as a dynamical system, not just consisting of pairwise interactions, and to take advantage of the traditional systems tools that already exist, (see [80] and [58]). Suppose that our researcher measures the expression level for genes G_j ($j = 1, \dots, M$) at times t_i ($i = 1, \dots, M$), denoted by $x_j(t_i)$. We also suppose that the expression level of a gene A_k at time t_q ($q > 0$) is a linear combination of the expression levels of all other genes G_j from the previous time step t_{q-1} [or more generally from all previous time steps t_r ($0 < t_r < t_q$)],

$$x_k(t_q) = \sum_{j=1}^M w_{k,j} x_j(t_{q-1}) + \beta_k \quad (4.13.1)$$

or rewriting in a matrix format,

$$\begin{pmatrix} x_1(t_q) \\ x_2(t_q) \\ \vdots \\ x_M(t_q) \end{pmatrix} = \begin{pmatrix} w_{1,1} & w_{1,2} & \cdots & w_{1,M} \\ w_{2,1} & w_{2,2} & \cdots & w_{2,M} \\ \vdots & \vdots & \cdots & \vdots \\ w_{M,1} & w_{M,2} & \cdots & w_{M,M} \end{pmatrix} \begin{pmatrix} x_1(t_{q-1}) \\ x_2(t_{q-1}) \\ \vdots \\ x_M(t_{q-1}) \end{pmatrix} + \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_M \end{pmatrix}$$

where $w_{k,j}$ is a weight term encoding the influence of the expression of gene G_j on the expression of G_k , and β_k is a parameter which may be stochastically distributed which could encapsulate *a priori* knowledge about regulation. Note that Eq. (4.13.1) is linear in the expression quantities and is first-order Markov, i.e., the state of the system at time t_q is entirely determined by the time t_{q-1} state.

We could also write Eq. (4.13.1) as a difference equation in keeping with our systems viewpoint,

$$\Delta x_k(t_{q-1}) = \sum_{j=1}^M w'_{k,j} x_j(t_{q-1}) + \beta_k \quad (4.13.2)$$

$$w'_{k,j} = \begin{cases} w_{k,j} - 1, & \text{if } k = j, \\ w_{k,j}, & \text{otherwise,} \end{cases}$$

where " $x_k(t_{q-1}) \neq x_k(t_q) \neq x_k(t_{q+1})$ ". Unlike the classic system of difference equations, $w_{i,j}$ here are the unknowns and $x_j(t_q)$ are the known measured quantities. In general, $w_{k,j} \neq w_{j,k}$. Recall from linear algebra that in order to solve for the M^2 unknowns $w_{i,j}$ uniquely, one needs to have M^2 different equations Eq. (4.13.2); in other words, there should be enough data time points. If the number of unknowns exceeds the number of equations, then the system is underdetermined and may have infinitely many solutions for $w_{i,j}$. Conversely, if there are more unknowns than there are equations, then the system is overdetermined and it may be algebraically insoluble. In this situation, one can attempt to find a weight matrix $w_{i,j}$ that "solves" the system in the sense of minimizing an *ad hoc* energy functional. A typical example of this approach is the least squares fitting of data to a line where the energy functional is the sum of distances from the fitting line. Once the weights $w_{k,j}$ in Eq. (4.13.2) have been computed and we suppose as modelling assumption that $w_{k,j}$ are time-independent, then Eq. (4.13.2) is a well-defined difference system and one may apply standard dynamical systems or Markov chain techniques to explore interpolation and prediction-type questions such as:

1. As a reality check: Does the computational result or prediction agree with the outcomes from the original physical experiment?
2. How would the model Eq. (4.13.2) evolve under a different set of initial states—simulating a different experimental setting? Are there particular initial conditions which lead to dramatically different subsequent behavior in the presence of a small perturbation? Continuous dependence on initial state or bifurcations?
3. If we removed the time-series data for a gene G_k and recalculate the weights $w_{i,j}$, how different is the resulting model from the one with G_k present?
4. What is the asymptotic or steady-state behavior of Eq. (4.13.2)? Attracting sets, hysteresis cycles, recurrent states?

The state of a system at any time t_q refers to the collective expression profiles $x_k(t_q)$ for $k = 1, \dots, M$. In a developing hippocampi system modeled with Eq. (4.13.2), Somogyi *et al.* found that the weight matrices were sparse *i.e.*, most of the entries were 0, in agreement with the generally accepted intuition that genes are not regulated equally by every other gene [164]. From a computation-theoretic point of view, it may also be interesting to study the time-continuous analog of the difference equation (4.13.2), which would be a system of linear differential equations,

$$\frac{dx_k(t)}{dt} = \sum_{j=1}^M w'_{k,j} x_j(t) + \beta_k$$

or stochastic differential equations, if we allow β_k to be a specific probability distribution as in [109]. When one has a deterministic system of linear ordinary differential equations, Fourier or wavelet methods could be used to transform into a system of algebraic equations.

An attendant complication with these methods is the aliasing phenomenon. One could potentially be undersampling the respective RNA levels below the fundamental frequency (probably cell cycle-related) of the biological system. As is well-known in signal processing, it is not possible to reconstruct the originating signal from samples that were obtained less than twice the threshold rate

known as the Nyquist frequency. The model can easily be generalized to become a nonlinear one by letting $w_{j,k}$ be both a function of time and expression, or to a higher-order Markov by assuming that the expression level of A_k at t_q depends upon the expression levels of A_j 's for other previous times t_r , $0 < t_r < t_q$. Weaver *et al.* [186] and Wahde and Hertz [185] studied a modified version of Eq. (4.13.1),

$$x_k(t_q) = g \left(\sum_{j=1}^M w_{k,j} x_j(t_{q-1}) + \beta_k \right)$$

where $g(y) = (1 + e^{-\gamma y})^{-1}$ is a modified step function where $\gamma > 0$ is the rate that the value of g changes from 0 to 1 at $y = 0$. When $\gamma \rightarrow \infty$, $g(y) = 0$ (resp. 1) for $y < 0$ (resp. $y > 0$).

There are several biologically based reasons to motivate this step function-like form of g (e.g., upon binding of a transcriptional repressor or initiation of a kinase cascade). Note that g has a well-defined inverse function, g^{-1} . The weights of the system above can be solved via linear algebra after applying g^{-1} on both sides of the equation. Operating within the systems paradigm, Hanke and Reich [83] and Somogyi and Sneigowski [164] have also modeled Eq. (4.13.2) as neural networks which have the advantage that one could train the parameters—in this case, the weights $w_{i,j}$ —to match the data set. Potential issues here include the selection of the type of neural network that is optimally suited to the questions being studied, the length of time required for training, and the characteristics of the training set.

Thus far we have considered the expression levels x_k to be essentially continuous variables interacting in a linear combinatorial manner. One may as validly let x_k be a discrete-valued variable taking values 0 and 1, where 0 indicates that gene A_k is off and 1 on, to produce a Boolean network as in [119]. State transitions are captured by a truth table which we have to discover using *ad hoc* rules optimizing mutual information.

Recently, there has been active work in the area of analyzing genetic networks within a probabilistic framework; of particular note is the application and development of bayesian models of gene expression and interaction which will be detailed in the next section.

It probably bears repeating here that no mathematical treatment of observational data will be sufficient to be convincing about genetic causal relationships. For that to occur, the tried-and-true methods of scientific hypothesis testing (see figure 2.2) and the concomitant "wet" laboratory work will remain necessary. Therefore it should not be too disappointing to note that we have still not adequately addressed the issue of how one may infer causal relationships from expression data, in particular limited expression data. From the purely computational point of view, it is not possible to do this without the aid of established biological facts and experimental intuition. This in turn requires a productive functional genomic pipeline of the sort diagrammed in figure 1.3.1.

4.13.3 Bayesian networks for functional genomics

Bayesian networks (BBNs) [139], also known as causal probabilistic networks, represent one of the most successful advances of artificial intelligence (AI). Born at the confluence of statistics and AI, BBNs provide a sound way to discover models of interaction among variables. When applied to functional genomics, they offer the opportunity to discover complex pathways of activation and interaction among genes. Although still in an early stage, the applications of BBNs to functional genomics already displays the common problems statistical methods encounter in the realm of functional genomics: the large number of variables met by a small number of cases and the

consequent under-determination of the models learnable from data. Nonetheless, because BBNs provide at least the potential of determining genetic networks on the basis of probability theory, and therefore all the desiderata that flow from such a grounding, they are beginning to attract increasing interest in this field. This section first reviews the basic concepts underpinning BBNs and then describes the early attempts to apply this methodology to functional genomics.

Bayesian networks Bayesian methods provide a sound and exible framework to reason about the interplay between conditions and effects under uncertainty. The theoretical core of bayesian theory itself—the well-known Bayes' theorem—is built around the intuition that the path leading to an effect can be reversed and a probability can be assigned to the set of possible conditions responsible for that effect [148]. The resulting approach also provides a straightforward way to integrate human-derived domain knowledge and domain information coming from field operations (*i.e.*, empirical biology) or simulated data.

The bayesian approach shares, with all sound knowledge representation formalisms based on probability theory, two main drawbacks. The first is representational in nature and arises from the fact that a naïve encoding of knowledge as a joint probability distribution of all the variables in the domain is very difficult to acquire from human experts. The second is computational, but is still based on the same naïve interpretation of the probabilistic representation: The probability distribution over the domain variables, so difficult to acquire from experts, can grow so large to be very costly to store and very expensive to use for reasoning.^[15]

Blending together probability and graph theory, BBNs provide a way to render the probabilistic representation of knowledge representationally amenable and computationally feasible. A BBN uses conditional independence assumptions encoded by a Direct Acyclic Graph (DAG) to break down an otherwise unmanageable joint probability distribution over the domain variables into a set of smaller components, easier to define and cheaper to use. In a BBN, the nodes in the graph represent variables and directed arcs represent stochastic dependencies among the variables. As a simple example, consider the scenario displayed in figure 4.19 composed of three variables, **Initial_Condition**, **External_Action**, and **Effect**, each having two states "True" and "False." The joint effect of the the two parent variables on the **Effect** is represented by the DAG in figure 4.19, with the directed links from the variables **Initial_Condition** and **External_Action** pointing to the variable **Effect**. Following the direction of the arrows, we call the variable **Effect** a child of **Initial_Condition** and **External_Action**, which become its *parents*.

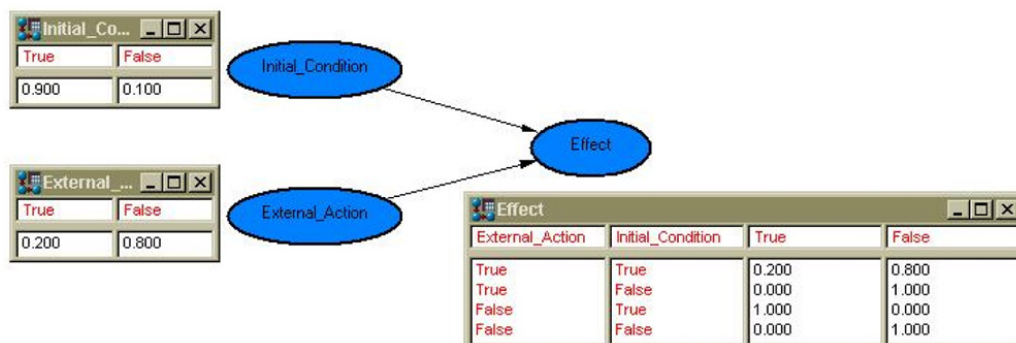


Figure 4.19: Example of a bayesian network. This network describes the impact of **Initial_Condition** and **External_Action** on **Effect**. This network is equivalent to the underlying joint probability distribution over the three variables, shown in table 4.1. In particular, figure 4.19 shows a BBN representing the combined action of a quite frequent **Initial_Condition** (90% chance of being true) and a rarely taken **External_Action** (20% chance of being true) over an **Effect**. The probability table associated with the node **Effect**

shapes the interplay between **Initial_Condition** and **External_Action** so that, when only the **Initial_Condition** is in place, then **Effect** will be present for sure, unless **External_Action** is deployed to dilute the impact of the **Initial_Condition**. This mechanism is described by the fact that, when **Initial_Condition** is true and **External_Action** is false, the probability that the **Effect** is true is 1, whereas when **Initial_Condition** is true and **External_Action** is also true, then the probability that the variable **Effect** is true is only 0.2. On the other hand, when the **Initial_Condition** is not in place, then the **Effect** is always absent, regardless of the presence or absence of the **External_Action**. The BBN in figure 4.19 decomposes the joint probability distribution of the three variables in table 4.1 into three probability distributions, one for each variable in the domain. This decomposition is the key factor for providing both a verbal and a human-understandable description of the system, like the one used for the BBN in figure 4.19, and for efficiently storing and handling this distribution, which grows exponentially with the number of variables in the domain.

Table 4.1: Joint probability distribution over the domain variables

Initial_Condition	External_Action	Effect	
True	True	True	0.036
True	True	False	0.144
True	False	True	0.720
True	False	False	0.000
False	True	True	0.000
False	True	False	0.020
False	False	True	0.000
False	False	False	0.080

Notwithstanding these differences, both the BBN and the joint probability distribution represent the same stochastic model of the combined **Effect of Initial_Condition and External_Action**. The BBN can therefore be used to perform all the kinds of reasoning operations provided by a probabilistic representation of the system. As mentioned before, bayesian theory provides naturally a variety of methods to reason about the behavior of a system and to understand the interplay between its conditions and its effects. Assuming sufficient microarray data (a difficult assumption at present, but likely to become less so in the near future), this is sufficient to model cellular physiology.^[16] Given a set of initial conditions, a BBN can forecast their effects (prediction—particularly useful for hypothesis testing). Conversely, a BBN can walk backward a chain of dependencies and determine the probability of the Initial Condition given the observation of an effect (explanation—essential in understanding complex genetic networks). Even more interesting, a BBN can identify the best configuration of initial conditions to achieve a particular effect (optimization—again useful for generating testable hypotheses as well as identifying pharmacological targets), and can identify the individual contribution of an Initial Condition to the realization of an effect. This ability to propagate information along the network without a prespecification of the inputs allows the analyst to "drop" evidence and conjectures into the network and figure out the effect of different initial conditions by "flickering" the input of the system or demanding different outcomes (sensitivity analysis).

Although, in their original concept, BBNs were designed to encode knowledge of human experts, their statistical roots soon prompted the development of methods able to learn them directly from databases. The advantages of modularity and compactness did not go unnoticed in the statistical community, so that BBNs and, in general, graphical models, enjoy a growing popularity in statistical practice, disclosing new directions for research and applications. The process of learning a BBN

from a database consists of the induction of its two different components:

1. The graphical structure of conditional dependencies (*model selection*)
2. The conditional probabilities quantifying the BBN (*probability estimation*)

Here, we take a bayesian approach to "learning" BBNs, so that the process of learning probabilistic models is regarded as the process of updating prior information on the basis of available evidence. This framework fosters the integration of expert domain knowledge and data analysis; when available, expert knowledge can be integrated in the discovery process but it is not necessary, as in traditional knowledge-based systems.

On the other hand, lack of human expert knowledge can be represented by uniform prior distributions of probability, and the discovery process is totally left to data. As far as AI is concerned, this line of research was pioneered in [168, 52] and further developed in [36, 88]. A parallel line of research is going on in statistics, in the field of graphical models in [190, 115]. The deep theoretical connection between probability theory—the foundation of BBNs—and statistics creates a powerful link able to provide scientific meaning and development methods.

Current bayesian techniques for learning the graphical structure of a BBN from data are based on the evaluation of the *posterior probability* of network structure, *i.e.*, the probability of the graphical model conditional on the data. We can take this measure as a scoring metric to compare alternative models of interactions, *e.g.*, to identify the most probable set of variables which are *parents* of a phenotype. This probability is then used as scoring metric to discriminate among a set of possible models. We are therefore looking for a way to compute the probability $p(M|\Delta)$ of a model of interaction M given the data Δ . By Bayes' theorem we have

$$p(M|\Delta) = \frac{p(\Delta|M)p(M)}{p(\Delta)}. \quad (4.13.3)$$

As we are using this quantity to compare models and all models are compared over the same data, $p(\Delta)$ turns out to be a constant and can be removed. If we assume that all models are equally likely, as is appropriate for an unsupervised method, then the quantity $p(M|\Delta)$ becomes proportional to the quantity $p(\Delta|M)$, known as *marginal likelihood*, which can be efficiently computed in closed form.

The decomposition of the joint probability distribution induced by the graphical model decomposes the marginal likelihood $p(\Delta|M)$ of the entire model into the product of the marginal likelihoods of each node and its parents, so that we can learn a model locally, by maximizing the marginal likelihood node by node. Still, the space of the possible sets of parents for an effect grows exponentially with the number of parents involved, but successful heuristic search procedures exist (both deterministic and stochastic) to render the task feasible. The decomposition of the marginal likelihood allows the user to assess locally the strength of the dependency of an effect upon its parents as a comparative measure against rival models (including the model in which the effect is not influenced by any variable reported in the database). While the marginal likelihood *per se* is difficult to interpret, it can be used to compute a more intuitive measure, known as the *Bayes factor*, to assess the strength of evidence of one model against another model. Since the Bayes factor is the ratio

$$\frac{p(M_1|\Delta)}{p(M_2|\Delta)}$$

between the posterior probability of two alternative models M_1 and M_2 , under the same assumptions used in the reduction of Eq. (4.13.3), the Bayes factor can be computed as the ratio of the marginal likelihood

$$\frac{p(\Delta|M_1)}{p(\Delta|M_2)}$$

of the two alternative models. This more intuitive measure will tell the analyst how many times the model M_1 is more probable than model M_2 , and the confidence in the selected model will be a function of the distance between the selected model and the most probable alternative models [105]. Once the graphical model is known, the decomposition of the underlying joint probability distribution induced by the graphical model provides, once again, efficient methods to learn the conditional distributions that quantify the dependencies in the network by taking advantage of local computations and conjugate analysis.

Bayesian networks applications in functional genomics BBNs are not new to genetic research. In fact, networks based on directed acyclic graphs actually originated from the genetics studies of Sewall Wright [194], who developed a method called *path analysis* [195, 196], a recognized ancestor of BBNs. The application of BBNs to functional genomics is, on the other hand, very recent. bbnns hold the promise of answering very interesting questions in functional genomics and, in principle, they seem to be the right technology to take advantage of the massively parallel analysis of whole-genome data to discover how genes interact, control each other, and align themselves in pathways of activation.

BBNs offer a different view than do the more popular clustering algorithms currently used for the analysis of massively parallel gene expression data [63, 167]. While these algorithms attempt to locate groups of genes that have similar expression patterns over a set of experiments to discover genes that are co-regulated, BBNs dive into the regulatory circuitry of genetic expression to discover the web of dependencies among genes.

The promise of BBNs in functional genomics goes even further, as intensive research efforts have been addressed, during the past decade, to define conditions under which BBNs actually uncover the *causal* model underlying the data [140, 89]. The most ambitious question is therefore the following: Given a set of microarray data, can we discover a causal model of interaction among different genes? The challenge is the common problem of sound statistical methods when faced with microarray data: a large number of variables with a small number of measurements. In the context of BBNs, this situation results in the inability to discriminate among the set of possible models as the small amount of data is not sufficient to identify a single most probable model.

Friedman *et al.* [71] address these problems using partial models of BBNs and a measure of confidence in a learned model. The strategy they follow is to search a space of underspecified models, each comprising a set of BBNs, and select a class of models rather than a single one. They also adopt a measure of confidence based on bootstrapping to evaluate the reliability of each discovered dependency in the database to avoid the risk of ascribing a causal role to a gene when not enough information is actually available to support the claim. Hartemink *et al.* [85] are tackling the underdetermination problem by tuning the unsupervised search of the most probable network structure. They leverage on established biological knowledge to select a small number of networks and then limit their comparisons to these networks only.

The use of BBNs in functional genomics is still in its infancy and the common problems of functional genomics data have still to be solved and much work is still left to render the technology fit for the task. Nonetheless, even these early efforts make clear the potential of these methods to dissect the inner structure of the regulatory circuitry of life.

^[14]In practice, sufficiently high-resolution time series of microarray measurements performed over an adequate interval are rare if not absent.

^[15]See the discussion of the appetite for hard-to-get probabilities in Szolovits and Parker [173].

^[16]With the caveats about false reductionisms articulated in section 1.4.1.

Chapter 5: Bio–Ontologies, Data Models, Nomenclature

Overview

Suppose that you have mastered normalization of microarray data sets, performed experiments to ensure reproducibility, and then gone on to acquire several large data sets and have now completed your clustering or classification experiments. Remarkably, after that preparation, you have what looks like a very nice set of results where the clusters of genes you found on one set of experiments have reproduced beautifully in another set of experiments that were designed to be identical. Furthermore, every time you slightly permute the conditions from those in the control or baseline condition, the sets of genes in the clusters seem to have changed in reproducible ways. And furthermore, the clusters appear to be compact and distinct so that you suspect that you have identified truly significant, functionally related genes.

If one steps back a little at this point, it becomes less obvious what the meaning is of all the effort that one has invested to date, because what one actually has generated is a list of microarray accession numbers. Even an experienced functional genomicist looking at this list of numbers will not have the vaguest idea of what it means. Some of these accession numbers cluster two or three genes, while other clusters include hundreds. At this point, the functional genomicist has very little alternative other than to look up the genes corresponding to each microarray accession number one by one and then peruse the literature regarding each one of the of the genes to determine

- whether that gene has a known function, and if so, in what class (*e.g.*, transcriptional factor, metabolic enzyme, structural protein, combinations of these, *etc.*);
- whether the genes found clustered together have been described in the literature as being functionally similar or related, or perhaps share promoter motifs, or a subset of the cluster are transcription factors for the rest of the cluster;
- whether homologs or orthologs have been found to be functionally related in any known physiological or pathological state;
- whether the resultant genes are known to be associated with the experimental conditions tested.

To the degree that the functional genomicist understands basic biology, this task is somewhat more tractable. A competent biologist will acquire in the course of his or her training a large framework of functional dependencies of different biological processes and the genetic machinery that underlies these dependencies. This kind of knowledge is unlikely to be present in any single journal article. Even a review article presumes a large shared body of biological knowledge that will be opaque to a researcher unfamiliar with a particular corner of the large space of biological knowledge. Without this rich context, many insights that the analyses might suggest will be missed. Consequently, a biologist familiar with the specific biological phenomenon being studied will be an invaluable collaborator in interpreting the meanings of the results from analysis of these massively parallel data sets. However, even a competent biologist armed with a very efficient search engine of the biomedical literature will find that determining whether the functional dependencies found in these clusters make any particular sense will take a very long time, often more than any other part of the analytic methodology of functional genomics.

Consequently, after having undertaken this laborious, rote lookup process several times, the following fantasy occurs to most genomicists: "Wouldn't it be nice if I could look at the cluster and automatically see that one of the clusters contained both genes coding for known transcriptional

factors and genes coding for structural proteins, and under two pathological conditions, those structural proteins were only translated if the transcriptional factors in that same cluster were expressed at their highest levels. Also, six researchers have developed mouse models in which four of these genes in that cluster have been misexpressed, and those mouse models can be obtained by filling out a form on a specified website..."

Enticing as this fantasy is, unfortunately, it does not represent the current state of the art. However, there are many early efforts to achieve this goal. Some skeptics will note that this is an "AI complete problem" [7] in the sense that reaching the goal outlined in the above scenario would really be equivalent to significant progress in the development of machine intelligence that can reason in a commonsensical fashion about a wide range of heterogeneous knowledge. However, as we shall see from some of the early efforts below, even a much more modest set of goals might be quite useful for the purposes of the functional genomicist. The technological solutions provided to achieve these goals fall into three increasingly circumscribed tasks: development of bio-ontologies, development of common data models, and development of common nomenclatures.

5.1 Ontologies

The term *ontology* is in its current usage substantially different from its original philosophical meaning, which was the "study of the nature of being." However, as most modern linguists will acknowledge, correct usage in the end is determined by widespread current practice and currently, the term ontology encompasses the study of formalized descriptions of all entities within the area of interest or research and all the relationships between those entities. In the discipline of functional genomics, bio-ontologies refer to organized and formalized systems of concepts of biological relevance in which the relationship between these concepts can be qualified. Before we proceed to some of the technical aspects of ontologies and examples of the more successful bio-ontologies, we list some notable efforts at creating usable bio-ontologies. First, we list here some additional motivations for the development of bio-ontologies.

- The need for a communal memory in the context of exponential growth in the amount of genomic information. That is, at the very least, we want to be able to share the biological knowledge that we are able to derive from our experiments, both to avoid duplication and to build on the experience of others.
- If we are to automate the process of sharing knowledge and use the prior inferences of functional dependencies, then the ontology must have sufficient formalization to enable a computational process that is both sound and efficient. As will be discussed briefly below (section 5.2), there is a natural tension between developing ontologies that are sufficiently expressive and those that are sound and efficiently computable.
- In order for a bio-ontology to be truly useful, it needs to be able to communicate the definition of terms in a reader- and context-independent way. This is particularly important in the domain of functional genomics because of the multiplicity of function of genes. For example, a gene's products will often have diverse functions within the same cell or even within the same subcellular compartment, and most often in different tissue types and under divergent environmental conditions. Stated in the converse, a gene's function will be highly determined by the subcellular context of its protein and the other coexpressed genes. For example, medium-chain dehydrogenase serves as a metabolic catalyst in one context, and in another it serves as a component of the structure of the eye lens (one of the crystallins) in some animals.
- An ontology should only represent the terms or relationships that are absolutely necessary (minimal ontological commitment). The lesser the ontological commitment, the greater the number of purposes for which a particular ontology can be used. For instance, consider a

bio-ontology that is specifically designed to capture the structural protein motifs associated with a gene and to capture the potential sites of interaction between proteins. This ontology may be extremely useful in the screening of genes that might be coding for proteins that are discovered to physically interact, but it may be less useful in determining whether or not a particular protein is a known transcriptional factor for a particular gene. This latter information might best be represented in a different formalism than the protein structure representation.

- There is a significant need for *organism-specific* ontologies, as genes do not always have the same function across different species. A gene in the human genome may have a single homolog in yeast, but the way in which it is regulated, the biological systems in which it participates, and the nature of the additional structural complexity of its protein in the human may be obscure without a species-specific bio-ontology.
- There also needs to be *cross-organism* ontologies because many genes have identical or analogous function across organisms. If a finding is made in a yeast experiment, *e.g.*, then the ontology should inform the functional genomicist of the potential for finding a human gene with the same function, if one is found with similar sequence homology. Conversely, a gene in one species may have similar function to that of another nonhomologous gene in another species. Understanding the parallels in the regulation and function of different genes across species may provide important biological insights.

5.1.1 Bio-ontology projects

We will now cover a few examples of successful bio-ontologies. For a more comprehensive and current compendium of ontologies such as these, the reader is referred to issue no. 1 of *Nucleic Acids Research*, of any recent year.

The Gene Ontology Consortium The Gene Ontology (GO) consortium is one of the more high-profile efforts that has developed in response to the pressing need for a unifying framework. As summarized by Ashburner *et al.* [13], the goal of the GO consortium is to provide a controlled vocabulary for the description of three independent ontologies:

1. *A description of molecular function.* This describes the biochemical activity of the entity such as whether it is a transcriptional factor, a transporter, or an enzyme, without making any further commitments as to where the functions occur or whether the function occurs as part of a larger biochemical process.
2. *The cellular component within which gene products are located.* This provides a localization of a gene product's molecular function, such as the ribosome, the nucleosome, or mitochondrion. Localization can help determine whether a purported function could occur through direct physical interaction between gene products or as a result of an indirect mechanism.
3. *The biological process implemented by the gene products.* Biological process refers to a higher-order process, such as pyrimidine metabolism, protein translation, or signal transduction.

The goal is to allow queries across databases using GO terms providing linkage of biological information within and across species. The three ontology domains are presented in figures 5.1, 5.2, and 5.3.

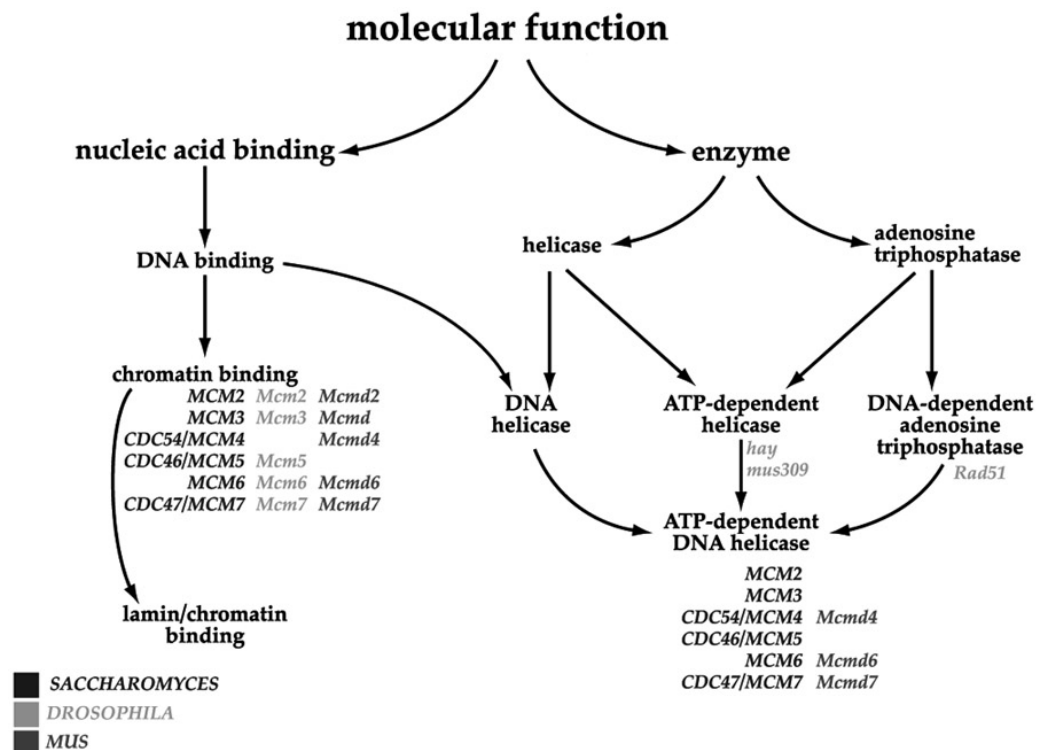


Figure 5.1: The Molecular Function Ontology of GO. (From Ashburner *et al.* [13].)

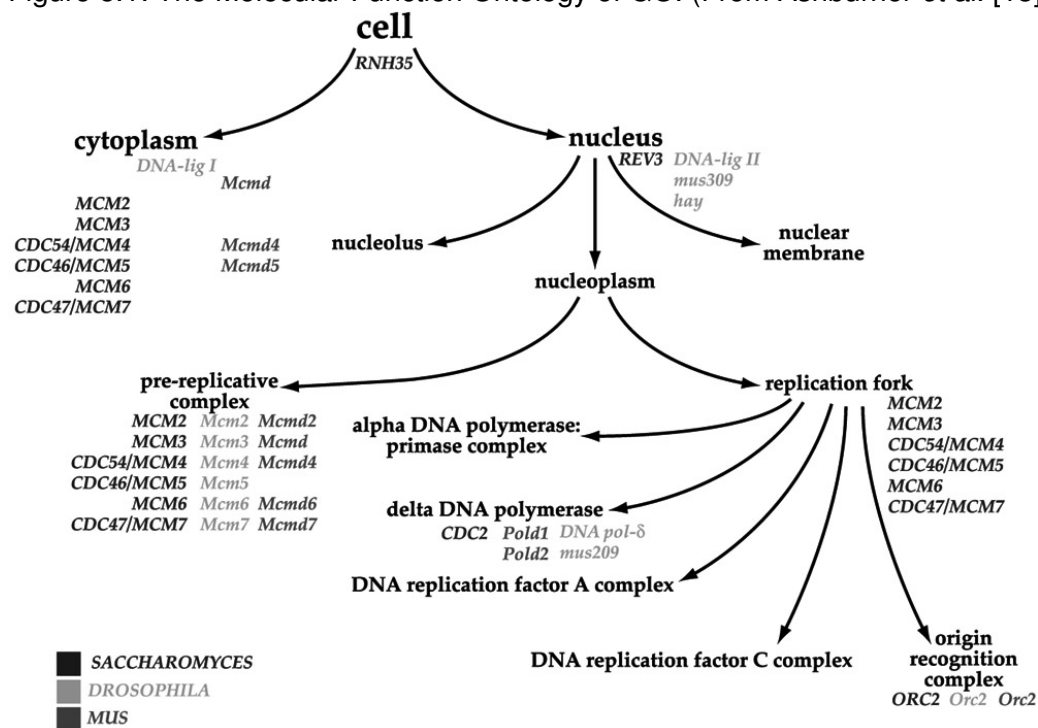


Figure 5.2: The Cellular Component Ontology of GO. (From Ashburner *et al.* [13].)

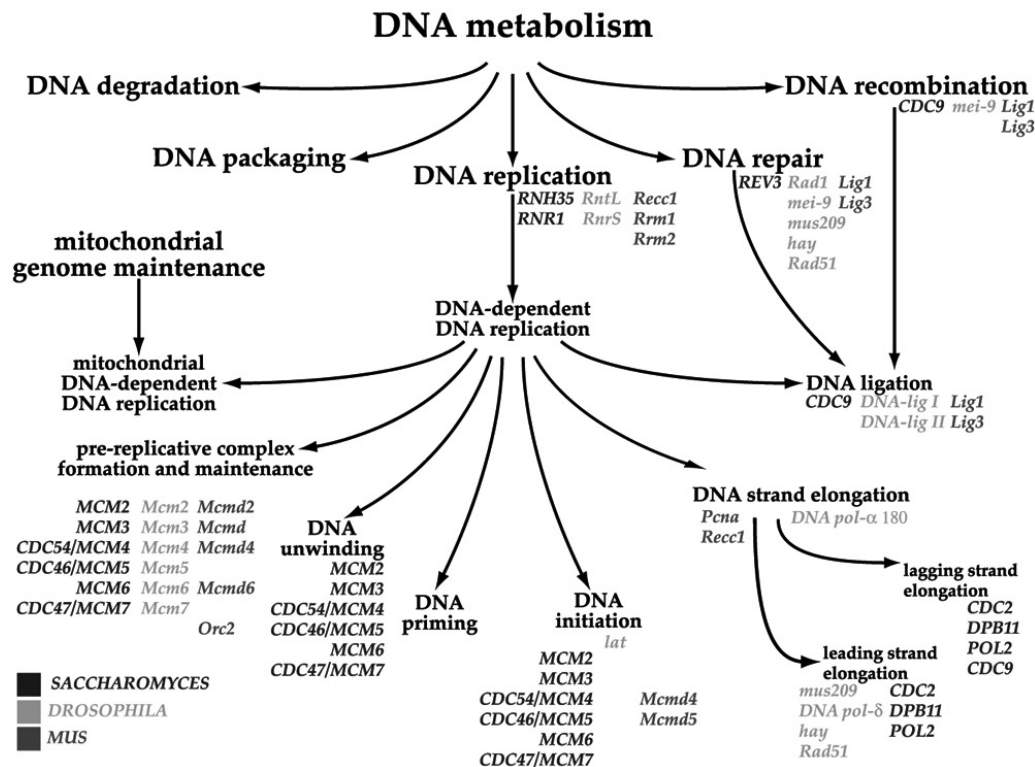


Figure 5.3: The Biological Process Ontology of GO. (From Ashburner *et al.* [13].)

These three ontologies are a result of a practical consensus of the kinds of distinctions that are important at present in functional genomics. If the GO consortium efforts were to populate this form of annotation across all genes for all sequenced organisms, this would be an invaluable resource even though the ontology is of so little commitment that it may not enable several categories of useful inferences. Examples of the latter include causal transitivity, temporal ordering, assertion of disjoint subconcepts, and many more. Representations which might support these more elaborate inferences are discussed in section 5.1.2. Again, there are some significant trade-offs in using ontologies of much more ontological commitment and expressivity (see section 5.2). Despite the limitations of the simple ontologies of GO, early efforts to annotate the organismal genomes using this infrastructure appear very promising, notably the GO annotations in the LocusLink site of the National Center for Biotechnology Information (NCBI).^[1]

Kyoto Encyclopedia of Genes and Genomes One of the more comprehensive attempts to describe the molecular and functional biological knowledge (all the way down to the detail of interacting molecules) for all known genes is the Kyoto Encyclopedia of Genes and Genomes (KEGG).^[2] Rather than generating annotations per gene, KEGG attempts to curate and elaborate on metabolic pathways, large molecular assemblies, and regulatory pathways. These larger abstractions are invaluable to biological researchers and allow the rapid identification of how a gene fits into cellular physiology. The KEGG ontology has five components:

1. *Pathways* with a corresponding graphic pathway map
2. *Ortholog groups*, which represent highly conserved functional groups annotated by the genes belonging to these groups across species
3. *Molecular catalogs* providing functional annotation of proteins, RNA, and small molecules
4. *Genome maps*, which mirror much of the positional information at the NCBI
5. *Gene catalogs* which have similar content to the Molecular Function Ontology of GO

The richness and utility of the KEGG ontology can be seen in figure 5.4 which provides a snapshot of one pathway maintained in its Pathways ontology. It documents our increasingly complex state of knowledge about the genetic regulation of programmed cell death—apoptosis.

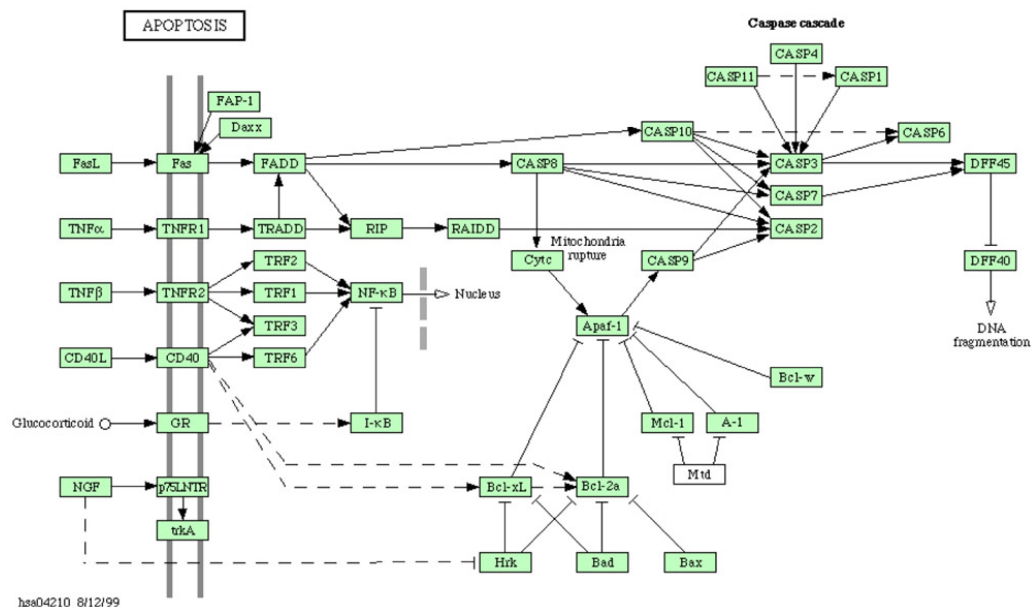


Figure 5.4: The KEGG representation for apoptosis.

5.1.2 Advanced knowledge representation systems for bio-ontology

The AI community has developed a set of tool kits for large and highly portable ontologies over the past two decades. It has not escaped that community's notice that the bio-ontologies developed to date do not exceed the representational capabilities of extant knowledge representation languages. In particular, languages such as Ontolingua [142] and the Ontology Mark-up Language/Conceptual Knowledge Mark-up Language (OML/CKML) [131] have been evaluated for their ability to represent biological knowledge and specifically functional genomic relationships. Ontolingua is a spin-off of the Knowledge Interchange Format (KIF) and is a result of research funded by the Advanced Defense Research Project Agency Knowledge Sharing Effort [137] targeted to support knowledge re-use in ontology building efforts, a target shared by the bioinformatics community. The OML/CKML effort has been led by Washington State University. These two representation languages have similar capabilities as described in a report by McEntire *et al.* [131], although at this time, the latter has a more compatible syntax with other meta-data languages through its use of eXtensible Mark-up Language (XML). The reader is referred to [131] for a description of the issues that had to be addressed when representing existing genomic knowledge bases.

Another knowledge representation system that has been useful in the bio-medical domain is PROTÉGÉ, designed by Mark Musen at Stanford. In practice, none of these more advanced knowledge representations has had nearly the impact and usage of the simple ontologies such as that of GO or Proteome.^[3] Part of the reason is that the need for any kind of functional annotation is so dire that the capabilities of the advanced knowledge representations (*e.g.*, automated inferences) are not a priority and therefore unnecessary overhead when applying them over tens of thousands of genes. It is also the case that even though these advanced representations have significant expressive power, they remain fundamentally limited as described below. We wish to emphasize this point because it is only through the implementation of bio-ontologies with greater expressive power that many of the properties desired for such ontologies can be achieved. The alternative is the creation of a multiplicity of special-purpose, limited expressivity representations that can only be made interoperable with significant effort and without guarantees of semantic or computational soundness.

^[1]<http://www.ncbi.nlm.nih.gov/LocusLink/>.

^[2]<http://www.genome.ad.jp/kegg/kegg2.html>.

^[3]<http://www.proteome.com/>.

5.2 Expressivity versus Computability

Since the 1980s, it has been recognized that there is a natural trade-off between the expressivity of a representation used for an ontology and its computability. In an illuminating paper by Haimowitz *et al.* [82], an evaluation was made of the adequacy of a formal description language—NIKL—to represent medical concepts used in a physiological diagnostic program. The authors of this paper described the requirements of a knowledge representation language to include the following desiderata:

- The system should have precise semantics, either based on the semantics of first order predicate calculus, or having at least comparable precision.
- The knowledge representation should automatically provide logical inferences. This is what sets a knowledge representation apart from a conventional database. It can answer questions that go beyond what was explicitly told to the system.
- It should include some form of taxonomical inheritance of characteristics by more specialized concepts from more general ones. Variations on the inheritance mechanism include whether it is a pure tree (only one parent or generalization for each concept) or more general graph (multiple potential generalizations of each concept). For example, an ontology could represent the fact that a protein can be both part of a signal transduction pathway and a transcriptional factor. Some inheritance mechanisms allow for exceptions and others do not (*e.g.*, although most RNA molecules code for proteins, some do not.).
- The inference mechanism should be sound and complete; *i.e.*, it should derive no false conclusions from true knowledge and guarantee that all true conclusions within the class automatically promised will in fact be made.
- Whatever automatic inference is provided by the representation system itself should be efficient.
- Because of the requirements of soundness, completeness, and efficiency, the expressive power (the kinds of things that can be said with it and inferred in the knowledge representation) should be limited to avoid the representation of undecidable or "NP-hard" concepts.

These desiderata apply to all the existing bio-ontologies and, therefore, the limitations that Haimowitz *et al.* found in their evaluation are likely to hold in the construction of bio-ontologies. Among the troublesome aspects of applying NIKL to the medical domain, Haimowitz *et al.* found the representation to be inadequate to capture spatial relationships so that, *e.g.*, the transitivity of the part-whole or contained relationships was not automatically preserved. Furthermore, causation has a variety of flavors (*e.g.*, a gene may be permissive for a particular process, or a gene product may be necessary but not sufficient for that process, or a gene product might be necessary and sufficient) and these could not easily or accurately be represented in NIKL. Also, the transitivity relationships that would then be expected of a causal representation were not suitably automatically inferred. Similarly, in the temporal representation of the different phases of a process, the dependencies of overlapping phases of a process were not adequately captured. These limitations would certainly create problems for representing large numbers of cellular processes germane to functional genomics.

Subsequently, several special-purpose representations were developed to create ontologies that met the list of desiderata listed above. These specialized knowledge representations pertained to temporal reasoning or spatial reasoning or causality. A few of them had limited capabilities across all of these aspects. None of the current bio-ontologies are represented in languages as rich as these efforts dating back to the 1980s. The implication is that, for our foreseeable future, the expressivity that can be reasonably expected from bio-ontological representation languages is going to be limited and we will not be able to express all the kinds of relationships that a biologist

wishes to assert. Nonetheless, for the reasons noted at the beginning of this section, the utility of having even a relatively inexpressive set of attributes for concepts associated with genes and gene interactions would be immense. The alternative is manual review of the unstructured literature, and that is unlikely to be a scalable solution without further efforts in natural language-based information retrieval techniques. As we have stated before, notwithstanding their limitations, these bio-ontologies are likely to grow in size and in use. Indeed there are several commercial efforts now that have implemented very low ontological commitment ontologies such as that of Proteome in which very simple ontologies have been populated by hundreds of knowledge workers to supply the biotechnology industry with annotations for the genome. Similar efforts are under way by the other gene discovery groups and companies in this area such as Celera, Curagen, and Rosetta.

A comparison of the various ontologies mentioned in this section is shown in table 5.1.

Table 5.1: Overview of representative ontology technologies. "Consumer report" of instances of some of the dominant ontologies used to represent genomic knowledge. Some of the instances are general purpose ontology building tools (*e.g.*, Ontolingua) and others are the sole instance of a particular representational technology (*e.g.* KEGG).

Ontology	Defines a nomenclature for genomics/bioinformatics	Specialization	Expressivity	Interoperability / Integration	Human expertise required
Ontolingua	Multipurpose tool; no nomenclature included or recommended. User community can define the nomenclature of choice.	Support of multiple inheritance allows representation of some of the most complex dependencies in genomics	Broad and richly typed representation with several add-on packages for specialized domains (<i>e.g.</i> temporal and spatial reasoning). Can assert disjointness, negation, etc. This expressivity unsurprisingly is coupled with minimal reasoning support.	Several tools from within the Artificial Intelligence community are interoperable. Less so with standard WWW tools.	Considerable knowledge acquisition required for proper use.
GO	Specifies a controlled vocabulary for GO descriptions.	Limited specialization without formal Inheritance mechanisms.	Limited expressivity (no causal transitivity, temporal ordering, assertion of disjoint subconcepts.)	Tightly integrated into the suite of databases and tools of the NCBI.	Minimal ontological commitment and restricted expressivity enable easy adoption and use.
KEGG					

	Specifies standard nomenclature for each entity (<i>e.g.</i> , EC for enzymes).	Very limited form of specialization without inheritance.	Limited expressivity (no assertion of disjoint subconcepts). However, the extensive use of binary relations allows primitive inference of causal transitivity and representation of pathways.	Intergrated with other pathway representations (<i>e.g.</i> , WIT ^[a]).	Minimal ontological commitment and restricted expressivity enable easy adoption and use.
OML/CKML	Multipurpose tool; no nomenclature included or recommended. User community can define the nomenclature of choice.	Supports multiple inheritance and automated subsumption, <i>i.e.</i> , inference from the logical constraints defined in the term of concept C' logically imply those of the more general concept C.	Missing some of Ontolingua's constructs (<i>e.g.</i> , collections) but relatively high expressivity.	Uses XML as syntax and therefore easily interoperates at least at the syntactic level with several e-commerce/WWW tools. As yet, though, few instances of integration with other tools.	Significant experience required but XML syntax brings it quicker to wider community of developers. However, still a work in progress.
[a] http://www.wit.mcs.anl.gov/WIT/					

5.3 Ontology versus Data Model versus Nomenclature

In the sections below, we describe the nomenclatures and data models that are used to represent genomic data. Often, however, data models and nomenclatures are described as being ontologies. In some very abstract sense, a data model is an ontology with very little ontological commitment (and little in the way of automated inferential mechanisms), and a nomenclature has even less. A *nomenclature* simply refers to a method of naming or identifying concepts. From the ontological perspective, a nomenclature allows the representation by name of the concepts in an ontology (whether implicit or explicit) without describing the relationships between these various concepts.

A *data model* defines a hierarchy of classes, attributes and relationships of those attributes without providing the expressivity based on first-order logic. It also does not have the other inferential capabilities that we have mentioned in the desiderata of section 5.2 and the description of the NIKL effort of Haimowitz *et al.* For instance, a data model describing the components of a microarray (see

section 5.4) might be useful for creating a shared structure for storing the results of a microarray experiment. This data structure does confer the capability to provide even simple inferential relationships such as negation or disjunction. However, a data model would be awkward to state that a gene can code for a transfer RNA or an rRNA or a protein but only two of these three classes. Certainly, elaborate data structures can be developed to capture representational detail, but for every disjunction or exception that the ontology architect wishes to add, the complexity of the data model will grow rapidly. That is, while the data model is necessary to provide a structure in which to maintain the properties of the concepts, it is no substitute for the descriptive power and inferential power of the kind of ontologies previously mentioned. This may appear obvious, but at the time of this writing there are several efforts which claim to be ontologies, but which are in fact no more than glorified nomenclatures and data models.

5.3.1 Exploiting the explicit and implicit ontologies of the biomedical literature

Instead of devising new ontologies specifically for use in functional genomics, others have taken advantage of the existing very low ontological commitment representations used to maintain the biomedical literature. The terminology structures of MEDLINE, the Medical Subject Headings (MeSH), are an outgrowth of the original paper-based *Index Medicus*. As illustrated in figure 5.5 below, MEDLINE's key words are organized into major headings within a few fixed hierarchies or trees (*e.g.*, physical findings, diseases, chemicals). This limited taxonomical structure makes it evident that MeSH has limited expressive capabilities. It has the immense benefit of having been carefully curated by extremely well trained librarians over decades, and therefore it represents an invaluable depository of annotation of biomedical knowledge. It includes approximately 20,000 major headings (and many more synonyms corresponding to the concepts represented by these headings) and 105,000 chemical names. Each article encoded in MEDLINE (and therefore accessible via PUBMED^[4]) has on average 10 MeSH annotations.

MeSH Tree Structures - 2001

1. **⊕ Anatomy [A]**
2. **⊕ Organisms [B]**
3. **⊖ Diseases [C]**
 - Bacterial Infections and Mycoses [C01] +
 - Virus Diseases [C02] +
 - Parasitic Diseases [C03] +
 - Neoplasms [C04] +
 - Musculoskeletal Diseases [C05] +
 - Digestive System Diseases [C06] +
 - Stomatognathic Diseases [C07] +
 - Respiratory Tract Diseases [C08] +
 - Otorhinolaryngologic Diseases [C09] +
 - Nervous System Diseases [C10] +
 - Eye Diseases [C11] +
 - Urologic and Male Genital Diseases [C12] +
 - Female Genital Diseases and Pregnancy Complications [C13] +
 - Cardiovascular Diseases [C14] +
 - Hemic and Lymphatic Diseases [C15] +
 - Neonatal Diseases and Abnormalities [C16] +
 - Skin and Connective Tissue Diseases [C17] +
 - Nutritional and Metabolic Diseases [C18] +
 - Endocrine Diseases [C19] +
 - Immunologic Diseases [C20] +
 - Disorders of Environmental Origin [C21] +
 - Animal Diseases [C22] +
 - Pathological Conditions, Signs and Symptoms [C23] +
4. **⊕ Chemicals and Drugs [D]**
5. **⊕ Analytical, Diagnostic and Therapeutic Techniques and Equipment [E]**
6. **⊕ Psychiatry and Psychology [F]**
7. **⊕ Biological Sciences [G]**
8. **⊕ Physical Sciences [H]**
9. **⊕ Anthropology, Education, Sociology and Social Phenomena [I]**
10. **⊕ Technology and Food and Beverages [J]**
11. **⊕ Humanities [K]**
12. **⊕ Information Science [L]**
13. **⊕ Persons [M]**
14. **⊕ Health Care [N]**
15. **⊕ Geographic Locations [Z]**

Figure 5.5: The MeSH terminology hierarchy. A portion of the MeSH terminology hierarchy starting at the top level and showing the first level of detail at the disease subhierarchy. The entire hierarchy may be browsed at <http://www.nlm.nih.gov/mesh/MBrowser.html>.

The MeSH nomenclature could be taken advantage of in functional genomics. For example, the work of Dan Masys *et al.* in the HAPI [127] project processes microarray data and clusters from

functional genomics analyses. The genes are looked up and the corresponding MeSH terms are found in databases maintained by the NCBI (see the nomenclature section 5.5 below). The MEDLINE structure associated with each publication is then used to categorize the genes found in the clusters, as shown in figure 5.6.^[5] That is, in addition to finding the publications corresponding to each gene, the MeSH hierarchy permits the identification of broader functional and pathological categories. In the instance of the clusters of genes characterizing acute myelogenous leukemia and acute lymphocytic leukemia [78], this demonstrates that the genes predictive of these leukemias are also known to be involved in polycystic kidney disease, inherited immunodeficiencies, and multiple sclerosis. Many of these relationships were only obtainable through the MeSH hierarchy and not directly by searching using only the gene names.

Diseases Associated with ALL-predictive genes

```

Virus Diseases (1) {>.3}
Neoplasms (5) {>.3}
  Neoplasms by Histologic Type (4) {>.6}
    Leukemia (4) {<.001}
      Leukemia, Lymphocytic (3) {<.001}
        Leukemia, B-Cell (1) {<.001}
          Leukemia, B-Cell, Acute (1) {<.001}
        Leukemia, Lymphocytic, Acute (1) {<.001}
          Leukemia, B-Cell, Acute (1) {<.001}
        Leukemia, T-Cell (1) {<.001}
      Precancerous Conditions (1) {<.001}
        Preleukemia (1) {<.001}
Nervous System Diseases (2) {>.3}
  Autoimmune Diseases of the Nervous System (1) {<.001}
    Demyelinating Autoimmune Diseases, CNS (1) {<.001}
      Multiple Sclerosis (1) {<.001}
    Demyelinating Diseases (1) {<.001}
      Demyelinating Autoimmune Diseases, CNS (1) {<.001}
        Multiple Sclerosis (1) {<.001}
Female Genital Diseases and Pregnancy Complications
(1){>.6}
  Genital Diseases, Female (1) {>.6}
    Infertility (1) {<.001}
      Infertility, Female (1) {<.001}
Hemic and Lymphatic Diseases (1) {>.6}
  Hematologic Diseases (1) {>.6}
    Preleukemia (1) {<.001}
Neonatal Diseases and Abnormalities (3) {>.3}
  Hereditary Diseases (2) {>.13}
    Werner Syndrome (1) {<.001}
  Infant, Newborn, Diseases (1) {<.001}
    Severe Combined Immunodeficiency (1) {<.001}
Immunologic Diseases (4) {<.01}
  Autoimmune Diseases (1) {>.13}
    Autoimmune Diseases of the Nervous System (1){<.001}
      Demyelinating Autoimmune Diseases, CNS (1){<.001}
        Multiple Sclerosis (1) {<.001}
    Immunologic Deficiency Syndromes (3) {<.001}
      Common Variable Immunodeficiency (1) {<.001}
      Severe Combined Immunodeficiency (1) {<.001}
  Pathological Conditions, Signs and Symptoms (1) {>.3}
    Pathologic Processes (1) {>.13}
      Disease Attributes (1) {<.001}
        Acute Disease (1) {<.001}

```

Figure 5.6: Transforming a cluster of genes into an annotated structure automatically using MEDLINE. Summary of concept hierarchy matches with the MeSH hierarchy terms for genes described by Golub *et al.* [78] that fell into the acute lymphoblastic leukemia (ALL) cluster. Alternatively, the textual content of the biomedical literature itself has an implicit (if muddy) ontological structure which can be exploited. That this can be done without solving the natural language challenge is illustrated by the work of Altman and Raychaudhuri [6]. These authors represented each publication as a word vector of the titles and abstracts, with each vector encoding

the frequency of those words for those articles. Using these vectors they were able to apply the same clustering techniques^[6] used to find the proximity of gene expression patterns to find how the publications clustered. It may be that in the future, combinations of simple ontologies and a statistical analysis of the content of publications will allow efficient mining of the immense biomedical literature for genomicists seeking to obtain the meaning of the gene or protein patterns they have observed.

^[4]<http://www.ncbi.nlm.nih.gov/>.

^[5]A similar functionality is also provided by the PubGene program at <http://www.pubgene.org/>.

^[6]With the word frequency used in the same way as a gene expression value.

5.4 Data Model Introduction

Data models are the logical structures used to represent a collection of entities and their underlying one-to-one, one-to-many, and many-to-many relationships. The main motivations for creating data models is usually to be able to implement them within a database management system, usually as a relational database management system (e.g., Oracle, SQL Server, Sybase, MySQL, etc.). As the relational calculus [48] is based on the first-order predicate calculus, it is possible, although unrewarding, to express a wide range of formal relationships within the data model. The reasons for the unwieldy nature of such an effort are articulated in the prior section. Presently, in the domain of bioinformatics and functional genomics, the most relevant heuristic distinction between bio-ontologies and data models is that bio-ontologies are usually developed in order to provide a shareable framework to describe attributes of genes, whereas data models at this time are mostly developed to store the measurements of genomic systems. For DNA sequences, this corresponds to the GenBank data model and for the domain of microarrays, it corresponds to a way of organizing the pertinent values obtained on measurement of a microarray hybridization experiment.

Among the goals often stated for the development of standardized microarray data models is the capability to allow researchers across multiple laboratories to be able to take advantage of the precious RNA samples obtained from various sources and hybridized to microarrays in several, initially unrelated, experiments. Ideally, these researchers should be able to perform aggregate analyses across all these various experiments independent of the provenance or manufacture of each of the microarray systems. In this way, we may create a shared set of repositories of expression data that might reveal subtle gene interactions across a larger sample than would be observable within a single institution's data set.

Unfortunately, expression measurements are human artifacts rather than ground "truths" in biology.^[7] A consequence of this fact is that there are many such artifacts. There are several microarray measurement systems in use, and many more under development, such that convergence upon a particular data model involves the arduous task of obtaining a sufficiently general representation of all the measurement types across different microarray systems, while at the same time providing sufficient specificity for each measurement system so as to avoid loss of detail of a particular study. Unfortunately, a large part of this specificity reflects differences in the measurement techniques of a particular microarray platform that result in nonlinear relationships with the results from other platforms. The implication of this is that even if the data models are shared across microarray platforms, the results encoded in these shared data models may not be directly comparable.

Let us be a little bit more explicit regarding the challenge of a standardized data model for microarray experiments results. Unlike DNA sequences, which are by nature digital and whose interpretation should be independent of the modality by which the sequence was acquired, expression data are analog signals and their value and interpretation are very much dependent on the technology used to obtain that value. Consider just two of the technologies used to measure

expression: photolithographically constructed oligonucleotide microarrays with several probes per gene hybridized against a single color-stained sample, and robotically spotted microarrays with one probe per gene hybridized against two samples each colored with a different dye. In the first case, the expression level is reported as a function of a trimmed mean of intensities, each probing for a particular part of a gene, and the "average difference" between intensities of oligonucleotides that perfectly match against that gene part, and oligonucleotides that have a central base mismatch with that same part. In the second case, the gene expression level is measured as a ratio of hybridization of one specific cDNA from one sample against the cDNA from another sample. Although there is some relationship between the two measurements, it is not clear or empirically determined what that relationship is. For example, two-dye microarray measurements are directly related to the second or control sample used, whereas single-dye oligonucleotide microarray measurements are in theory an absolute expression level.

As a result, there are at least as many microarray expression data models available as there are microarray expression technologies and, in fact, considerably more. At the time of this writing, there has yet to be any consolidation of these different competing standards into a single one, although this is likely to happen in the near future. However, in the absence of such a consensus, we point out the most promising and successful of these efforts. Not all these efforts in data modeling are sufficiently general at the time of this writing to encompass all the current microarray gene profiling technologies. We will note this as we describe the current major candidates for standardized data models.

Gene Expression Omnibus (GEO) The NCBI has been developing the GEO data model to support the construction of a gene expression repository with the same inspiration and goals that were originally targeted for the GenBank repository for DNA sequence. The goal of GEO is to support spotted microarray technology, oligonucleotide microarray technology, hybridization filters, and Serial Analysis of Gene Expression (SAGE) data. The entities used in GEO are shown in figure 5.7 and can be accessed on-line.^[8] Examination of this figure reveals that, currently, Affymetrix style files are poorly supported, whereas the two-dye spotted microarray data type is better supported.

Sample fields

- **Experiment type:** Single channel (absolute value) such as radioactive labeled filters, or dual channel types such as a red/green labeled microarray.
- **Sample title:** Try to choose a short title which will be specific to your sample -- preferably greater than 8 and less than 20 characters. The title must be unique over all your previously accessioned GEO samples. You will be notified if such a name clash occurs when you try to submit, and then will be given a chance to change your sample title. Text is required in this field.
- **Lot/batch:** Any manufacturer defined lot and/or batch numbers for the platform used to derive this sample's data.
- **mRNA source:** It is best to use a short list of words to describe the source of the mRNA used to derive this sample's data. Do not list organisms in this field. Text is required in this field.
- **Ch1 mRNA source:** It is best to use a short list of words to describe the source of the mRNA used to derive this sample's data. Do not list organisms in this field. Text is required in this field.
- **Ch2 mRNA source:** It is best to use a short list of words to describe the source of the mRNA used to derive this sample's data. Do not list organisms in this field. Text is required in this field.
- **PubMed id:** PubMed id (PMID) references a publication which, perhaps, further describes your sample. A PubMed id is a numeric value which you may obtain from the PubMed record.
- **Web link:** World Wide Web link to reference a webpage which, perhaps, further describes your sample.
- **Keywords:** One or more terms, as a comma-delimited list, which you think might be useful terms in queries.
- **Image file:** Use the Browse button to the right to select a local file to be uploaded. The purpose of this image is to serve as a qualitative reference of your experiment. It is not meant to be image-analyzable. In order to be accepted into GEO, the image size must be less than 100 kilobytes, and must be in JPEG or GIF format.
- **Total # tags:** A whole, non-zero number is required. The reciprocal of this number is used for SAGE library normalization.
- **Norm. factor:** This number is meant to be multiplied to all of the background-corrected scalar values extracted from each platform feature referred to in the data table. A number is required.
- **Protocol:** Select the anchoring enzyme used in your SAGE protocol, e.g., NlaIII or Sau3A. Protocol selection in SAGE libraries replaces platform selection. A selection is required.
- **Platform used:** Submission of non-SAGE sample data requires explicit reference to a valid GEO platform accession number. Therefore, platform information must be submitted before sample data derived from that platform. A selection is required.
- **Data file:** Use the Browse button to the right to select a local file to be uploaded. This file should meet the validation criteria set for the platform type relevant to your sample. A content bearing, valid data file is required.

Figure 5.7: A subset of the definition of *sample* in the GEO. At the time of this writing, the sample definition in the GEO does not appear to support the full complexity of oligonucleotide high-density microarrays.

The Gene Expression Mark–Up Language (GEML) GEML syntax is based on XML. GEML was originally developed by Rosetta Inpharmatics, and subsequently, several other public and private institutions have become involved, such as Agilent Technologies and Nature America, Inc. Details on GEML can be found at <http://www.geml.org/>.

GEML starts with an XML format, as this is the most widely used meta–data syntax developed. XML is maintained by the World Wide Web consortium to define general–purpose mark–up languages, and is now used widely as the interchange format for a variety of electronic commerce functions and scientific interchange efforts. There is a multiplicity of tools for generating, parsing, and displaying XML content and consequently GEML leverages this existing effort. The goal of GEML is illustrated in figure 5.8 (taken from their website). At the top level are the various types of microarray measurement systems and in the middle is the GEML *lingua franca* into which all the values measurement systems are translated. Below the GEML layer are shown several applications that work with microarray data as abstracted into a standardized format in GEML. This architecture is similar to many other three–tiered architectures where there is a common data model abstracting away the heterogeneity and complexity of the various databases that it accesses. In theory, a programmer for microarray analysis software would only have to understand the GEML Document Type Definition (DTD) to write applications that worked with the results of measurements on all microarray platforms.

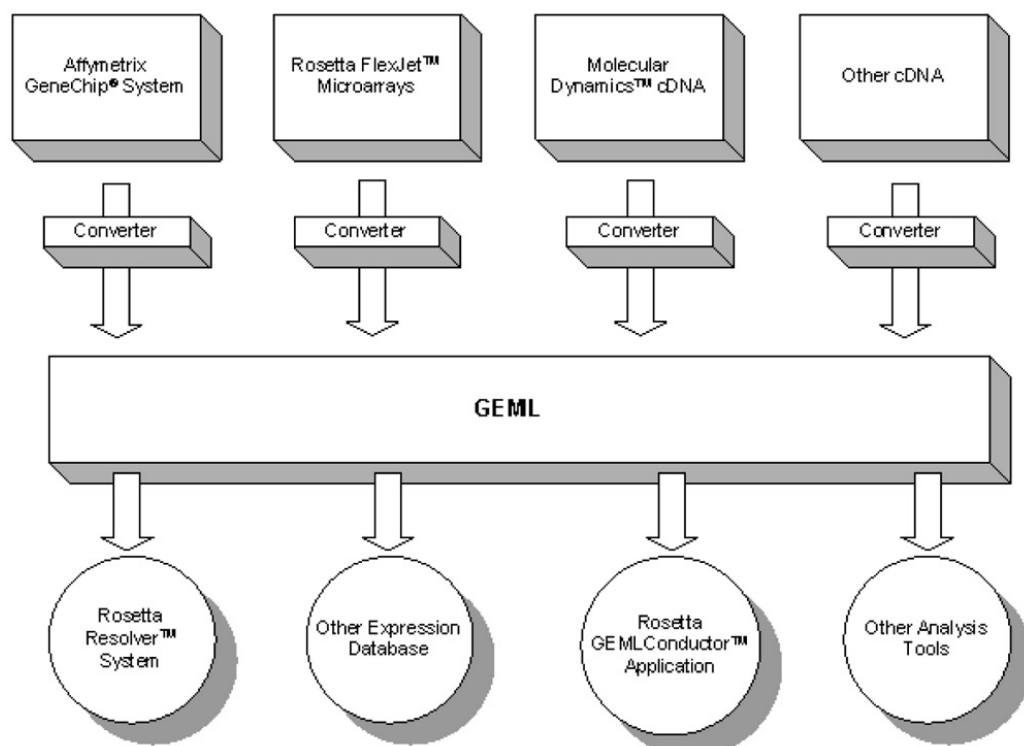


Figure 5.8: The abstraction layers of the Gene Expression Mark–up Language (GEML).

The GEML specification takes advantage of the DTD file. A DTD is the actual specification of the formal data model that any XML message that hews to that DTD must follow. DTD specifications are the World Wide Web consortium's prescribed mechanism for defining application–specific data models, and many industries (*e.g.*, finance and imaging) are in the process of defining a consensus DTD for their own applications.^[9] GEML specifies two DTDs: a pattern DTD, which describes the genes reported on a chip layout, and the profile DTD, which contains expression data, treatment, and hybridization information. That is, the latter DTD describes the experimental conditions and the resultant values. figure 5.9 shows the pattern DTD that has been quoted verbatim from the GEML site. Examination of this DTD reveals that the encoding of the experimental results of single or two–dye stained samples hybridized against cDNA microarrays would be fairly straightforward,

whereas encoding the 20 probe values that constitute a single probe set on an oligonucleotide microarray might be challenging and require some contortions in the way the data is squeezed into those existing attribute lists.

```

<!-- GEMPLPattern.dtd -->

<!ELEMENT project (pattern*,
                   printing*,
                   other*)>
<!--Project = group of patterns
                   and/or printings-->
<!ATTLIST project
    name CDATA #IMPLIED
    id CDATA #IMPLIED
    date CDATA #IMPLIED
    by CDATA #IMPLIED
    company CDATA #IMPLIED >
<!ELEMENT pattern (reporter+,
                   other*)>
<!--Pattern = collection of
                   one or more features-->
<!ATTLIST pattern
    name CDATA #IMPLIED
    type_id CDATA #IMPLIED
    species_database CDATA #IMPLIED
    description CDATA #IMPLIED
    access CDATA #IMPLIED
    owner CDATA #IMPLIED>
<!ELEMENT reporter (feature+,
                    gene?,
                    other*)>
<!ATTLIST reporter
    name CDATA #REQUIRED
    systematic_name CDATA #REQUIRED
    accession CDATA #IMPLIED
    deletion CDATA "false"
    control_type CDATA "false"
    fail_type CDATA #IMPLIED
    active_sequence CDATA #IMPLIED
    linker_sequence CDATA #IMPLIED
    primer1_sequence CDATA #IMPLIED
    primer2_sequence CDATA #IMPLIED
    start_coord CDATA #IMPLIED
    mismatch_count CDATA #IMPLIED>
<!--Reporter = measures expression
                   of a gene-->
<!ELEMENT feature (position,
                  pen?,
                  other*)>
<!--Feature = location of a
                   reporter for a gene-->
<!ATTLIST feature
    number CDATA #IMPLIED
    ctrl_for_feat_num CDATA #IMPLIED>

<!ELEMENT gene (accession*,
                alias*,
                other*)>
<!--Gene = what the reporter
                   is reporting on-->
<!ATTLIST gene
    primary_name CDATA #REQUIRED
    systematic_name CDATA #REQUIRED
    species CDATA #IMPLIED
    chromosome CDATA #IMPLIED
    map_position CDATA #IMPLIED
    description CDATA #IMPLIED>
<!ELEMENT accession (other*)>
<!ATTLIST accession
    database CDATA #REQUIRED
    id CDATA #REQUIRED>
<!ELEMENT alias (other*)>
<!ATTLIST alias
    name CDATA #REQUIRED>
<!ELEMENT position (other*)>
<!ATTLIST position
    x CDATA #REQUIRED
    y CDATA #REQUIRED
    units CDATA #REQUIRED>
<!ELEMENT pen (other*)>
<!ATTLIST pen
    x CDATA #REQUIRED
    y CDATA #REQUIRED
    units CDATA #REQUIRED>
<!ELEMENT printing (chip+,
                    other*)>
<!ATTLIST printing
    date CDATA #IMPLIED
    printer CDATA #IMPLIED
    type CDATA #IMPLIED
    pattern_name CDATA #IMPLIED
    run_description CDATA #IMPLIED>
<!ELEMENT chip (other*)>
<!ATTLIST chip
    barcode CDATA #REQUIRED>
<!ELEMENT other (other*)>
<!ATTLIST other
    name CDATA #REQUIRED
    value CDATA #REQUIRED>

```

Figure 5.9: The GEML pattern document type description.

Microarray Gene Expression Database Group (MGED) MGED was developed in recognition of the need for standardization of microarray data models. It is an open discussion group that originally held its meetings in the United Kingdom, and subsequently has held two international meetings.^[10] The most notable output of the MGED group, which is constituted of multiple private and public institutions, is the Microarray Mark-up Language (MAML). The MAML DTD is shown in figure 5.10. The MAML DTD is sufficiently comprehensive that even at the time of this writing, it appears to cover all the currently available microarray technologies without requiring each site implementing the MAML database to put their own data through too many contortions in order to adhere to the data model. In particular, the abstract way in which a composite element has been defined in this DTD allows for the representation of a wide range of physicochemical means for assessing gene expression.

<pre> <!ELEMENT creation_info (contact, software*, hardware*) > <!--ATTLIST creation_info date DATE #REQUIRED--> </pre>	<pre> <!ELEMENT array_platform (array_def reference) > <!--ATTLIST array_platform id ID #REQUIRED--> </pre>
<pre> <!ELEMENT contact (parameter*) > <!--ATTLIST contact last_name CDATA #REQUIRED first_name CDATA #IMPLIED middle_name CDATA #IMPLIED lab CDATA #IMPLIED department CDATA #IMPLIED institution CDATA #REQUIRED street CDATA #IMPLIED city CDATA #IMPLIED province_state CDATA #IMPLIED country CDATA #REQUIRED postal_zip_code CDATA #IMPLIED phone CDATA #REQUIRED fax CDATA #IMPLIED email CDATA #REQUIRED href CDATA #IMPLIED --> </pre>	<pre> <!ELEMENT array_def (creation, description?, comment*, reference*, treatment*, parameter*, seq_feature*) > <!--ATTLIST array_def name CDATA #REQUIRED surface_type (non-absorptive absorptive other)#REQUIRED other_surface_type CDATA #IMPLIED reporter_type (single-multimer multiple-oligomer other)#REQUIRED other_reporter_type CDATA #IMPLIED model CDATA #IMPLIED version CDATA #IMPLIED href CDATA #IMPLIED --> </pre>
<pre> <!ELEMENT hardware (contact?, description?, parameter*) > <!--ATTLIST hardware make CDATA #REQUIRED model CDATA #REQUIRED year CDATA #IMPLIED href CDATA #IMPLIED --> </pre>	<pre> <!--ELEMENT seq_feature (bio_seq?, ref_bio_seq?, ref_clone?, gene?, reference?, treatment*, parameter*, coordinate?, description?, comment*) > <!--ATTLIST seq_feature id ID #REQUIRED name CDATA #IMPLIED --> </pre>
<pre> <!ELEMENT software (contact?, hardware*, description?, parameter*) > <!--ATTLIST software id IDREF #REQUIRED name CDATA #REQUIRED version CDATA #IMPLIED year CDATA #IMPLIED operating_system CDATA #REQUIRED href CDATA #IMPLIED --> </pre>	<pre> <!--ELEMENT bio_seq (reference CDATA) > <!--ELEMENT ref_bio_seq (reference CDATA) > <!--ELEMENT gene (reference CDATA) > <!--ELEMENT ref_clone (reference CDATA) > <!--ELEMENT coordinate EMPTY > <!--ATTLIST coordinate horizontal CDATA #IMPLIED vertical CDATA #IMPLIED --> </pre>

Figure 5.10: The MAML pattern document type description. Shown are three fragments of the MAML DTD. The first column demonstrates the generic description of a data set creator. The second column demonstrates the capability of maml to support composite expression measurements and analyses of the sort found on the Affymetrix platform.

Attempting a consensus model Obviously, if there is such a wide variety of standard microarray models, there is no standard. Without a single standard, combining expression data across microarray experiments is arduous, if not impossible. Fortunately, the Object Management Group (OMG) appears to be creating a common ground where proponents of these various data models will be able to begin to reconcile their various proposals and create a consensus data model. Although this consensus data model has not yet been finalized, the OMG has already incorporated some of the proposals that we listed previously. As the OMG has successfully created agreed-upon standardized data models in many different application domains outside biology, there is some reason to hope that they will be similarly successful in this domain. For the latest information on the OMG efforts regarding a standardized microarray data model, we will refer the reader to the following website: <http://www.geml.org/omg.htm>.

In addition to the OMG efforts, there is the Interoperable Informatics Infrastructure (I3C), a group that is tasked to produce technical recommendations for the interchange of data with the goal of

creating practicable solutions.^[11] The membership of I3C is constituted of 60 companies and organizations committed to addressing the balkanization of data modeling efforts illustrated by table 5.2.

Table 5.2: List of freely available data models and databases for microarrays.

Data model or database	URL
Affymetrix Analysis Data Model (previously Genetic Analysis Technology Consortium)	http://www.affymetrix.com/support/aadm/aadm.html
Another Microarray Database	http://www.microarrays.org/software.html
ArrayDB	http://www.genome.nhgri.nih.gov/arraydb
ArrayExpress	http://www.ebi.ac.uk/arrayexpress/Design/design.html
ExpressDB	http://www.twod.med.harvard.edu/ExpressDB
GeneX	http://www.ncgr.org/research/genex
MicroArray Database	http://www.pompous.swmed.edu/exptbio/microarrays/mad
Stanford Microarray Database	http://www.genome-www4.stanford.edu/MicroArray/SMD

Again, it is interesting to note how much more difficult it has been to arrive at a consensus model for microarray expression measurements than it took to create a standard data model for genetic sequences in GenBank. The reason, as stated before, is that gene expression measurement is an analog measurement rather than the digital measurement of DNA sequence and therefore the measurements can only be understood with respect to a particular measurement system, a human artifact. The goal of the standardized microarray data model can only be realized once a common set of abstractions can be reliably found across all these different measurement technologies and artifacts.

^[7]That is, this compares with the case of DNA sequence, where the way to represent the ordering of a particular sequence of nucleotides is, at least in theory, independent of the measurement system.

^[8]<http://www.ncbi.nlm.nih.gov/geo/info/fields.cgi>.

^[9]In the latest iteration of specifications for XML document specifications, the XML schema has superseded the DTD, although there is sufficient overlap to allow DTDs to be easily converted into XML schemas.

^[10]<http://www.mged.org/>.

^[11]<http://www.i3c.org/>

5.5 Nomenclature

The issuing of consistent names to genes is an apparently mundane chore. Unfortunately, at this time, its importance may exceed that of shared data models and ontologies for the purposes of the immediate progress in our functional genomics efforts. Why might this be?

First, we want to make sure that we compare the behavior of the same genes and their properties across all experiments. When a microarray manufacturer reports that the n th element of its array interrogates the expression of a particular gene, the following questions should be asked: Which

particular polymorphism (s) of that gene, which alternative splicing product(s), and which other gene(s) are being reported on?

Consequently, the nomenclature (*i.e.*, naming scheme) used to identify gene expression measurements on microarrays has to be able to capture the increasingly polymorphic definition of genes. As we learn more about single nucleotide polymorphisms (SNPs), alternate splicing products, pseudogenes, frame-shift mutations, and other variations in just the primary structure of the genes and the proteins they produce, a nomenclature has to be able to distinguish between these various forms of the same gene and yet group them as belonging to the same coding gene. This goes beyond the UniGene^[12] effort which attempts to cluster known experimentally derived sequences into a cluster with a single consensus sequence, ideally corresponding to a single particular gene.

The task of finding a standardized and universal nomenclature for each gene is just the tip of the iceberg of a recurrent challenge in microarray technology: the task of linking a spot on a microarray to a specific gene or expressed sequence tag (EST). This is challenging in both mundane technical ways and in deep fundamental ways. The mundane ways in which it is challenging is that if one does not have the exact sequence for each microarray spot, at the very least, one has to obtain the correct sequence that each spot on the microarray was engineered to represent. For instance, on a spotted cDNA microarray, one has to determine the gene sequence of the clone used for each spot. With an oligonucleotide microarray, one has to determine from the manufacturer what sequence each probe set was designed to correspond to. In cDNA microarrays, these probes are typically linked to an IMAGE (see the description of the IMAGE consortium on p. 245) sequence ID. These are typically issued with the cDNA microarrays or with the probes that are spotted onto the microarray. Unfortunately, there has been a high error rate in many of the cDNA clones obtained from various manufacturers and so although the name mapping may be relatively straightforward, it may not actually represent what is being hybridized. Consequently, researchers may have to resequence a clone if they want to verify that the probe is assaying for the presence of the gene they think it is. The challenge is somewhat different with proprietary oligonucleotide microarray manufacturers, where the manufacturer's catalog states what gene each probe set has been engineered to hybridize to but the actual oligonucleotide sequences themselves are unknown.^[13] This is problematic for several reasons.

- The gene sequence that the probe set may be engineered for may be a fusion gene, *i.e.*, an abnormal composite of two different genes. Consequently, since we do not know which part of the fusion gene is actually being represented by the probe set, it is unclear whether that probe set measures just the presence of the fusion gene transcript or whether it can also measure the presence of either one of the constituent genes separately.
- Furthermore, at the time of this writing, some oligonucleotide probe sets are designed against large sections of the genome such as a cosmid (see glossary, p. 277). Knowing the genomic sequence does not provide very much information about which gene in particular is being expressed from that region of the genome. If the specific oligonucleotides were made public, then one could perform a BLAST search to find which specific gene in that region the oligonucleotides corresponded to.
- As the genome efforts in the human, mouse, and other species increase in accuracy and completeness, the mapping between the microarray oligonucleotides and the genes they correspond to is likely to improve. Specifically, as we gain more knowledge of polymorphic variants of genes and their alternate splicing products, the investigator will be able to verify which particular variant a probe set is measuring. However, in the absence of such information, researchers cannot take advantage of increased knowledge of the genomic sequences to provide a more accurate labeling of the intensities that they are measuring on the microarrays.

- A related problem occurs with oligonucleotide microarrays where sequences that were thought to be part of one gene at one time are found to be components of other genes as the UniGene (see p. 244) clustering become more refined. Practically, this means that in one generation of Affymetrix microarrays, a probe set is said to correspond to one gene, whereas in another generation it may correspond to two or more genes based on the new assignments of the particular GenBank subsequences to different genes. In the absence of knowledge of the exact oligonucleotide sequences, or in the absence of a version history of Affymetrix accession numbers, the analysis of oligonucleotide microarray data, particularly across generations of microarrays, becomes problematic.
- Occasionally, manufacturers of oligonucleotide sequences just make mistakes in their design process and engineer oligonucleotides that do not interrogate for the expression levels of the specified genes. The most recent and notorious example of such an error was in the design of Affymetrix oligonucleotide microarrays for murine systems that was unreported for months. If these sequences had been made public, then the error would likely have been discovered much earlier.

Pragmatically then, in the year 2002, some relatively laborious cross-indexing of terms across nomenclatures needs to be done in order to first determine which gene(s) a particular spot on a microarray was engineered to assay, and secondarily, to compare results across different microarrays. In practice, this means looking up in a table provided by each microarray manufacturer each accession number corresponding to each probe or probe set on a microarray, whether it be a tentative consensus (TC) number from the Institute for Genomic Research (TIGR) organization, a GenBank accession number, a UniGene accession number, an IMAGE clone ID, or some other accessioning scheme, finding the representative genetic sequence corresponding to that probe or probe set, and then running a Basic Local Alignment Search Tool (BLAST) search against one of the standardized nomenclatures such as LocusLink (see p. 243), if it is a gene with a known function, or UniGene, if a more comprehensive search is desired. Because this lookup function has to be done hundreds of times per analysis due to the large number of genes present, it requires one to write a script that submits the appropriate sequences to a BLAST program. The submission can be over the Internet using the batch version of BLAST at the NCBI. For better performance, if one has sufficient storage space locally, the entire GenBank repository and BLAST programs can be downloaded and run locally, using instructions at the NCBI. It is prudent to re-run these scripts on a regular basis as the UniGene and LocusLink databases are frequently updated.

An alternative approach is to develop the infrastructure to track each microarray probe or probe set and automatically determine the latest meanings for each. We have developed such a system that automatically parses the latest tables converting GenBank to UniGene, and UniGene to LocusLink, as well as the content files for these databases. Using these tables, one can then connect each of these references to the original publicly available accession numbers provided by the microarray manufacturers. The advantages of this approach are that:

- One can immediately find official names and symbols for probes and probe sets, as well as known synonyms.
- One can take the opposite route and search for probes and probe sets by gene meaning or other characteristics. For instance, if one has prior information that a gene on human chromosome 4 is associated with a particular phenotype, one can optimize functional genomics analysis by restricting to only those genes known to be in the appropriate location. One could also restrict analysis by protein domain, such as restricting to only those genes containing a zinc finger or DNA binding domain.
- One can find and search by functional categorization, such as a term from GO or a disease from Online Mendelian Inheritance in Man (OMIM).
- Because UniGene is essentially formed by applying BLAST to GenBank entries, it spares

- one from performing those BLAST operations.
- The tables can be updated regularly automatically.

We maintain a website at <http://www.unchip.org/> which uses these tables and allows the translation of a microarray-specific accession number (currently mostly Affymetrix) to one of the more globally used nomenclatures described below. A similar service is available for Affymetrix customers^[14]. A commercial product offering similar, but more broadly applicable functionality is called GeneSpider and may be available soon from Silicon Genetics.

5.5.1 The unique gene identifier

Given the aforementioned desiderata that motivate the need for a standard nomenclature for genes, the quest for such a standardized nomenclature has generated widespread international efforts. There are several candidate nomenclatures, the most popular and useful of which we summarize here. Optimally, these nomenclatures should capture polymorphisms, splice variants, and mutations of the genes as they appear in one species and in their orthologs in all other sequenced genomes of other organisms. We list some of the more popular nomenclatures used for microarray experiments. In practice, microarray repositories will use several nomenclatures to reference the measurements made in gene expression experiments. For a more comprehensive and current compendium of nomenclatures such as these, the reader is referred to issue no. 1 of *Nucleic Acids Research*, of any recent year.

LocusLink (<http://www.ncbi.nlm.nih.gov/locuslink/>) Perhaps the most extensively curated and possibly the broadest nomenclature is maintained through the LocusLink database [145]. LocusLink identifiers are stable, in that they are designed not to change over time, and are genome independent. There are other databases which have a much larger set of expressed sequence tags (see glossary, p. 277) or putative genes, but for which either the complete coding region is not known, or for which the function of the gene product is unknown. The curated nature of LocusLink is important because, unlike GenBank records, which are under the full editorial control of only those who submitted the sequence, the sequences and annotations in LocusLink are under an extended and distributed editorial process.

Broadly speaking, the LocusLink record is comprised of three components. The first is a stable unique identifier called the locus ID. The second component is the reference sequence which is obtained from the REFSEQ^[15] database, which in turn is seeded by a GenBank source coding sequence. This reference sequence is left as provisional until the full length is obtained according to the editorial process. The third component includes all the annotations such as the completeness indicator for the gene (indicating that the entire coding component of the gene is complete), accession numbers providing links to a protein record, links to the OMIM,^[16] links to other databases such as UniGene and DBSNP, and an extremely useful English language summary that describes the gene. Of note, LocusLink is involved in an active collaboration with the Human Gene Nomenclature Committee (HGNC)^[17] to provide an overall unifying international standard. Most recently, LocusLink also maintains links to functional annotation databases such as those supplied by Proteome^[18] and publicly maintained ontologies such as GO.

It appears likely that LocusLink will continue to be the locus for much annotation of organismal genomes and the provider of stable identifiers for their genes.

UniGene (<http://www.ncbi.nlm.nih.gov/UniGene/>) Although LocusLink provides the most stable identifier for known genes, or at least those for those which at least one function is at least presumed, at this early stage of the discipline of functional genomics, the majority of genes do not have any known function.^[19] For this reason, other related databases are useful. Chief among these

may be the UniGene system at the NCBI [28]. UniGene attempts to be comprehensive; to catalog all the characterized genes as well as hundreds of thousands of uncharacterized and novel expressed sequence tags, while attempting to reduce the redundancy inherent in those individual catalogs. The content of each UniGene record includes a set of sequences with GenBank identifiers that have been clustered such that they are presumed to correspond to a single specific gene. As this clustering is done mostly automatically, and the clustering procedures have evolved over time and been augmented with more data, the UniGene clusters are not guaranteed to be stable, unlike LocusLink identifiers. The strength of UniGene is that it attempts to be comprehensive and includes not only the GenBank accession numbers for sequences in each cluster but also known alternate splicing variants. Currently, UniGene describes human, rat, cow, mouse, and zebra fish organisms. Each UniGene record has a unique ID called a cluster ID. Because these cluster identifiers are unstable, a cluster ID can be retired for a variety of reasons.

- The sequences that congregate to a cluster might be found to be contaminated.
- Two or more clusters may be joined to form a single cluster in which case the original cluster IDs are retired.
- A cluster may be found to be composed of more than one gene and therefore must be split into two or more clusters. Thus the original cluster ID has to be retired and new ones generated for the subclusters.

Because of this terminological instability most reports using the UniGene nomenclature will also report the "build" (*i.e.*, version) number from which the cluster IDs were drawn.

GenBank (<http://www.ncbi.nlm.nih.gov/>) An overview such as this would be incomplete without mentioning GenBank, the public database of all known nucleotide and protein sequences from which many of the other nomenclatures are derived [22]. At the time of this writing, GenBank contains over 12 million annotated sequences and is growing at the rate of over 5 million sequences per year. GenBank sequences are divided into divisions, roughly corresponding to taxonomy (*e.g.*, bacteria, viruses, *etc.*) as well as for specific projects, such as ESTs, genome sequencing, sequence-tagged sites, and others.

Each GenBank record contains taxonomic information regarding the organism from which the sequence was obtained, bibliographical references, and biological features found within the sequence. Each record has an accession number that is stable and unique; each entry is also assigned a unique unstable GI number so that revisions may be tracked. Beyond this, very little additional annotation is provided, given the low-commitment nature of this database.

GenBank is a member of the International Nucleotide Sequence Database Collaboration, including the DNA Data Bank of Japan and the European Molecular Biology Laboratory, which defines common ontologies for taxonomy and features allowing daily movement and translation of data between these databases.

IMAGE (<http://www.image.llnl.gov/>) The Integrated Molecular Analysis of Genomes and their Expression (IMAGE) consortium was formed to create, collect, and characterize cDNA libraries from a variety of different tissues and was initially spearheaded by the efforts by the Washington University Genome Sequencing Center and Merck & Co. [118]. Several microarray databases refer directly to the IMAGE clone identifiers rather than to any one of the more precisely characterized sequences described above. Those IMAGE cDNA clones that have been sequenced are available in the dbEST database maintained at the NCBI.^[20] Perhaps the best feature of the IMAGE clone identifiers is that they refer to physical instances of these clones on the master plates from which the clones are generated. In that sense, the clone identifiers are stable. However, considering that some of the clones obtained from these master plates have been known to suffer contamination or

have not been sequence-validated, they are not stable.

The Institute for Genomic Research (TIGR, <http://www.tigr.org/>) TIGR also maintains a fairly extensive list of human and other organism nomenclatures. TIGR has its own consensus curation and assembly process that it uses to maintain a number of databases for organisms, including human, mouse, rat, zebra fish, rice, Arabidopsis, soy, tomato, maize, potato, cattle, and many others. The nomenclature specific to TIGR is the tentative consensus sequence, or TC. TC clusters are created by a process similar to UniGene by assembling ESTs into clusters based on sequence. However, alternate splice forms are built into separate TC clusters, as opposed to UniGene, where these are kept in the same cluster. Many manufacturers of microarrays refer to TIGR accession numbers or TC identifiers in their descriptions of the probe sets or cDNA probes on their microarrays. TIGR itself uses TC identifiers to create a useful set of mappings of orthologs across all the species indexed in the TIGR database.

Enzyme nomenclature database The Nomenclature Committee of the International Union of Biochemistry and Molecular Biology maintains a catalog of known enzymes and their functions. Enzyme nomenclature is based on the reactions that are catalyzed, and not the genes that make up the enzymes, or the protein structures of those enzymes. Thus, simple one-to-one translations between Enzyme Commission (EC) numbers and GenBank or LocusLink accession numbers are neither possible nor accurate. For example, the enzyme described by EC 1.1.1.27 (lactate dehydrogenase, LDH) assists in the conversion of lactic acid to pyruvate. However, that enzyme is made of four independently chosen subunits of two types, either LDHA (LocusLink 3939) or LDHB (LocusLink 3945). Note that these genes have different LocusLink entries in other species, whereas the enzyme function keeps the same EC accession number.

This nomenclature can be searched using the Expert Protein Analysis System maintained by the Swiss Institute of Bioinformatics^[21].

Other identifiers Other identifiers used on microarrays include the Expressed Gene Anatomy Database identifiers^[22], and the Munich Information Center for Protein Sequences Yeast Genome Database^[23]. In addition, other identifiers of which a functional genomicist should be aware include Ensemble^[24], the Genome Database^[25], the Mouse Genome Database^[26], the Saccharomyces Genome Database^[27], and the Database of Transcribed Sequences^[28].

^[12]<http://www.ncbi.nlm.nih.gov/UniGene/>.

^[13]That is, they are kept as confidential proprietary information.

^[14]<http://www.netaffx.com/>

^[15]<http://www.ncbi.nlm.nih.gov/LocusLink/refseq.html>.

^[16]<http://www.ncbi.nlm.nih.gov/Omim>.

^[17]<http://www.gene.ucl.ac.uk/nomenclature>.

^[18]<http://www.proteome.com/>.

^[19]And a larger majority only have a few of their functions known.

^[20]<http://www.ncbi.nlm.nih.gov/dbEST>.

^[21]<http://www.ca.expasy.org/enzyme/>

^[22]<http://www.tigr.org/tdb/egad/egad.shtml>

^[23]<http://www.mips.biochem.mpg.de/>

^[24]<http://www.ensembl.org/>

^[25]<http://www.gdb.org/>

^[26]<http://www.informatics.jax.org/mgihome>

^[27]<http://www.genome-www.stanford.edu/Saccharomyces>

^[28]<http://www.allgenes.org/>

5.6 Postanalysis Challenges

5.6.1 Linking to downstream biological validation

When we introduced the functional genomics pipeline of figure 1.5 (see p. 17) we made it clear that without biological validation, the hypothesis generation and test procedure for which the pipeline was intended will be arrested. A small but important subtask of bridging the flow from computational to biological validation is the assurance that the genes referred to in the analysis (*e.g.*, the lists of genes belonging to each cluster) should refer to the appropriate physical clone regardless of whether the nomenclature used for analysis was GenBank, UniGene, TIGR, Stanford Yeast ORL codes, or any other internal or local numbering schemes. Specifically, biological validation will involve performing biological "wet bench" experiments, which may well depend on obtaining clones of the genes identified in the analysis.

Thus, some degree of translation is necessary to follow up on the results. It has been our experience that this process takes many fold longer than the actual bioinformatics analysis. Since many of the genes or ESTs that appear on the lists after analysis are typically unknown, one can argue that a bioinformatics analysis of microarrays is never actually complete. It may only be after another week, month, or year, that another group may finally assign a meaning to one of those unknown genes. Only after the analyst sees that result might a hypothesis be generated. Thus, an infrastructure that periodically re-queries the accession numbers or genes to see if anything new is known about each oligonucleotide, cDNA, or gene is necessary. Furthermore, appropriate analysts should be informed when such an event happens. Such a system might be similar to how PubCrawler operates on the biomedical literature [96].

Here we touch upon some of the conceptually shallow but pragmatically thorny challenges that face the researchers as they attempt to make sense of the results of their data mining forays.

5.6.2 Problems in determining the results

Several problems can develop during the postanalysis interpretation of these results.

- No genes BLAST to finding: Some microarray results may be given as accession numbers for ESTs. Thus, to follow up these results, it is necessary to take the nucleotide sequence for the EST and BLAST that sequence against GenBank or UniGene looking for a "full-length" gene. In translating these accession numbers, however, there may be no gene that matches strongly enough. This can happen because the full-length gene may not be known yet, or due to inaccuracies in the EST consensus sequence. Gene findings with this problem may not be able to be followed up.
- Many genes BLAST to finding: Sometimes, many full-length genes may be found for a single EST, meaning the EST is nonspecific. At the time of the microarray construction, the EST may have been thought to be specific for one gene, but with progress in sequencing, this may no longer be true. To follow up these findings, many clones may need to be used in biological experiments.
- EST is no longer unique: Often, microarrays may have been designed with a particular sequence thought to be unique for a single gene, EST, or EST cluster. With progress in sequencing, however, it may be later realized that what was thought to be a single EST or gene cluster actually represents two or more unique clusters. It is often unclear as to which of the new clusters best maps to the older set of probes. Unfortunately, revision control is often not comprehensive in these accession number databases, and it becomes very difficult to link new clusters with old designators.

- Genomic sequence BLASTs, but not genes: Sometimes, gene sequences from positive microarray findings can be found in stretches of genomic sequence using BLAST. Additional bioinformatics (and biological) work then needs to be done to determine the gene boundaries before biological validation can take place.

Chapter 6: From Functional Genomics to Clinical Relevance—Getting the Phenotype Right

Functional genomics experiments can only have biological or clinical relevance if information (or better yet, knowledge) external to the system being studied is employed. Even in the case of so-called unsupervised algorithms or clustering techniques, the early investigations (*e.g.*, [56, 63]) made sense of the clusters of expression only by reference to external knowledge: the genes known to be involved in cell cycle regulation, protein translation, *etc.* Without such external *a priori* knowledge, the observed clusters are of little use in generating or testing new hypotheses about biology. In the experiments that had the most obvious clinical relevance, such as finding genes that aid in the distinction between acute myelogenous leukemia versus acute lymphoblastic leukemia [78], the prognosis of large B-cell lymphoma [3], or pharmacogenomic prediction of gene targets for chemotherapy [39], it was central to the import of these investigations that they integrated well-characterized external biological or clinical data with the expression data.

If functional genomics is to lead to a qualitative change in the way in which clinical medicine (diagnostics, prognostics, and therapeutics) is practiced, then most studies will similarly have to incorporate high-quality and relevant external data. That is, if we want to know how a gene expression pattern may predict mortality, a particular disease profile, or drug responsiveness, we need to be at least as meticulous in characterizing the patients, their histories, their tissues, and the contexts surrounding how data were acquired, as we are in obtaining the gene expression profile. Perhaps because the need has only become recently apparent, to date there has been little in the way of systematic approaches to the acquisition of such extragenomic data. We discuss here why this is and what steps can be taken toward a more systematic approach.

6.1 Electronic Medical Records

A central intellectual and technological asset to functional genomics has been GenBank (see section 5.5.1) and related genomic and protein databases. Their standardized data models have allowed research laboratories throughout the world to rapidly populate them with the very latest information. In turn, these databases are freely available throughout the world via the Internet and have seeded, accelerated, and inspired thousands of research projects.

In contrast, there are few, if any, consequential shared national clinical databases. Specifically, patient data in one information system can only rarely be transferred to another system in clinical practice. Within a single healthcare system, annotations such as billing diagnoses will suffer from distortions imposed in an attempt to maximize reimbursement for care. Despite decades of research and development of clinical record systems, they remain problematic. If properly constructed and maintained, electronic medical records could provide an invaluable set of phenotypic annotations with which to bring clinical relevance to genomic data.

The marked contrast between genomic and clinical care databases is deceptive. The Human Genome Project has benefited from the elegant simplicity of the genetic code. Consequently, there are relatively few items that GenBank requires to be submitted for an entry to be a valid (and useful) component to its database. The clinical care of human beings is a far more complex process, requiring at the minimum a detailed record of the history of multiple clinical interventions and outcomes, relevant life history, and clinical measurements that span several modalities, from serum chemistry to brain imaging. It is not surprising that the data model required to capture all this information is extremely complex as is evidenced by the Health Level Seven (HL7) Reference Information Model [147]. It is a remarkable tribute to the persistence of the individuals involved in

these clinical data model standardization efforts that they have been able to arrive at a reasonably adequate standardized representation of not only the aforementioned descriptors but much of the process and business of clinical care.

This challenge relates back to our prior discussion of the complexity of microarray expression data^[1] which is much more akin to that of clinical laboratories than that of DNA sequences. Microarray measurements reflect the particular artifactual qualities of the measurement system used, in contrast to the (at least theoretical) invariance of sequence with sequencing technique. It is therefore not surprising that even prior to encompassing the entirety of clinical annotation, the genomics community has faltered in developing shared and standardized data models where the simplicity of the genome no longer dominates. However, the complexity of microarray data models is easily dwarfed by the complexity of clinical data, and the effort to generate a standard human phenotypic data model for genomics would similarly demand orders of magnitude more effort than has been invested in data modeling by functional genomicists to date. It is therefore somewhat surprising that several efforts to develop clinical phenotype data models have been proposed without much reference or attempt to co-opt the efforts of the clinical informatics community.

A more efficient mechanism might be for functional genomicists and bioinformaticians to appropriate that segment of an existing clinical data model, such as HL7.

This might require that members of the bioinformatics community participate in an HL7 committee in order to fully accommodate the annotation needs of functional genomics. This would be a far smaller cost than having to duplicate the existing massive efforts of HL7.

^[1]As we described in section 5.4.

6.2 Standardized Vocabularies for Clinical Phenotypes

Standardized clinical vocabularies are just as essential as standardized phenotypic data models. The lack of widely accepted standardized vocabularies for clinical care has greatly hampered the development of automated decision support tools and clinical research databases. The impossibility of guaranteeing that a serum sodium or systolic blood pressure has the same code or term throughout our hospital system is troublesome. Fortunately, several efforts in the private and public realm, e.g., Logical Observation Identifiers, Names, and Codes (LOINC) [99], are addressing this issue. The National Library of Medicine has invested large resources in the Unified Medical Language System (UMLS) to enable these different vocabularies to be interoperable, at least at a basic level [120, 158]. Most of the most widely used vocabularies, such as Reed codes [84], Medical Subject Headings (mesh) [51], and Systematized Nomenclature of Medicine (SNOMED) [53] are represented in the UMLS.

The same problems are not unknown in bioinformatics as is pointed out in the section on nomenclature (section 5.5). Many microarray technologies use their own system of accession numbers, which then have to be translated into a more widely used nomenclature such as LocusLink.

As is the case for data models, as functional genomic investigations venture into the domain of clinical correlates of gene sequence, expression, and protein concentrations, the lack of standardized clinical vocabularies will prevent large-scale accumulation of useful data at the national or even regional level. For example, if a set of expressed genes are thought to be associated with a particular set of cardiac arrhythmias, this association will only be found if the nomenclature used to describe the arrhythmias in the clinical databases is drawn from one of the standardized vocabularies. Additionally, these vocabularies would have to be used with the same semantics at each data collection site. Rather than inventing yet another competing standard for a

clinical vocabulary, bioinformaticians should examine which of the existing component vocabularies within the UMLS best suits the needs of a particular set of studies.

6.3 Privacy of Clinical Data

In 1997, the Institute of Medicine [47] reported on significant lacunae in both technology and policy in protecting confidential patient data. Among the problems of most concern emphasized in this report was the relatively unrestricted access by third parties to these data for secondary uses and the lack of the adequacy of the anonymization process (in practice and theory [170, 171]). Subsequent to this effort, the clinical informatics community developed several model confidentiality policies [153] and cryptographic identification systems [111, 174].

As the fruits of the Human Genome Project are translated first into clinical research protocols and then into clinical practice, personally identifiable genomic data will find their way into several instances of the information system. The challenges posed to security and privacy of such data might seem to dwarf any encountered to date with conventional clinical data. At first blush, genomic information is likely to be much more predictive of current and future health status than most clinical measurements. And, with very few exceptions, an individual's genome is uniquely identifying and applicable to that individual's entire life. This identifiability is apparently more reliable, persistent, and specific than typically cited identifiers, including a person's name, social security number, date of birth, or address. In practice, however, the widespread nature of clinical databases and the ease with which a unique individual can be identified [170] allow for breaches of privacy to be effected relatively trivially, even without genetic information. Furthermore, as demonstrated by insurance companies, highly accurate actuarial and prognostic knowledge can be developed solely on the basis of clinical history, habits such as smoking, family history, a physical examination, and a handful of laboratory tests. It remains to be demonstrated, given all the genomic reductionist fallacies reviewed at the beginning of this book, just how much additional prognostic accuracy will be delivered by the use of personal genomic data.

Nonetheless, at the very least, the architects of information systems storing genetic data should learn from all the mistakes and designs developed for the security architectures and privacy policies of conventional clinical information systems. Conversely, the dire concerns voiced over the storage of personal genetic data will likely generate new policies and security architectures that will enhance the confidentiality of clinical information systems. Moreover, when personal genetic data become incorporated into routine medical practice, the safeguards in place for the medical record's confidentiality will have significant import for the confidentiality of the genetic data that will be referenced in the medical record. It also may further drive the increased degree of patients' control of their own medical record [125].

Why should we bring up these concerns at all in a book on functional genomics? Simply because there are two strong imperatives to keep the privacy of the patient or tissue donor intact. First, on the simple grounds of human rights, there is a basic or implicit contract between researchers and patients. The second and less altruistic motivation is that if the confidentiality of the patient is breached, even the perception of the loss of confidentiality (particularly if it involves a disease for which there is attached social stigma or risk of loss of employment or income) could jeopardize an entire study. Or it could well lead to a broad societal reaction against genomic studies. Even now, there exist multiple recourses for the patient to obtain both civil and criminal penalties against the researcher. These penalties have further been formalized by the recent components of the Health Insurance Portability and Accountability Act (HIPAA) as amended in the final days of the Clinton administration in 2000 [32].^[2] In this brief section we outline the major obligations involved in protecting the confidentiality of genomic patient data and common pitfalls in implementing them.

6.3.1 Anonymization

In theory, if a database were to be truly anonymous, then, even if it fell into the possession of an unscrupulous party, no harm could come of it. Therefore, a first attempt at the design of a functional genomics database will typically involve one of the following two strategies for anonymizing a database.

The first strategy is to delete or replace common identifying attributes, such as name, date of birth, or address, which could be used very easily with external databases to identify the patient.^[3] The risk with this approach is that although it does indeed reduce the ease of identification to eliminate these common "face recognizable" identifiers, the records can still be readily identifiable. The underlying insight is that there will be a list of attributes, such as which doctor the patient visited, when the patient was first diagnosed, or what kind of cancer the patient had, which can serve to uniquely identify the patient. At the very least, it can be used to reduce the set of patients matching these attributes to a handful of individuals. Then, if there is an incentive to do so, there is little or no barrier to reidentification of the patient record. This approach has been most extensively developed and documented by Latanya Sweeney [169,170], who documented that even meticulously deidentified records such as those from the billing systems of Medicare could be used to re-identify patients, including prominent politicians. She developed a collection of programs, called SCRUB, which enable an increasing degree of anonymity due to elimination of data fields which could be used to particularly identify a single patient. Nonetheless, one of the clearer messages of that research was that to the degree that a clinical database is useful in providing clinical detail to characterize the patient, there is an increasing risk of breach of the anonymity of that record. With this in consideration, designers or architects of functional genomic databases should be mindful of being parsimonious in their choice of attributes used to annotate these clinical databases, such that only the minimum needed for the purposes for which the databases are designed should be implemented.

The second, alternative approach is to segregate those fields, including name and addresses, that would be the most revealing of the patient's identity and place them into a separate identification database controlled by a third trusted party (trusted by the patient) who would only allow the patient data to be "joined" to the nonidentifying data with explicit permission of the patient or the institutional review board of a research institution. That is, the reidentification of the patient would be determined by consent and disclosure policies and practice. We note that the anonymity of the nonidentifying data could still be compromised by the same reidentification techniques applied to the scrubbed databases described above.

In our own investigations of multicenter large-scale genomic databases, we have investigated the use of cryptographic health identification systems which could be used locally or nationally to create systems that enabled either the patient, the provider, or the institution to determine very specifically under which circumstances the patient record could be joined to the identifying information [111]. Furthermore, we have argued for patient-controlled encryption of their record [125] to prevent the application of reidentification techniques.

In the end, however, it should be recognized that no matter how secure the means used to encrypt the database or to anonymize it, if there is a sufficiently large financial or personal incentive to breach the privacy of a particular individual, it will be done, whether through a direct assault on the protection measures used, or through a corruption of internal controls be they technical or sociological. In this light, it is most reasonable to ensure that when patient samples are entered into a database, they be obtained only upon very extensive debriefing of the patient of the measures that will be used to protect the record, as well as the potential risks. With full disclosure and consent, there is much less likely to be acrimony and *post hoc* disruptions that may affect the

integrity of the entire functional genomic investigation undertaken by a particular investigator.

6.3.2 Privacy rules

By passing the HIPAA, Congress established standards governing the privacy of identifiable health information. Though this legislation covers many aspects of privacy in clinical medical care, there is a significant area of overlap with clinical research and the practice of obtaining consent from human beings involved in such research.

Those organizations covered by the HIPAA legislation and the privacy rule can continue to use properly deidentified health information for research purposes. However, those organizations can only use and disclose *protected health information* for research purposes with permission from each individual, or without individual authorization in a restricted set of cases.

1. An institutional review board has approved a waiver of research participant's authorization. This could be done if all of the following conditions are met:
 - ◆ Use of the information involves no more than minimal risk to the individuals;
 - ◆ The waiver will not adversely affect the privacy rights of the individuals;
 - ◆ The research could not be done without a waiver or without use of the information;
 - ◆ The risk to the research subjects is low compared to the potential benefits to those subjects;
 - ◆ A plan is in place to protect against improper use;
 - ◆ Identifiers to the information will be destroyed as early as possible;
 - ◆ The information will not be re-used.
2. The researcher documents that the data will be used only to design or assess the feasibility of a research project.
3. The researcher documents that the protected health information of deceased individuals is being used and is necessary for research on their decedents.

This section is not meant to provide specific legal advice on this subject. More information about the implications of HIPAA on clinical research is available at <http://www.aspe.dhhs.gov/>. There are other U. S. laws to protect research subjects, including the Common Rule from the Federal Drug Administration. Privacy rules in other countries, especially in the European Union, will differ. The reader is strongly encouraged to consult his or her own institutional review board, or equivalent organization, for further guidance on this subject.

^[2]And as pursued through at least the early months of the George W. Bush administration.

^[3]A list of such identifiers that have been deemed of particular risk by the U.S. Department of Health and Human Services as part of HIPAA can be obtained at <http://www.aspe.hhs.gov/>

6.4 Costs of Clinical Data Acquisition

"Getting the data in" has often been cited [130] by authorities in clinical informatics as being among the most difficult challenges in successfully deploying clinical information systems. In particular, the costs of acquiring detailed and structured data from the clinical care process have been daunting. Voice and handwriting recognition information systems have not been broadly adopted for a variety of performance and usability issues. These cost and practicality issues will continue to present an obstacle to clinical information system utility and deployment until better solutions are arrived at. In contrast, the Human Genome Project has managed to achieve significant economies of scale in data acquisition in sequencing and expression measurement. Gene microarrays alone have dropped in cost by a factor of 10 in just the last 2 years.

Here again, once genomic investigators attempt to bridge the gulf from purely genomic data sets to phenotypically (*i.e.*, clinically) annotated data sets, they will be confronted with the same challenges of clinically oriented, codified data acquisition. The questions of which user interfaces are the most cost-efficient, reliable, and generalizable to multiple clinical domains are among the implementation and design challenges that they will face. Although they have yet to arrive at definitively successful answers, clinical informaticians have already completed several decades worth of engineering and ethnographic studies [49, 50] addressing the very same questions.

We have personally witnessed the difficulties of some genomic studies in which the genotyping portion of the study was readily completed but the phenotyping was costly, slow, and ultimately rife with errors. This experience is likely to be increasingly common and therefore the need to solve the clinical annotation challenge is pressing.

Chapter 7: The Near Future

Overview

In this chapter, we take the risky road of futurology with respect to functional genomics. We will make it even riskier by taking a short-term view, so that we will soon know just how far off the mark we were. The reason that we are at all engaging in this exercise is that the technological, biological, and analytic aspects of functional genomics investigations are changing every year. This leads to the, possibly vain, conceit that anticipation of future trends will allow a laboratory to invest in the most high-yielding technologies and personnel. We find ourselves asking the same questions many of our collaborators have asked us in this regard:

- What technology should we invest in for gene profiling measurement?
- What software platforms should we license, install, and become proficient in using?
- Even if we consistently use the same platform, will we be able to compare results of expression profiling experiments across different generations of the same microarray technologies?
- What are the most important issues we should plan for in the next 10 years to best leverage our efforts in functional genomics?

We attempt here to begin to answer these questions using the data on hand, our intuition, and our experience with past technological revolutions. *Caveat lector*, prognostication is all too cheap.

7.1 New Methods for Gene Expression Profiling

In the last 5 years of the twentieth century, and subsequently, there has been rapid growth in the development of alternative techniques for measuring gene expression on a large scale. There are two principal drivers behind the growth of diverse measurement techniques. The first is simply the commendable attempt to provide technologies that are more reliable, reproducible, precise, and cost-effective. The goal, after all, is to be able to make the measurement of a set of gene profiles no more expensive than a simple serum chemistry drawn routinely during a doctor's visit. It should be robust and affordable enough to be deployable widely for purposes of microbial genome detection for public health surveys and for routine screenings for a variety of diseases during clinical care and well-care checkups. Unfortunately, the world of commercial intrigue and murkiness around intellectual property has been the second important driving force in developing these technologies. Affymetrix, for example, has a patent that, under some interpretations, might include a range of microarray gene expression measurement technologies on a two-dimensional surface. Consequently, much of the development of alternative technologies has been focused on avoiding the potential conflict with this intellectual property (see figure 7.1).

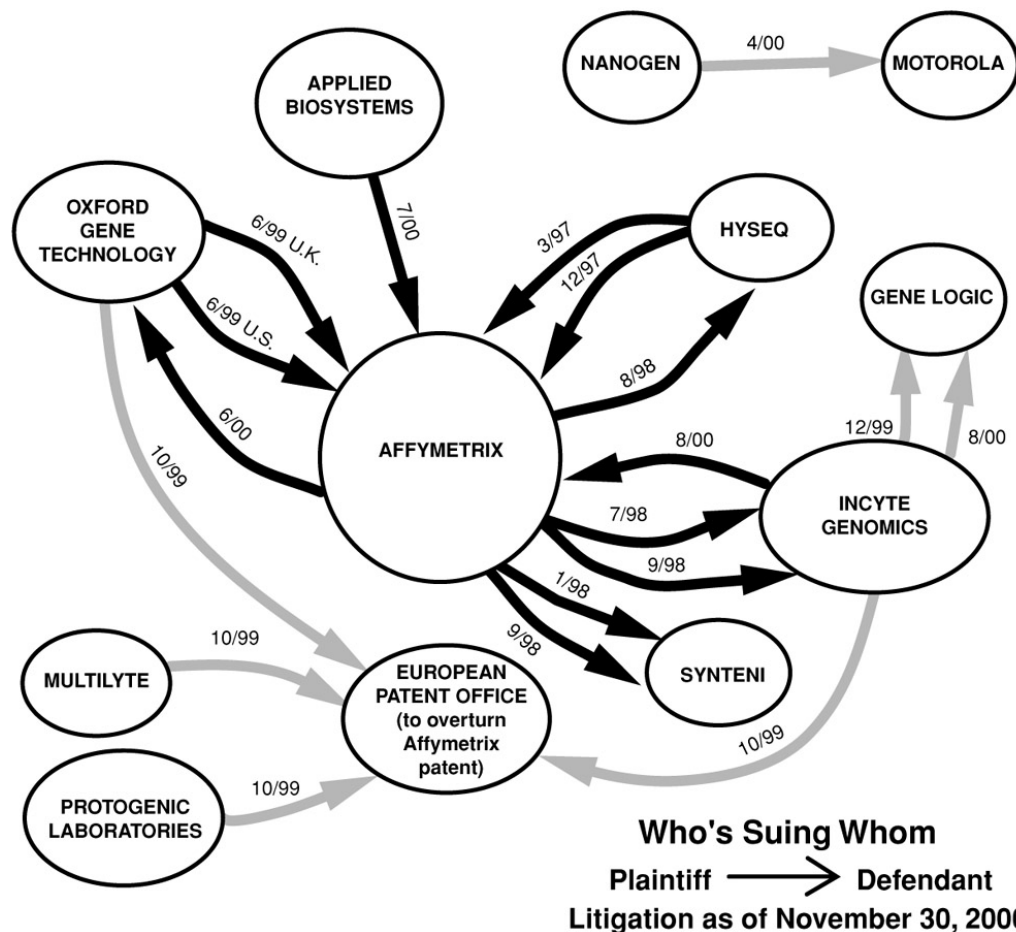


Figure 7.1: Lawsuits in the microarray field. With the number of lawsuits already in this area, it should be no surprise that companies in this industry are striving to determine new and unique intellectual property. (From Gibbs *et al* [77].)

We review here some of the major and upcoming technologies. Perhaps the best way to think of these technologies is within the following framework. Some of them use full-length cDNA probes, others use oligonucleotide probes. Independently of what kind of probe they use, the technologies may use a solid two-dimensional surface, a three-dimensional matrix, or optically coded beads.

We are hardly going to be able to even enumerate all the emerging technologies for expression measurement. For the latest up-to-date list, the reader is directed to <http://www.gene-chips.com/>. Also, for an excellent review of the scientific underpinnings of many of these technologies, we refer the reader to a review by Southern *et al.* [166] which provides a survey of the physicochemical basis of molecular interactions of hybridization on microarray surfaces. We would also impress upon the reader that most of the technologies described either have only begun early production lines or are still under development. Consider that it has taken well over 3 years, and counting, for the mainstays of microarray expression measurement, oligonucleotide microarrays, and robotically spotted microarrays to achieve minimal levels of accuracy and reproducibility. Then, as interesting and appropriate as some of the following methodologies may appear, they bear close scrutiny and testing prior to any large scale attempts for use in scientific experimentation.

Additionally, each of the technologies described is, of necessity, undergoing iterative processes of improved quality, higher density, and lower cost. This makes it all the more challenging to decide which microarray platform to adopt. This is very much analogous to choosing a home video gaming system: Nintendo, Sony, Microsoft, each has its own game console. As each manufacturer rolls out its next-generation platform at different times, consumers have to agonize over whether it is better to buy the platform that best suits their needs today or the one that will do twice as much in 4

months. As the refinement of these platforms is an iterative, continual, and competitive process, there is no stable or consistent "correct" response to the consumers' dilemma.

The video gaming console analogy is also relevant to the commoditization of microarray platforms. In the 1980s, real-time three-dimensional rendering was the province of highly specialized, expensive computer systems available in only a few laboratories. The cost of the hardware and staff far exceeded the cost of the system software. Now, with commodified video consoles, over the short life of a consumer system, the major cost is the acquisition of software and game modules. Similarly, the current microarray systems are labor-intensive and the hardware (including the microarrays themselves) and staff drive the high costs of expression profiling laboratories. As a result of the competition and improvements in manufacturing technology, the costs of hardware and attendant staff will diminish and likely become dominated by the costs of tissue acquisition, banking, and annotation.

With these issues in mind, we review some of the available and upcoming platforms.

7.1.1 Electronic positioning of molecules: Nanogen

Most biological molecules have a natural positive or negative charge. When these molecules are exposed to an electric field, those with a positive charge move to an area with a negative charge, and *vice versa*. Novel microarray technologies such as those of Nanogen (San Diego, CA) use this property to maneuver molecules to specific test sites on an electronic microchip. By carefully controlling which element of a grid of electrodes is charged, this technology can direct many test molecules to specific areas of the chip. Because of the more active nature of this technique, this could potentially deliver molecular binding on the microarray several orders of magnitude faster than passive hybridization methods.

Regarding gene expression measurement, a cDNA molecule that is negatively charged moves in an electric field to an area of net positive charge. The sample DNA is significantly concentrated over time in the area of positive charge. DNA that does not have the right mass/charge ratio is repelled from the area of the electrode under closely controlled electronic conditions (see figure 7.2). The current trade-off (pun intended) for all this configurability is that the density of features of these microarrays is at least one order of magnitude less than the current density offered by companies such as Affymetrix. However, the density of this technology is rapidly growing.

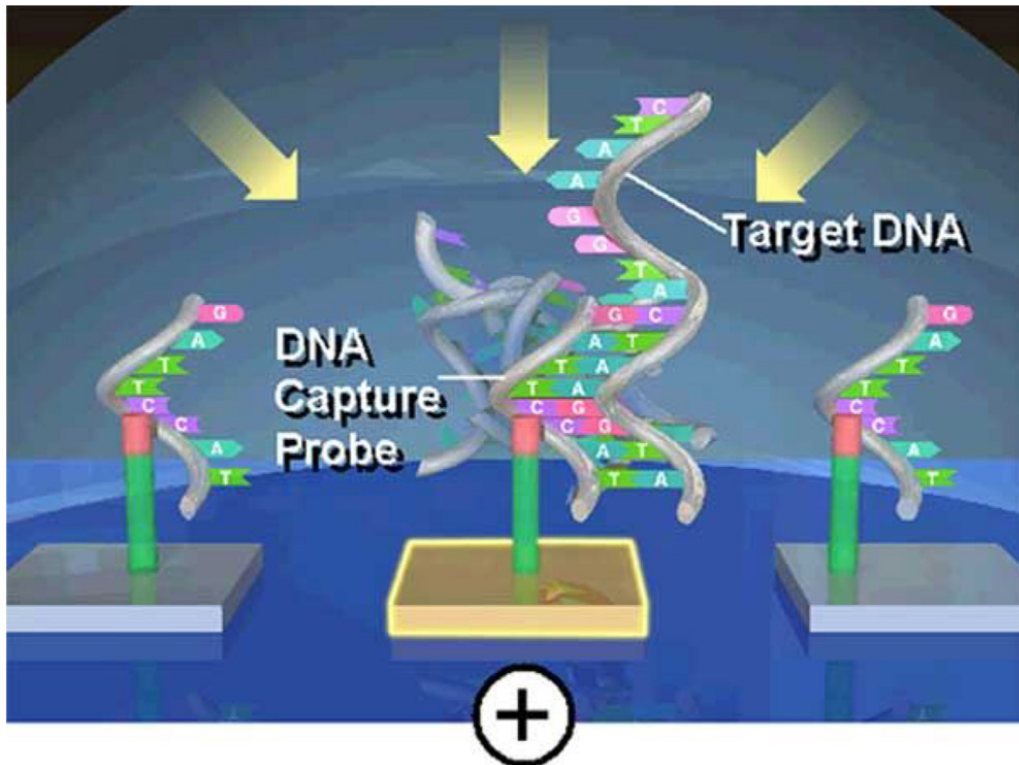


Figure 7.2: The Nanogen microelectronic array.

This system may be best suited for laboratories that are interested in a medium-throughput platform that will allow several genomic applications such as gene profiling, proteomics, and genotyping.

7.1.2 Ink-jet spotting of arrays: Agilent

Agilent (Palo Alto, CA) is building on its long history of highly engineered ink-jet delivery systems. It has licensed technology from Rosetta Inpharmatics (Kirkland, WA) so that it can synthesize and "spray" onto a glass slide oligonucleotide molecules in a highly localized fashion (see figure 7.3). That is, the nucleotides are "sprayed" separately by the ink-jet heads and linked *in situ* on the glass slide into oligonucleotides. At present, this technology permits up to 25,000 features per glass slide using volumes only in the picoliter range per feature. Because the features are built using longer nucleotides, there is typically only one feature per gene. This is in contrast to the 40 different features that are used in each probe set on Affymetrix oligonucleotide microarrays, using shorter and therefore individually less specific oligonucleotides.

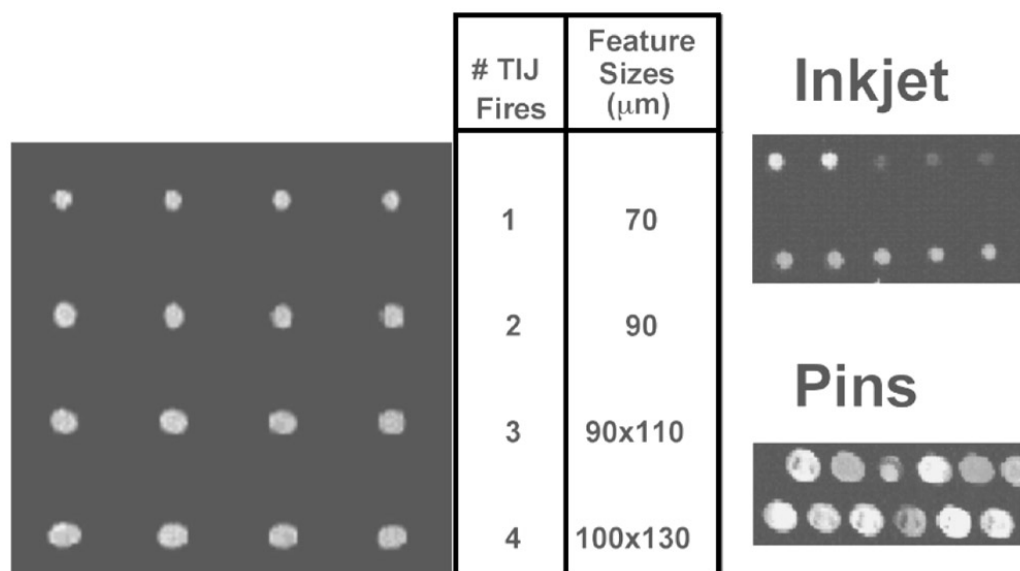


Figure 7.3: Flexibility of on-the-fly ink-jet arrays. On the right is a comparison of morphology of ink-jet versus pin arrays. Ink-jet printing produces uniform and consistent features. An artifact of on-the-fly printing is formation of oblong spots at maximum print speed (full fires) that can be made into circles upon slowing down. This underscores the theoretical flexibility of this system.

In theory, the contact-free nature of the ink-jet process could result in fewer spatial artifacts than the robotically spotted technique. Also, the synthesis technique used allows oligomers with a length up to 60 nucleotides to be constructed. Again, in theory, this could provide more specificity than the shorter oligonucleotides used by Affymetrix, and better normalization by GC content and correction for cross hybridization than longer cDNA molecules. The nature of this technology allows for a very rapid design and test cycle for these microarrays. Although it remains to be seen (at the time of this writing) whether Agilent does deliver on the promise of this technology, it has an opportunity to create a high-quality platform with manufacturing costs orders of magnitude less than others.

7.1.3 Coded microbeads bound to oligonucleotides: Illumina

Technology from Illumina (San Diego, CA) uses oligonucleotides linked to beads which are individually optically coded. These beads are then bound to the tips of individual fibers in a fiber-optic bundle (see figure 7.4). The company states that currently 2000 different genes can be represented on each fiber bundle, but appears to be scalable from 50,000 to 5 million beads per bundle. In addition, each array contains between 96 and 1536 bundles. Each bead can be optically interrogated by laser excitation via the fiber-optic bundle, using a proprietary technique. This technology is very much in the early testing stage and we will not have a good idea of its reproducibility and accuracy until after more independent testing.

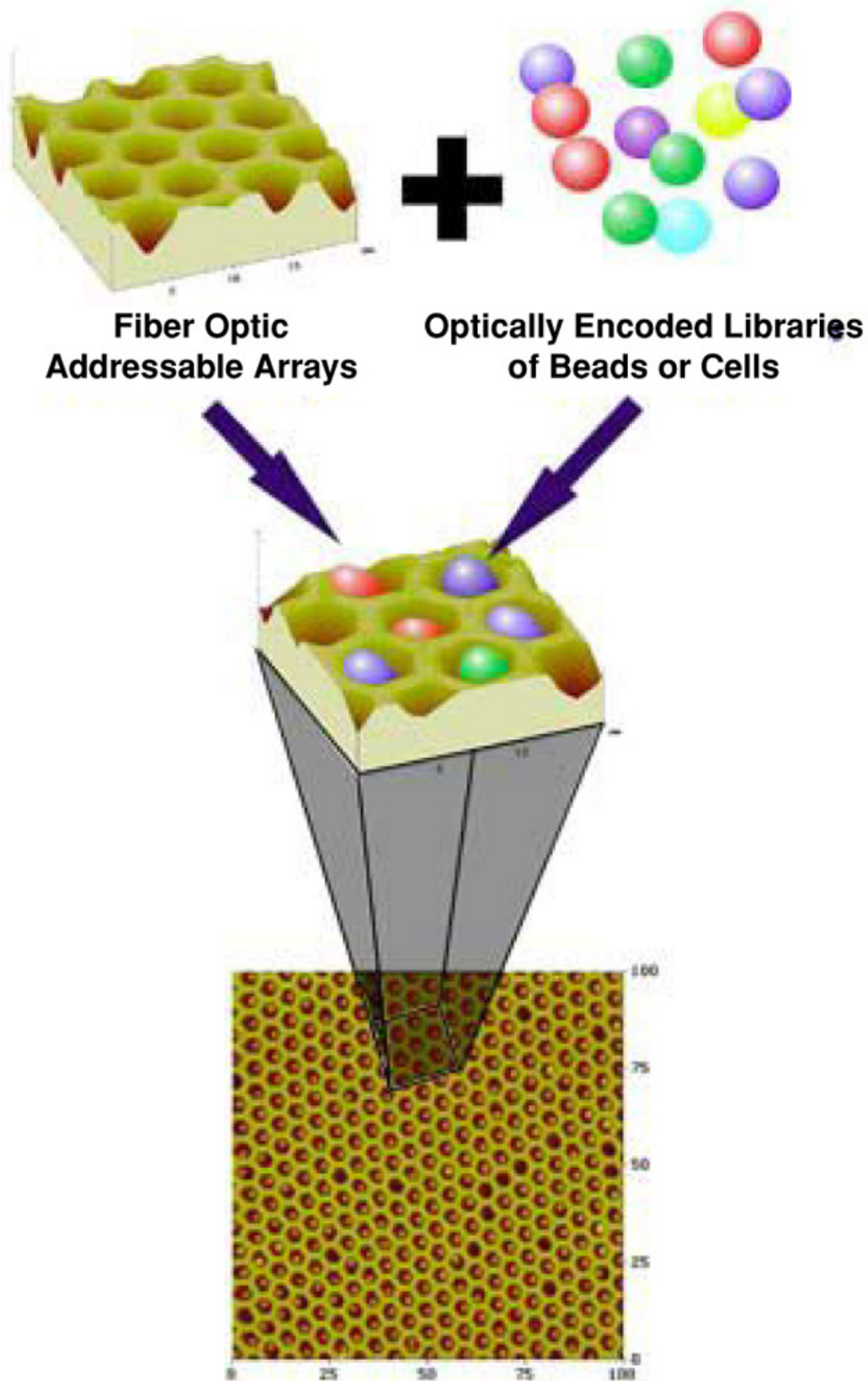


Figure 7.4: The Illumina BeadArray.

If this technology does in fact prove viable, then its potential scalability to millions of probes will allow new ground to be reached in profiling. For example, millions of probes would allow each gene in the human genome to be represented, as well as all known polymorphisms and alternative splicing products.

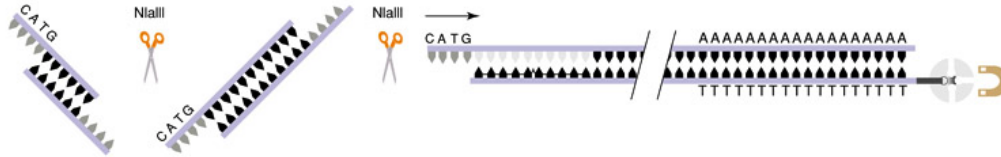
7.1.4 Serial Analysis of Gene Expression (SAGE)

SAGE is a technique designed to take advantage of high-throughput sequencing technology to obtain a quantitative profile of cellular gene expression [183]. Rather than directly measuring the expression level, the technique "counts" the number of transcripts or "tags" for each gene. A tag is a specific oligonucleotide of length 9 to 11 nucleotides corresponding to the transcript of interest which is selected on the 3' end of a transcript, relative to a site recognized by a restriction enzyme (usually *Nla*III). First, double stranded cDNA is made from mRNA, using a biotinylated oligo(dT) primer. The cDNA is then cut with the anchoring enzyme (*Nla*III), expected to cut every 256 base pairs. The resultant double-stranded cDNA is bound to streptavidin beads, divided into two pools, and ligated to one of two linkers. The tags are then spliced out using a tagging enzyme that cuts at a location away from its recognition site (such as *Fok*I). The tags are joined together and the resultant string of tags is sequenced. This sequencing allows each SAGE tag to be counted. Therefore, the number of tags for each gene that the tag represents should be proportional to the expression level of that gene. Figure 7.5 illustrates this process.

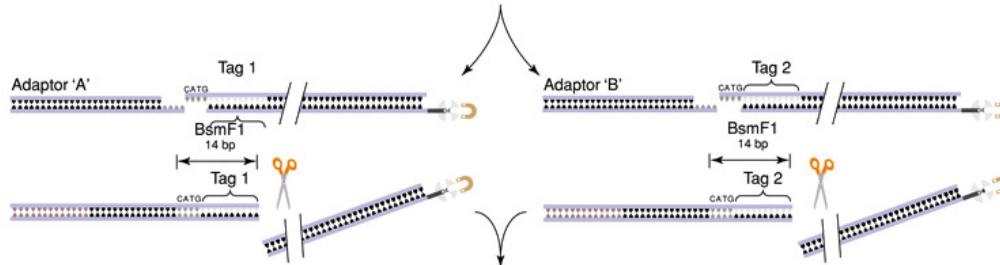
(a) Synthesis of biotinylated double-stranded cDNA.



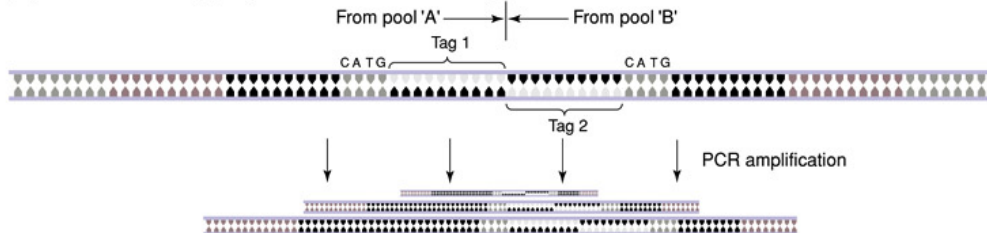
(b) Restriction enzyme digestion of cDNA and capture of 3' most NlaIII cDNA fragment.



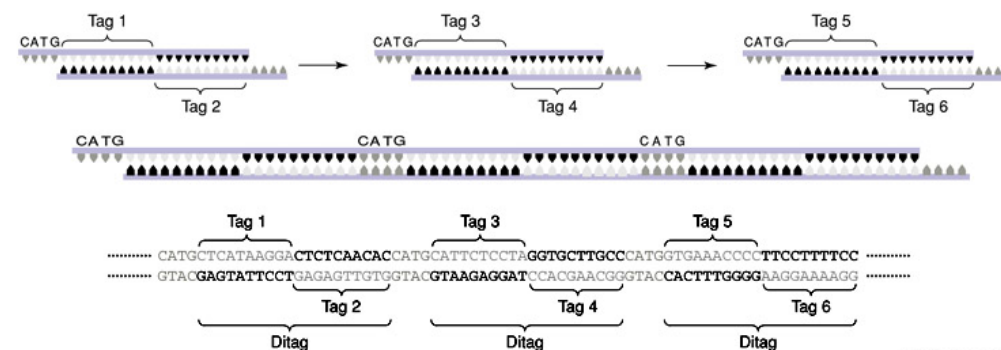
(c) Addition of specific adaptors and excision of each cDNA's 10-bp tag.



(d) Formation of ditags, amplification and removal of adaptors.



(e) Formation of concatemers and sequencing.



Drug Discovery Today

Figure 7.5: The Serial Analysis of Gene Expression. (Derived from Madden *et al.* [124].

SAGE suffers from the potential lack of specificity of these oligonucleotide tags, much in the same way as do the oligonucleotide and cDNA microarrays. It can be argued that the small length of the tags makes them even more susceptible to nonspecific interactions (i.e., multiple genes or gene products with the same sequence as the oligonucleotide). In fairness, the most recent gene profiling investigations use longer nucleotide sequences. Furthermore, and more problematically, sequencing technology has an error rate that might substantially degrade the correct assignment of a tag count to a gene. Although there have been some early, optimistic results for small numbers of genes, showing the correlation of *sage* results with microarray expression values [100], more comprehensive testing is necessary. Of course, as with all these technologies, the choice of a "gold standard" to rate such comparisons is hard to come by.

7.1.5 Parallel signature sequencing on microbead arrays: Lynx

Lynx Therapeutics (Hayward, CA) combines two techniques to assess gene expression: the ability to sequence without using a gel, and the ability to clone millions of templates onto microbeads [31]. Specifically, cDNA templates are first ligated into vectors containing a set of different 32-mer oligonucleotide tags. A 1% subset of these ligated vectors is taken, such that each template is ligated to a unique tag and each template is represented at least once. The subset is then PCR-amplified, made single-stranded, then hybridized against a set of microbeads each separately containing a sequence complementary against a 32-mer tag. The microbeads are then fixed in place within a flow cell. Sequencing can take place directly on the microbeads through successive use of restriction enzymes exposing 4 base pairs, which in turn can be successively bound by fluorescently labeled probes.

Since the source templates can be generated from mRNA, this can effectively be used as a massively parallel method to find tags associated with each mRNA. Also, since presumably each copy of each mRNA is uniquely labeled with a different tag, each copy is sequenced separately, and thus this system is not only quantitative, but may be more sensitive at lower gene expression levels, like SAGE. At the time of writing, commercialization of this technique has just started.

7.1.6 Gel pad technology: Motorola

Using technology developed at the Argonne National Laboratory, Motorola Life Sciences (Northbrook, IL) has developed a novel "gel pad array" technology and has begun early production rounds of these arrays. Within each "gel pad," or gel matrix, several different kinds of reaction technologies can be embedded, including oligonucleotides. Although the oligonucleotides attached to the gel pads (themselves 100 $\frac{1}{4}$ m in width) might not seem to have the same access to the free-floating complementary DNA strands in a biological sample as they might have on a glass slide, in practice they appear to have similar specificity and sensitivity of hybridization. The use of oligonucleotides is preferable to full length cDNA as reproducibility and cost are kept lower and the probes can be suitably normalized for GC content. As is the case for other oligonucleotide technologies, care must be taken in selecting an oligonucleotide sequence so that it does not cross-hybridize with other genes.

As the gel pads allow oligonucleotides lengths of greater than the 25 $\frac{1}{4}$ m used in Affymetrix microarrays, in theory they can provide higher specificity. Also, although the feature size is relatively large compared to oligonucleotide arrays made by Affymetrix, current gel pad technologies can be arrayed up to 10,000 probes on a single glass slide. As for other platforms, higher densities are promised.

7.2 Respecting the Older Generation

It has already happened that laboratories have invested thousands of dollars and hours into one generation of microarray technology, and then have had to move on to the next generation of microarrays. After several experiments, it becomes apparent to them that results across the generations of microarrays are not identical. The question then arises of how to salvage the information from the first generation of microarrays. The concern is, of course, that they will have to treat the data sets of different microarray generations as distinct and incomparable and therefore they will lose "statistical power" in their classification or clustering studies. This concern is magnified when the earlier studies involved unique and irreplaceable tissue samples which may no longer be available to be studied with the newergeneration microarrays. We illustrate this challenge here with

some of our own experience with the Affymetrix platform. We hasten to add that the challenges we have had in combining results across generations of this platform have their correlates on other platforms.

7.2.1 The generation gap

The HuGeneFL and HG-U95A oligonucleotide microarrays from Affymetrix are two successive generations of the platform for human expression profiling. Because the latter microarray (one of a 5-chip set) is a superset of the former generation, one would expect that probe sets in one generation would have the same expression pattern as measured in the next generation of microarrays. However, as demonstrated in figure 7.6, there are a large number of probe sets that have very poor correlation when the same RNA sample is hybridized identically and in the same laboratory to the two generations of microarrays.

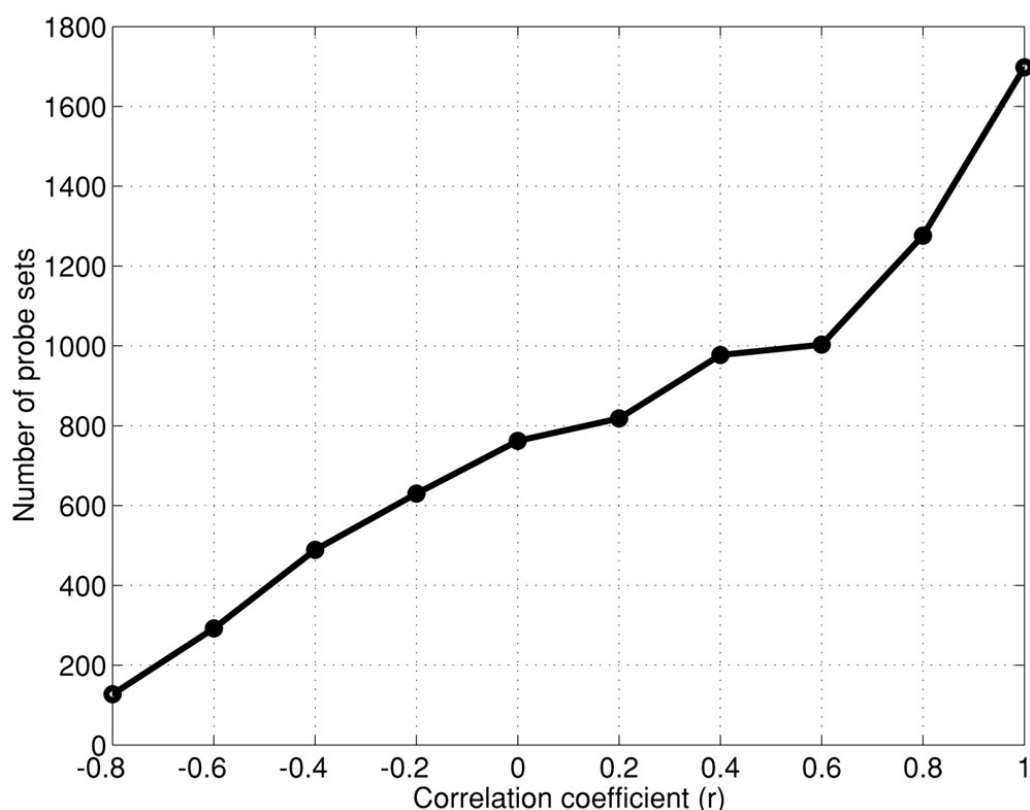


Figure 7.6: Poor measurement reproducibility across microarray generations. Two generations of Affymetrix microarrays were used to the gene expression of seven samples. Each of the 8075 common probe sets thus had seven measurements from both the older HuGeneFL and newer HG-U95A microarrays. The line represents the distribution of correlation coefficients of each probe set. Though the exact correlation coefficient may not be useful, it is notable that over 25% of the probe sets showed measurements that were zero or negatively correlated across generations. (Derived from Nimgaonkar *et al.* [136].)

The most likely explanation for this disparity is that Affymetrix designed the HG-U95A array to use a set of probes or oligonucleotides for each probe set that may only partially overlap the set of probes used in the HuGeneFL arrays. Presumably this reflects improved oligonucleotide selections based on greater knowledge of the genome and accumulated empirical results of the specificity and sensitivity of particular probes. However, because Affymetrix has continued to maintain secrecy around the sequence of each oligonucleotide in a probe set, we cannot assess this directly. Affymetrix does release a table mapping the probes (within a probe set) in one generation to the next. This allows a comparison of how well probe sets correlate as a function of the number of

shared probes (see figure 7.7). At a threshold of three or more probes in common per probe set, the correlation is substantially improved. This may explain in large part why the overall correlation between the two generations of microarrays is so poor.

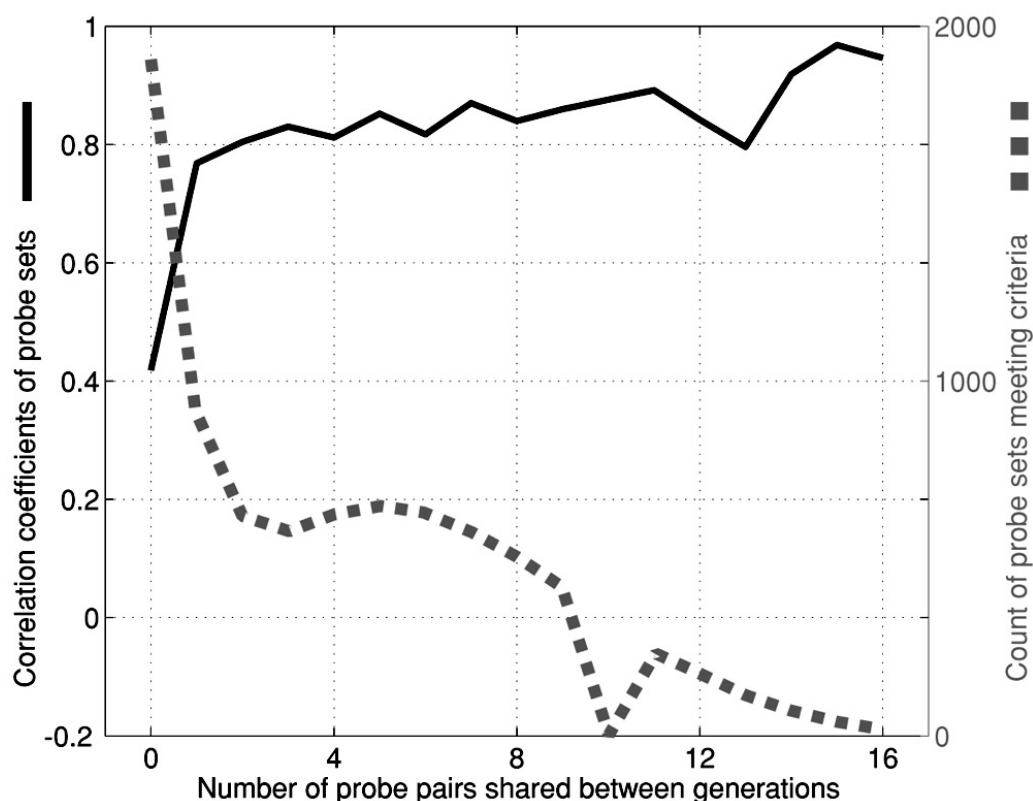


Figure 7.7: Measurement reproducibility of probe sets across microarray generations, by the number of probe pairs in common. A total of 8075 probe sets are deemed in common by Affymetrix between the older HuGeneFL and the newer HG-U95A microarrays. However, each probe set has a varying number of common probe pairs, from none to all 16. The bar graph represents the distribution of the 8075 probe sets, and the line represents the correlation coefficients of each set of probe sets across generations. For example, one finds a correlation coefficient of .77 when plotting all 6412 probe sets with one probe pair in common across microarray generations. (Derived from Nimgaonkar *et al.* [136].)

7.2.2 Separating the wheat from the chaff

We know then that the consistency of expression measurements for the overall set of probe sets across a generation of microarrays is poor. There also is a subset of probe sets for which consistency is high. To maximally salvage the data of the HuGeneFL arrays, then, the goal should be to determine which probe sets are members of that high-consistency subset. One reasonable heuristic is to pick a threshold correlation and then pick only those probe sets that have at least n probes in common such that the correlation for n as obtained from the histogram of figure 7.7 is above the correlation threshold. Just what this threshold should be will be determined by a decision analytic procedure of the kind illustrated on page 2.9.

However, the above procedure will only provide a small subset of the genes represented by the various probe sets on each of the microarrays. There may be a substantially larger set of genes that are consistently reported across the generations of microarrays that might be "salvageable." If a particular laboratory is in possession of duplicate hybridizations across these generations, then the technique described in section 3.2.2 can be adapted to provide an estimate of which probe sets are the most consistent.

That is, if the identical RNA extract has been hybridized to two generations of microarrays, then a simple program can be written to determine for any measure of gene expression (*e.g.*, the Cy3/Cy5 ratio, Average Difference, Log Average Ratio, Present vs. Absent Calls, *etc.*) the threshold (*e.g.*, minimal Cy3/Cy5 ratio) above which the expression levels reported have an acceptable level of consistency across the microarray generations. These thresholds will not only vary with the particular microarray platform employed but also with the laboratory performing the hybridizations and the tissue types employed. Consequently, this procedure must be done for each laboratory and the results may not be safely generalizable across laboratories.

7.2.3 A persistent problem

This problem of the comparability of expression measurements across generations of microarrays is likely to be with us for several years. Microarray technology is in such flux that a definitive platform that only suffers small incremental changes is not in the immediate future. Given that laboratories will be studying the same biological system for several years, and therefore over several generations of microarrays, the most prudent and conservative advice would be to:

- Decouple the hybridization schedule from tissue sampling and banking. Preferably all the hybridizations should be done within a few months, if not days. The effect of the different time of hybridization is often visible in subsequent analysis. For example, careful review of the clustering of the acute myelogenous leukemia (AML) and acute lymphocytic leukemia (ALL) data set of Golub *et al.* [78] reveals that the test and training sets automatically cluster separately. That is, the differences between the gene profiles in the test and training sets are of the same order as the differences between AML and ALL. As reported in [78] the only difference between the two sets was the date on which the hybridization was done. In addition to environmental and procedural variation, it is likely that each batch of microarrays was manufactured on a different date.
- Purchase or construct sufficient microarrays to last your entire study. Even if a newer and better generation of microarrays becomes available, the quality of the study is best served by staying with a single generation of microarrays. Otherwise, the investigator will have to wander into the uncertain territory of estimating consistency across generations of microarray technology for each gene interrogated by the microarray, as outlined above. Unfortunately, some studies necessarily are so long as to make the logistics of this tactic unrealistic.
- Maintain a reference RNA pool. This can be used to determine the consistency of measurements across microarray generations when all else fails. The selection of the reference pool should be informed by the need to maintain a large amount of reference RNA over years with minimal degradation. In addition, the pool should contain a distribution of transcript abundance that is appropriate for the experiments envisaged. Too much or too little of a transcript will cause skewed noise or variation profiles due to different degrees of cross-hybridization against different biological samples.

7.3 Selecting Software

Any high-throughput functional genomics investigation will require that there be an armamentarium of analytic tools on hand. Even a large laboratory that has several bioinformaticians creating customized tools will still have to integrate these tools with the best-of-breed available commercially or through academic licenses. More often, there is insufficient time or personnel to develop any software locally, so external software tool kits must be obtained. The decision of how to go about acquiring these is inevitably weighed down by considerations of the local environment, local resources, budgetary constraints, and the need for customizability of these tool kits. It is well outside

the scope of this book to address this process. Nonetheless, we provide here an abbreviated framework with which to consider acquisitions of bioinformatics software for use in functional genomics.

In order to provide the analytic services required, there are two broad categories of function that must be served: a collaborative data-sharing environment, and a set of robust analytic tools. The former will allow a research group to share common materials such as microarray data sets, annotations of these data sets, publications, and workflow tools such that the progress of any particular study can be monitored, and the latter will provide for the extraction of clinical or biological knowledge from genomic data sets.

Primary concerns in the first category, collaborative software platforms, will be

1. the scalability of the software to satisfy large numbers of local and remote users;
2. the security of the site to protect both the intellectual property of the data as well as the privacy of any patients whose data may be stored in the database.

A highly nontrivial aspect of the data-sharing or collaborative software tool kit is to maintain a data model for the microarrays that is capable of encompassing all the measurements that are likely to be obtained through the available microarray platforms (see section 5.4). Similarly, the collaborative software should have a sufficiently detailed and flexible phenotypic data model to allow structured annotations using controlled vocabularies to support the phenotypic-genotypic correlations that are an essential component of a successful functional genomics pipeline.

The second category of bioinformatics software includes the analytic tools that provide the implementation of the techniques that we have described in the data-mining chapter (chapter 4) and in the measurement techniques chapter (chapter 3) such as fold difference, self-organizing maps, dendrograms, support vector machines, relevance networks, k -means clustering, matrix incision trees, gene shaving, and others. They also provide means of visualizing analytic results and their relationships and linking these with the published literature, annotations, and links to relevant biomedical databases.

In the late 1990s software packages fell cleanly into one of those two aforementioned categories. Subsequently, in order to capture additional market share, the analytic tool vendors have added collaborative data-sharing capabilities and likewise the collaborative data sharing software has become increasingly equipped with a suite of analytic tools commonly used in the investigations reported in the literature. At present, however, few of the solutions are comprehensive. That is, the suite of analytic tools provided by the collaborative software vendors are limited or limiting in their analytic capabilities, and the collaborative, data-sharing, and warehousing facilities of the analytic tools are embryonic and of unproven scalability. Some of the best software packages are free; they have been developed often with federal funding by the best and the brightest and most public-minded of bioinformaticians. As a result, an investigator may not always get what he or she pays for in terms of value in quality software. Nonetheless, with respect to support for installation, debugging, and technical advice, these will be very limited with any of these free packages. Also, publicly available software packages typically require some knowledge of operating systems in order to be able to install them correctly on the computational hardware platforms. This is particularly true of Unix- and Linux-based computational platforms which are typically the most stable and reliable of platforms but also require the most expertise in technical support and management.

Some of the more notable free analysis solutions are listed and coarsely taxonomized in table 7.1. Note that many of these are available for free only to academic users, and charges may apply for

commercial users. Also, refer to table 5.2 for free data models and databases.

Table 7.1: List of freely available analysis software for microarray

Tool name	Description	URL http://
AMADA	Dendrograms, principal components analysis	http://www.web.hku.hk/~xxia/software/AMADA.htm
ANOVA	Matlab software programs for microarray data	http://www.jax.org/research/churchill/software/anova/
Array Viewer	Differential gene expression	http://www.tigr.org/softlab
BRB array tools	Excel software, provides scatterplots, dendrograms, and class prediction	http://www.linus.nci.nih.gov/BRB-ArrayTools.html
CAGED	Clustering by expression dynamics	http://www.kebab.tch.harvard.edu/caged
Cleaver	k -means clustering, principal components analysis, and classification	http://www.classify.stanford.edu/
Cluster and TreeView	Dendrograms, self-organizing maps, k -means clustering, principal components analysis	http://www.rana.lbl.gov/EisenSoftware.htm
Cyber-T	Differential gene expression	http://www.genomics.biochem.uci.edu/genex/cybert
dChip	Differential gene expression	http://www.dchip.org/
Equalizer	Microarray normalization	http://www.organogenesis.ucsd.edu/TheEqualizer.htm
Expression Profiler	Dendrograms, with many dissimilarity measures	http://www.ep.ebi.ac.uk/
GeneCluster	Self-organizing maps	http://www-genome.wi.mit.edu/MPR/GeneCluster/GeneCluster.html
J-Express	Dendrograms,	http://www.ii.uib.no/~bjarted/jexpress/main.html

	self-organising maps, principal components analysis, <i>k</i> -means clustering	
MAExplorer	Differential gene expression, scatterplots, <i>k</i> -means clustering, dendrograms	http://www-lecb.ncifcrf.gov/MAExplorer
Multiple Experiment Viewer	Many normalization, clustering, dissimilarity measures, and graphical options	http://www.tigr.org/softlab
RelNet	Relevance networks	http://www.book.chip.org/
SAM	Excel software, compares expression to clinical parameters	http://www-stat.stanford.edu/~tibs/SAM/index.html
ScanAlyze	Microarray spot detection	http://www.rana.lbl.gov/EisenSoftware.htm
SpotFinder	Microarray spot detection	http://www.tigr.org/softlab
XCluster	Dendrograms, self-organizing maps, <i>k</i> -means clustering	http://www.genome-www.stanford.edu/~sherlock/cluster.html

Recent reviews of commercially available software tools in this domain include [29, 16, 35].

7.4 Investing in the Future of the Genomic Enterprise

If microarray technology is going to be commoditized such that it can be used routinely as a measurement system, even in relatively unskilled hands, then what will distinguish that particular genomic enterprise with aspirations after excellence?

The most pressing need and most obvious quality that will distinguish the best functional genomics investigations from the rest will be the nature of the investigators participating. As to which investigators these might be, the genomic pipeline outlined in section 1.5 points to the kinds of expertise that will be most in demand. There will need to be top-flight basic biologists, pathologists, clinicians with access to populations of interest, all capable of and motivated toward posing the high-yield clinical questions that are likely to drive many of these investigations. Based on the experience of several of the more successful laboratories, it seems that at least for the near future,

the largest and most pressing need for expertise is in the area of bioinformatics. A typical successful laboratory will have approximately two thirds of its investigators drawn from the ranks of bioinformaticians. This proportion is likely to be maintained for at least the next 5 to 10 years, until the analytic tools are much more robust and the analytic approach is systematized and standardized. At that juncture, it may be that bioinformatic analyses will be packageable in the same way as common biostatistical analyses for clinical trials are packaged in programs such as SAS (Statistical Analysis System) or SPSS (Statistical Package for the Social Sciences). Of course, the availability of such packages allows neophytes to perform uninformed, flawed, but superficially good-looking analyses, a problem that most biostatisticians are all too aware.

For the foreseeable future then, the most significant investments in expertise will probably have to be made in the area of bioinformatics. As individuals with training in biology and computer science and and medicine are rare and hard to come by, most laboratories are currently hiring individuals with strong computational skills, such as computational physicists, mathematicians, and statisticians, and providing them with the intellectual tools and background to become bioinformaticians. This process is not without its problems and uneven results.

A more lasting and productive process is to develop training programs that will produce the necessary interdisciplinary education in biology, computer science, statistics and probability theory, to mention just a few desirable knowledge bases that those individuals should be equipped with. The National Institutes of Health have recognized this need and are beginning to fund several training programs. Locally, the Division of Health, Sciences, and Technology (HST) at Harvard University and the Massachusetts Institute of Technology has developed a Bioinformatics and Integrative Genomics (BIG) training program. BIG is designed to take students with strong, quantitative undergraduate backgrounds such as chemical engineering, mechanical engineering, physics, mathematics, or computer science and then provide them with a curriculum that involves clinical exposure at Harvard Medical School during clinical rotations, joint classes with Harvard Medical School students, as well as basic biology courses. This and advanced topics in probability theory, computer science, and other genomics-related courses will provide these students with the basis to complete their doctoral thesis at HST in genomic science and subsequently to assume leadership roles in genomic research.

Even now, when many of the genomic measurement techniques have in fact not yet been commoditized, it is apparent that the most expensive and labor-intensive part of the genomic enterprise will be the accurate and large-scale phenotyping of human populations and model organisms. Specifically, it takes a lot more time and personnel to accurately document all the characteristics of a human pedigree with all the attendant clinical findings and measurements of each member of the pedigree, than to simply genotype or obtain expression measurements on each member of the pedigree. Consequently, any substantive forward-looking genomic effort will involve a carefully thought-out plan for the acquisition of detailed phenotypic information for all obtained samples. This activity has previously been most associated with genetic epidemiologists who have carefully obtained prospective perspective gathered phenotypic data in well-designed studies such as the Nurses' Health Study, the Physicians' Health Study, and the Framingham Heart Study.

With motivations similar to those of epidemiologists, prospective annotation and phenotyping are indeed what are required. Those who expect to be able to mine the existing clinical records of a patient's care for the necessary data are going to be disappointed. Even if they can manage the issues of patient consent and data release, the quality of the clinical record and its emphasis is such that most of the data needed for effective and reliable phenotyping of the clinical population are unavailable. Consequently, any serious attempt to achieve high-throughput phenotyping for a substantive study will require that a laboratory or institution invest in the necessary infrastructure for such activities. These activities are labor-intensive and therefore not inexpensive. Specifically, it

requires the use of trained personnel, such as genetic counselors, physician assistants, or highly trained research assistants, who are aware of the complexity of human phenotyping. As human subjects have relatively long lives as compared to microbial or murine models, these activities have to be supported adequately over decades. High-throughput human phenotyping also requires that the institutional laboratory put in place the consent and data release procedures that will enable the research to be done subsequently, and to ensure that these procedures are consistent with the aforementioned rules and regulations.

The importance of prospective phenotyping activity cannot be overemphasized. Without it, we will never be able to bridge the gap between the observed and measured phenotypic and genomic data and the clinical import of these data. This is a far less sexy activity than other aspects of the functional genomic enterprise, but it ultimately will be what distinguishes those efforts that are the most successful and clinically relevant. Who best performs such prospective high-throughput phenotyping will determine whether the locus of human-related genomics will be under the administrative 2gis of hospitals, medical schools, the industrial sector, or governmental organizations.

Glossary

Constructed in part from the glossary in Primer on Molecular Genetics, U. S. Dept. of Energy, Human Genome Project, 1992.

A

Alternative Splicing Product

Variation in the way pre-mRNA is spliced together. Introns are typically spliced from pre-mRNA and the remaining exons are pieced together to form a contiguous transcript of mRNA (see Exon). However, the same set of introns may not always be spliced out, and the resultant mRNA may have a different combination of introns and exons. These alternative versions of the mRNA result in different downstream proteins being formed (alternative splicing products). Alternative splicing products are only one reason why the number of gene products is much larger than the number of genes. It is also a reason why an oligonucleotide designed to hybridize with a subsegment of transcription products of a particular gene may fail to measure the transcription of that gene.

B

Base Pair

The chemical structure that forms the units of DNA and RNA and that encode genetic information. The bases that make up the base pairs are adenine (A), guanine (G), thymine (T), cytosine (C), and uracil (U) (see DNA).

Binding Site

Regions on macromolecules which are sequence and structure specific and complementary that enable any two or more macromolecules to mechanically bind and interact.

Bioinformatics

The collection, organization, and analysis of large amounts of biological data, using networks of computers and databases. Historically, bioinformatics concerned itself with the analysis of the sequences of genes and their products (proteins), but the field has since expanded to the management, processing, analysis, and visualization of large quantities of

data from genomics, proteomics, drug screening, and medicinal chemistry. Bioinformatics also includes the integration and "mining" (detailed searching) of the ever-expanding databases of information from these disciplines.

C

Chromosome

One of the physically separate segments that together forms the genome, or total genetic material, of a cell. Chromosomes are long strands of genetic material, or DNA, that have been packaged and compressed by wrapping around proteins. The number and size of chromosomes varies from species to species. In humans, there are 23 pairs of chromosomes (a pair has one chromosome from each parent). One pair are called the sex chromosomes because they contain genes that determine sex. The chromosome carrying the male determining genes is designated Y and the corresponding female one is the X chromosome. The remaining pairs are called autosomes. Chromosome 1 is the largest and chromosome 22 the smallest. Each chromosome has two "arms" designated p and q.

Complementary DNA (cDNA)

DNA that is synthesized from a messenger RNA template; the single-stranded form is often used as a probe in physical mapping or for detecting RNA. Since cDNA is constructed from messenger RNA (after introns have been spliced out), it does not contain introns.

Cosmid

Artificially constructed cloning vector containing the *cos* gene of lambda phage (a virus). Cosmids can be packaged in lambda phage particles for infection into *E. coli*; this permits cloning of larger DNA fragments (up to 45 kb) than can be introduced into bacterial hosts in plasmid vectors.

D

Deoxyribonucleic Acid (DNA)

The chemical that forms the basis of the genetic material in virtually all living organisms. Structurally, DNA is composed of two strands that intertwine to form a springlike structure called the double helix. Attached to each backbone are chemical structures called bases (or nucleotides), which protrude away from the backbone toward the center of the helix, and which come in four types—adenine, cytosine, guanine, and thymine (designated A, C, G, and T). In DNA, cytosine only forms optimal hydrogen bonding with guanine, and adenine only with thymine. These interactions across the many nucleotides in each strand hold the two strands together.

E

Electrophoresis

The use of electrical fields to separate charged biomolecules such as DNA, RNA, and proteins. DNA and RNA carry a net negative charge because of the numerous phosphate groups in their structure. In the process of gel electrophoresis, these biomolecules are put into wells of a solid matrix typically made of an inert substance such as agarose. When this gel is placed into a bath and an electrical charge applied across the gel, the biomolecules migrate and separate according to size in proportion to the amount of charge they carry. The biomolecules can be stained for viewing and isolated and purified from the gels for further

analysis. Electrophoresis can be used to isolate pure biomolecules from a mixture or to analyze biomolecules (such as for DNA sequencing).

Enzyme

A protein that catalyzes chemical reactions in a living cell. Enzymes are protein molecules whose function it is to speed the making and breaking of chemical bonds required for essential physiochemical reactions.

Exon

The protein-coding DNA sequences of a gene (see Intron).

Expressed Sequence Tags (EST)

A set of single-pass sequenced cDNAs from an mRNA population derived from a specified cell population (e.g., a tumor or a particular muscle type). ESTs provide one estimate of which DNA sequences are actually transcribed and therefore potentially functional as genes.

F

Functional genomics

The use of genetic technology to determine the function of newly discovered genes by determining their role in one or more model organisms. Functional genomics uses as its starting point the isolated gene whose function is to be determined, and then selects a model organism in which a homolog of that gene exists. This model organism can be as simple as a yeast cell or as complex as a nematode worm, fruit fly, or even a mouse.

G

Gene

The basic unit of heredity; the sequence of DNA that encodes all the information to make a protein. A gene may be "activated" or "switched on" to make protein—this activation is referred to as gene expression—by these proteins which control when, where, and how much protein is expressed from the gene. In the human genome, there are an estimated 30,000 genes (although recent studies suggest a larger number).

Gene product

The biochemical material, either RNA or protein, resulting from expression of a gene. The amount of gene product is used to measure how active a gene is; abnormal amounts can be correlated with disease-causing alleles. As gene products include all the alternative splicing products, there are estimated to be at least 100,000 distinct such products.

Gene therapy

The technology that uses genetic material for therapeutic purposes. This genetic material can be in the form of a gene, representative of a gene or cDNA, RNA, or even a small fragment of a gene. The introduced genetic material can be therapeutic in several ways: It can make a protein that is defective or missing in the patient's cells (as would be the case in a genetic disorder), or that will correct or modify a particular cellular function, or that elicits an immune response.

H

High-throughput screening

The use of miniaturized, robotics-based technology to screen large compound libraries against an isolated target protein, cell, or tissue in order to identify binders that may be

potential new drugs. In conjunction with genomics and combinatorial chemistry (the production of large numbers of medically relevant compounds), high-throughput screening has revolutionized the capacity of pharmaceutical and biotechnology companies to identify potential new drugs. Typically, high-throughput screening has relied on 96-well plates as the standard, although higher-density formats (384, 1536) are possible. Recently, advances in miniaturization and micro fluidics have allowed screening of up to 100,000 compounds against a target on a single chip daily.

Hybridization

The interaction of complementary nucleic acid strands. Since DNA is a double-stranded structure held together by complementary interactions (in which C always binds to G, and A to T), complementary strands favorably reanneal or "hybridize" to each other when separated.

I

Intron

Noncoding portion of the gene that is spliced out from the nascent RNA transcript in the process of making an mRNA transcript. Frequently includes regulator elements (*i.e.*, binding sites) in addition to those of the promoter.

M

Microfluidics

Microfluidics is the miniaturization of fluid-based biochemistry to very small volumes such that a large series of chemical reactions can be performed in parallel in a microarray format. The essential issue with developing microfluidics based devices is the ability to maintain adequate access to the solutions needed in each element of the array. Several companies have manufactured "labs-on-a-chip" that allow the high-throughput chemistries needed in screening, synthesis, and probing of biological molecules.

Mutation

Any alteration to DNA that can potentially result in a change in the function of one or more genes. Mutations can be a change in a single base of DNA (point mutation) or a loss of base pairs (deletion) affecting a single gene, or a movement of chromosomal regions (translocation) affecting many genes. Some changes in DNA occur naturally and lead to no harmful effects; these changes in a population are called polymorphisms.

N

Northern Blot

RNA from a sample is spatially separated and distributed by mass on a gel. Radioactively labelled DNA or RNA strands with sequence complementary to the RNA segments from the sample are used to locate the position of those RNA segments.

O

Oligonucleotide

A short molecule consisting of several linked nucleotides (typically between 10 and 60) chained together and attached by covalent bonds.

Open Reading Frame

Regions in a nucleotide sequence that are bounded by start and stop codons and are therefore possible gene coding regions.

P

Pharmacogenomics

Pharmacogenomics is the study of the pharmacological response to a drug by a population based on the genetic variation of that population. It has long been known that different individuals in a population respond to the same drug differently, and that these variations are due to variations in the molecular receptors being affected by the drug, or to differences in metabolic enzymes that clear the drug. Pharmacogenomics is the science of studying these variations at the molecular level. Applications of pharmacogenomics include reducing side effects, customizing drugs, improvement of clinical trials, and the rescue of some drugs that have been banned due to severe side effects in a small percentage of the eligible population.

Polymorphism

Difference in nucleotide sequence among individuals. Single nucleotide polymorphisms (SNP) are thought to be the principal component of inherited genetic variation. There is approximately one SNP every 300 to 1000 nucleotide of the human genome.

Probe

Any biochemical agent that is labeled or tagged in some way so that it can be used to identify or isolate a gene, RNA, or protein. Typically refers to the immobilized specified nucleic acid in a detection system [8].

Promoter

Regions of DNA typically upstream or before a gene which contain transcription factor binding sites (see Transcription Factor), but also include elements for specific initiation of transcription. That is, there are stretches of DNA to which transcription factors may bind and thereby modulate transcription but which are not promoters, because they do not include the elements of transcriptional initiation.

Protein family

A set of proteins that share a common evolutionary origin reflected by their relatedness in function, which is usually reflected by similarities in sequence, or in primary, secondary, or tertiary structure. A set of proteins with related structure and function.

Proteomics

The study of the entire protein complement or "protein universe" of the cell. Mirroring genomics, proteomics aims to determine the entire suite of expressed proteins in a cell. This includes determining the number, level, and turnover of all expressed proteins, their sequence and any post-translational modifications to the sequence, and protein-protein and protein-other molecule interactions within the cell, across the cell membrane, and among (secreted) proteins.

Polymerase chain reaction (PCR)

A technique used to amplify or generate large amounts of replica DNA or a segment of any DNA whose "flanking" sequences are known. Oligonucleotide primers that bind these flanking sequences are used by an enzyme to copy the sequence in between the primers. Cycles of heat to break apart the DNA strands, cooling to allow the primers to bind, and heating again to allow the enzyme to copy the intervening sequence lead to doubling of the DNA present at each cycle.

S

Serial Analysis of Gene Expression (SAGE)

The use of short 10–14 bp sequence tags linked together to identify sequences in a sample. The count of the instances of each sequence in a specific sample is a number that bears (in theory) a direct relationship to the number of transcripts of a particular genes (see section 7.1.4).

Single nucleotide polymorphism (SNP)

Variations in single base pairs scattered throughout the human genome that serve as measures of genetic diversity in humans. About 1 million SNPs are estimated to be present in the human genome, and SNPs are useful markers for gene mapping and disease studies.

Southern Blot

DNA from a sample is cut with restriction enzymes and the position of the fragments (*e.g.* on a gel) is determined by the fragment's molecular weight. Complementary strands of radioactively labelled DNA are used to identify the position of the DNA fragments on the gel.

T

Transcription Factor

A molecule, typically a protein, which binds to promoter (see Promoter) or other DNA binding sites with some regulatory role in transcription. The binding (or unbinding) of a transcription factor from a promoter eventually leads to a change of transcription activity in the gene controlled by that promoter.

Target

The free nucleic acid that is being interrogated [8].

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Chapter 2: Experimental Design

Figure 2.1: Three experiments within the multi-dimensional representation of experiment design space. Experiment design space defines all the possible stimuli or conditions to which a particular biological system could be subjected. Shown here is an experiment design space that is concerned with insulin signaling in different tissues, in different mouse "knockout" models, with different levels of insulin.

Figure 2.2: Expression space in which gene expression is loosely versus tightly coupled. If genes are tightly coupled in the expression space, they will tend to occupy a small subspace of expression space over any set of experiments. If, however, these genes are loosely coupled or even causally unrelated, then their expression levels will have little relation to one another and therefore tend to be scattered over a greater volume of the expression space.

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Chapter 3: Microarray Measurements to Analyses

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Figure 3.26: Dynamic relationships between genes. (a) The expressed product of Gene A binds an enhancer region that increases transcription of Gene B. (b) Gene B's initial expression level before being affected by Gene A can vary throughout the experiment. As a result, measuring the correlation between the absolute levels of Genes A and B will not reveal the underlying enhancement relationship between the two. Instead, this can only be done by analyzing the expression dynamics—the change in expression level of Gene B in relation to the expression level of Gene A. (Derived from Reis *et al.* [150].)

Figure 3.27: Negative dynamic correlation. The distribution of slopes of MAD3 and EXM2 plotted one against another.

Chapter 4: Genomic Data–Mining Techniques

Figure 4.1: Example of how a single point can distort overall correlation. On the left: the scatterplot shows a negative correlation. The scatterplot on the right is identical, save for the additional point added. If the values of even a single point are high enough, the correlation coefficient can be altered (though the variance of the correlation coefficient, if calculated, would be higher). This is primarily because using the correlation coefficient assumes values distributed normally.

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Figure 4.11: Graphical convention for relevance networks. To optimize the hypothesis generation process, relevance network display software contains ties from microarray accession number to GenBank, UniGene, LocusLink, Gene Ontology, and other databases

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Chapter 5: Bio–Ontologies, Data Models, Nomenclature

Figure 5.1: The Molecular Function Ontology of GO. (From Ashburner *et al.* [13].)

Figure 5.2: The Cellular Component Ontology of GO. (From Ashburner *et al.* [13].)

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